
Algebraic K-Theory

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

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Algebraic K -theory attaches to a ring R a sequence of abelian groups $K_0(R), K_1(R), K_2(R), \dots$ that measure, in successively subtler ways, how far linear algebra over R departs from linear algebra over a field. Over a field every module is free and every invertible matrix is a product of elementary ones; over a general ring neither holds, and the K -groups are precisely the record of the obstructions. The group $K_0(R)$ is the Grothendieck group of finitely generated projective modules, measuring how far projectives are from free; $K_1(R) = GL(R)/E(R)$ is the Whitehead group, measuring how far invertible matrices are from being products of elementary ones (the determinant, and more, when R is commutative); and $K_2(R)$ records the relations among the elementary matrices.

In each degree the construction springs from one impulse, the same that builds the integers from the natural numbers: complete a monoid to a group by adjoining inverses. This is transparent for K_0 , where the monoid is projective modules under direct sum. Quillen's insight was that the whole tower $K_0, K_1, K_2, \dots$ is governed by a single homotopy-theoretic construction: the higher groups are the homotopy groups of a space built from R , whether by the plus construction applied to $BGL(R)$ or by the Q -construction on the category of projective modules. The two agree, and they reproduce the classical K_0, K_1, K_2 in the bottom three degrees.

Unlike its topological sibling ([5]), algebraic K -theory is not periodic: the groups $K_n(R)$ form an infinite tower of genuinely different invariants whose computation, even for the ring of integers, runs to the frontier of mathematics. Their arithmetic connects to the heart of number theory, the K -groups of rings of integers encoding special values of L -functions and étale cohomology. The bridge to the topological side is the Serre–Swan theorem, under which $K_0(C(X)) = K^0(X)$ ([**NateCStarAlgebras**]), so that the algebraic and topological theories coincide exactly in degree zero before diverging above it.

This book builds the subject from the ground up. We assume comfort with rings and modules ([**NateCommAlg**], [**NateCoreAlgebra**]) and with homological algebra and exact categories ([7]); the higher theory additionally draws on the homotopy groups and classifying spaces of [10]. Throughout, rings are associative with $1 \neq 0$, and modules are right modules unless stated otherwise, so that a homomorphism $R^n \rightarrow R^m$ is an $m \times n$ matrix acting on column vectors. We begin with projective modules and the Picard group, build K_0 and its ties to homology and Morita equivalence, climb to K_1 and K_2 , and culminate in Quillen's higher theory.

Projective modules

If R is a field or a divisor ring, every R -module is a vector-space which is algebraically characterized up to dimension. More generally, a free R -module is a module that is R -module isomorphic to $R^{\oplus I}$ for some indexing set I . Any R -module homomorphism between two free modules is based on where chosen basis of the domain is represented by the chosen basis of the codomain; namely it can be represented by a matrix (or the generalization thereof for infinite dimension). These are the simplest modules that can be put over a ring. A natural question would be to see what are the possible “next weakest” conditions we can put, and what rings R “force” this condition. We shall explore a few of these conditions, one important one shall be for a module to be *projective* which we shall show how this could be thought of as being *locally free*. Measuring how this structure differs from being free will be the beginnings of the study of the K_0 -group in the next chapter.

1.1 IBP, Stably Free, and Projective Modules

If R is a field or a divisor ring, then $R^n \cong R^m$ implies $n = m$. If R is commutative this is still the case (consider $R/\mathfrak{m}$ by a maximal ideal), group rings (for finite groups), and any (left)-Noetherian ring satisfies this property as well (using a version of Nakayama’s lemma can prove this, see [1]). This property is *not* true for general rings:

Example 1.1.1: Failure of Invariant Basis Number

let R be a commutative ring and let M be the $\mathbb{N} \times \mathbb{N}$ matrices where each column has finitely many nonzero entries, this is sometimes denoted $CFM_{\mathbb{N}}(R)$. Then $CFM_{\mathbb{N}}(R) \cong CFM_{\mathbb{N}}(R)^2$ via the map:

$$\varphi : CFM_{\mathbb{N}}(R) \rightarrow CFM_{\mathbb{N}}(R)^2 \quad M \mapsto (\text{odd columns of } M, \text{ even columns of } M)$$

Another example the reader could work through is $\text{End}_k(k^{\oplus \mathbb{N}})$.

Definition 1.1.2: Invariant Basis Property

Let R be a ring. Then we say that R satisfies the (right) *invariant basis property* (IBP) if

$$R^n \cong R^m \quad \Rightarrow \quad n = m$$

If R satisfies IBP, then the rank of a free R -module M is an invariant, independent of the choice of basis.

Proposition 1.1.3: Necessary Condition For Ibp

Let R be a ring. Then R satisfies IBP if there exists an injective ring map $f : R \rightarrow F$ from R to a field (or division ring) F

Proof:

Given such a map, notice that any basis for a free R -module M is the basis for the vector-space $V = M \otimes_R F$, and V is independent of choice of basis, and hence any two bases must have the same cardinality.

Kernel's of a surjective R -module homomorphisms $\varphi : R^n \rightarrow R^m$ need not be free, however they must be stably free:

Definition 1.1.4: Stably Free

Let P be an R -module. Then P is called *stably free* of rank $n - m$ if there exists n, m such that

$$P \oplus R^m \cong R^n$$

If R satisfies IBP, then the rank of a stably free module is independent of the choice of n and m .

Certainly, the kernel stably free as R^m is free as we must have a split sequence (see [1]). Not all R -modules are stably free (or more interestingly, not all stably free modules are free). The notion of stably free seems very similar to *projective*, however we have not specified the condition for the 2nd summand in the definition of projective modules and hence it is a slightly weaker condition. We shall get to projective modules shortly.

Let us take a moment to consider the case where an R -module P is stably free of rank 1, that is $P \oplus R \cong R^n$. In this case, we may ask when can we “cancel” the R on each side, a fruitful discussion considering the K -theory we are building up to. In this case, let σ be a row vector of the right hand side, and call it the *unimodular row*. We get the following interesting set of equivalences:

1. σ is unimodular
2. $R^n \cong P \oplus R$ where $P = \ker(\pi)$ where $\pi : R^n \rightarrow R$ (given by σ)
3. $R = r_1 R + \cdots + r_n R$
4. $1 = r_1 s_1 + \cdots + r_n s_n$ for appropriate $s_i \in R$

Note that if P were free in $R^n \cong P \oplus R$, then a basis of P would give a new basis for R^n , which corresponds to an invertible matrix (say g) whose first row is the unimodular row $\sigma : R^n \rightarrow R$ corresponding to P . From this we get the following citation:

Proposition 1.1.5: Free and Unimodular

Let P be an R -module that stably free of rank 1. Then P is free if and only if the corresponding unimodular row may be completed to an invertible matrix.

Proof:

combine the above discussion.

If R is commutative, then every unimodular row of length 2 may be completed: if $r_1s_1 + r_2s_2 = 1$, then the invertible matrix is:

$$\begin{pmatrix} r_1 & r_2 \\ -s_2 & s_1 \end{pmatrix}$$

And so, if $R^2 \cong R \oplus P$, then $P \cong R$. We shall get stronger versions of this result later (note here though that commutativity was important, this fails for unimodular row of length two in a non-commutative ring).

In dimension 3, we can already find counter example:

Example 1.1.6: Stably Free But Not Free

Take

$$R = \frac{\mathbb{R}[x, y, z]}{(x^2 + y^2 + z^2 - 1)}$$

Note that every element $(f, g, h) \in R^3$ gives a vector-field in $\mathbb{R}^3$. Let σ be the unimodular row $\sigma = (x, y, z)$. Then $1s$ is the vector-field pointing radially outward. If P were free, a basis of P would yield two tangent vector fields on S^2 which are linearly independent at every point of S^2 (since together with σ , they span the tangent space of 3-space at each point). But this contradicts the infamous Hairy-sphere theorem^a.

^aOr more humorously *Hairy ball theorem*

Let us now return to the discussion of projective modules.

Definition 1.1.7: Projective Module

Let P be an R -module. Then P is *projective* if there exists an R -module Q such that $P \oplus Q$ is a free R -module.

See [1] for details on the equivalence and examples, on important one is that there is always a lift in the following commutative diagram:

$$\begin{array}{ccccc} & & P & & \\ & \swarrow \exists! & \downarrow g & & \\ M & \xrightarrow{f} & N & \xrightarrow{0} & 0 \end{array}$$

where f is surjective. Naturally, free and stably free module are projective. A more nuanced example is that all $M_n(k)$ -modules (with $n > 1$) are projective but not free (which can be deduced using Artin-Wedderburn). A more canonical set of examples comes from vector bundles. Take $R = C^0(X)$ to be the set of continuous real valued function on a compact topological space X , and let $\Psi : E \rightarrow X$ be a vector-bundle. Then the set $\Gamma(E)$ of continuous sections naturally forms a projective R -module. Let us show this (see [2] for the algebraic geometry perspective of the equivalence between vector bundle and locally free sheaves). The trivial bundle is $\mathbb{R}^n \times X \rightarrow X$ and we get $\Gamma(\mathbb{R}^n \times X) = R^n$. Now assume $\Psi : E \rightarrow X$ is a non-trivial vector-bundles (up to vector-bundle isomorphism). Then if $\Gamma(E)$ were free, there would be sections $\{s_1, \dots, s_n\}$ that would form a basis $f : T^n \rightarrow E$ which would give an isomorphism $\Gamma(T^n) = \Gamma(E)$. As the kernel and cokernel of bundles of f have no nonzero sections, they must vanish, and so f would be an isomorphism, and so $\Psi : E \rightarrow X$ would in fact be trivial.

If X is compact, then the famous *Swan's Theorem* gives an equivalence between the category of projective modules over $C^0(X)$ and the category of vector bundles over X , showing that vector bundles are the geometric interpretation of projective modules, which motivates similar discussions over sheaves of module for a scheme.

Proposition 1.1.8: Finitely Generated Projective Modules and Idempotents

Let P be a finitely generated projective R -module. Then P corresponds to an idempotent element in $M_n(R)$ for appropriate n .

Proof:

If P is finitely generated, then $P \oplus Q \cong R^n$, meaning there is an idempotent element in R which can be associated to an element of $M_n(R)$. To find this element, we can take the natural projection-inclusion

$$R^n \rightarrow P \rightarrow R^n$$

This map is an idempotent in $e \in M_n(R)$ (when representing this composition as a matrix under a choice of basis). Labeling this composition e , we see that $e(R^n) = P$.

If P is a projective R -module over a PID, then it is in fact free (as given by the structure theorem for modules over PID¹). Another famous class of modules where all finitely generated projectives are free comes from Quillen and Suslin stating that finitely generated modules over a polynomial ring (or a Laurent polynomial ring) over a PID is free². This then gives the result mentioned earlier about group rings over finite abelian groups, for $\mathbb{Z}[G]$ is the Laurent polynomial ring $\mathbb{Z}[x, x^{-1}, \dots, z, z^{-1}]$. On the other hand, if G is *not* abelian (but has the conditions of being torsion-free and nilpotent), then any projective $\mathbb{Z}[G]$ -module P is stably free but not free, in particular that:

$$P \oplus \mathbb{Z}[G] \cong (\mathbb{Z}[G])^2$$

As of writing this book, it is unknown if the condition “nilpotent” can be dropped and produce a similar (or the same) result.

Another important class of finitely generated projective R -module that are always free are those over *local rings*. For informative purposes, the result can in fact be extended to infinitely generated

¹Note this even works for infinitely generated projective modules, something briefly outlined in some exercises in [1]

²This result appears in [2] where it is used to show that all locally free $\mathcal{O}_X$ -modules on affine space must be free

projective R -modules over local rings (Kaplansky gave such a proof). The following is given as a reminder:

Proposition 1.1.9: Projective Over Local Ring, then Free

Let R be a local ring. Then every finitely generated projective R -module is free, where

$$P \cong R^n \quad p = \dim_{R/\mathfrak{m}}(P/\mathfrak{m}P)$$

Proof:

see [6, chapter 20.3]

Proposition 1.1.10: Free At Stalks

Let R be a commutative ring, $\mathfrak{p}$ a prime ideal, and P a finitely generated projective R -module. Then the localization $P_{\mathfrak{p}}$ is isomorphic to $(R_{\mathfrak{p}})^n$ for some $n \geq 0$. Furthermore, there exists some $s \in R \setminus \mathfrak{p}$ where the localization away from s is free:

$$P \left[\frac{1}{s} \right] \cong R \left[\frac{1}{s} \right]^n$$

And we further get that $P_{\mathfrak{p}'} \cong (R_{\mathfrak{p}'})^n$ for every prime ideal $\mathfrak{p}'$ of R not containing s

Proof:

As P is projective, let $P \oplus Q = R^m$. Then $P_{\mathfrak{p}} \oplus Q_{\mathfrak{p}} = (R_{\mathfrak{p}})^m$, hence $P_{\mathfrak{p}}$ is a finitely generated projective $R_{\mathfrak{p}}$ -module. As $R_{\mathfrak{p}}$ is local, $P_{\mathfrak{p}}$ is in fact free. Now, every element of $P_{\mathfrak{p}}$ is of the form p/s , for $p \in P$ and $s \in R \setminus \mathfrak{p}$. By choosing a basis for $P_{\mathfrak{p}}$, we may clear the denominators to get an R -module homomorphism $\varphi : R^n \rightarrow P$ that become an isomorphism when localizing at $\mathfrak{p}$.

Now, as $\text{coker}(\varphi)$ is a finitely generated R -module which vanishes upon localization, it must be annihilated by some $s \in R \setminus \mathfrak{p}$. Given this s , we get a surjective map:

$$\varphi \left[\frac{1}{s} \right] : \left(R \left[\frac{1}{s} \right] \right)^n \rightarrow P \left[\frac{1}{s} \right]$$

As $P \left[\frac{1}{s} \right]$ is projective (as localization preserves projectiveness), $\left(R \left[\frac{1}{s} \right] \right)^n$ is isomorphic to the direct sum of $P \left[\frac{1}{s} \right]$ and a finitely generated $R \left[\frac{1}{s} \right]$ -module M (where $M_{\mathfrak{p}} = 0$). As M is annihilated by some $t \in R \setminus \mathfrak{p}$, we get:

$$\varphi \left[\frac{1}{st} \right] : \left(R \left[\frac{1}{st} \right] \right)^n \xrightarrow{\cong} P \left[\frac{1}{st} \right]$$

completing the proof.

We should next define the “vector bundle” structure that we can put on modules. To that end, we will have to assume some algebraic geometry, nothing more than some knowledge on spectra (see [6, chapter 21.7] or [2, chapter 2.1]). Let $\text{Spec } R$ be the collection of prime ideals of R , and define the

Zariski topology with basic open sets given by:

$$D(s) = \{\mathfrak{p} \in \operatorname{Spec} R \mid s \notin \mathfrak{p}\} \cong \operatorname{Spec} R \left[\frac{1}{s} \right]$$

Definition 1.1.11: Rank At Prime Ideal

Let R be a commutative ring and M be a finitely generated R -module. Then the *rank* at $\mathfrak{p}$ is:

$$\operatorname{rank}_{\mathfrak{p}}(M) = \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$$

Note that $M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}} \cong \kappa(\mathfrak{p})^{\operatorname{rank}_{\mathfrak{p}}(M)}$, hence $\operatorname{rank}_{\mathfrak{p}}(M)$ is the minimal number of generators for $M_{\mathfrak{p}}$. If P is a finitely generated projective R -module, then $\operatorname{rank}(P) : \mathfrak{p} \mapsto \operatorname{rank}_{\mathfrak{p}}(P)$ is a continuous function between the $\operatorname{Spec} R$ with the Zariski topology and $\mathbb{N}$ with the discrete topology.

If $\operatorname{Spec} R$ is connected, then $\operatorname{rank}(P) : \operatorname{Spec} R \rightarrow \mathbb{N}$ must always be a constant function, hence every finitely generated projective R -module has constant rank. In the simplest case, R is an integral domain, then the rank is constant with:

$$\operatorname{rank}(P) = \dim_{\operatorname{Frac}(R)}(P \otimes_R \operatorname{Frac}(R))$$

Note that if P is of constant rank, then it is finitely generated³. If M is not projective, then $\operatorname{rank}(M)$ need not be continuous: consider $R = \mathbb{Z}$ with $M = \mathbb{Z}/p\mathbb{Z}$.

We conclude this section with an important observation about the “lack” of importance of higher dimensions. For the following, recall that M, M' are stably isomorphic if $M \oplus R^m \cong M' \oplus R^m$ for some $m \geq 0$.

Theorem 1.1.12: Base Serre Cancellation

Let R be a commutative Noetherian ring with $\operatorname{kdim}(R) = d$, and let P be a projective R -module of constant rank $n > d$. Then:

1. $P \cong P_0 \oplus R^{n-d}$ for some projective R -module P_0 of constant rank d
2. If P is stably isomorphic to P' , then $P \cong P'$
3. For all M, M' , if $P \oplus M$ is stably isomorphic to M' , then

$$P \oplus M \cong M'$$

The idea is that any extra structure when considering higher dimension vector bundles only adds “trivial” information, making the information from vector-bundles ties to the local information of the base space.

Proof:

see H. Bass, Algebraic K-theory, Benjamin.

³Prove it for the case of rank 1, then for the rank r case show that $\Lambda^r(P)$ is finitely generated

1.1.1 *Milnor Square

The following material is going to be more useful in section 2.5 when looking over Mayer-Vietoris sequences, however it appears when studying the Picard group of a ring. The idea is that the Milnor square gives us a way of gluing together different parts of a ring akin to the Mayer-Vietoris sequence from algebraic topology (see [10]). Recall from algebraic topology that this construction was express done to be computationally rather simple. We “algebras” this technique by looking at a particularly nice way to “patch” two rings in such a way that will allow for some simple computations of various invariance (ex. Picard group, K -groups, etc)

Let R be a ring, $I \subseteq R$ an ideal, $\varphi : R \rightarrow S$ a ring homomorphism that maps I isomorphically to an ideal in S , which we shall also label I . Then R is the pullback of S and R/I in the following diagram:

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R/I & \longrightarrow & S/I \end{array}$$

This square is called the *Milnor square*. In algebraic geometry, this represents R as the fibered product of S and R/I over S/I .

We can patch modules given a Milnor square. Let M_1 be an S -module and M_2 be an R/I -module. The Milnor square defines the natural S/I -module isomorphism:

$$g : M_2 \otimes_{R/I} S/I \cong M_1/IM_1$$

Then M can be thought of as a triple (M_1, g, M_2) and is a subset of $M_1 \times M_2$ given by the kernel of the R -module homomorphism:

$$M_1 \times M_2 \rightarrow M_1/IM_1 \quad (m_1, m_2) \mapsto \bar{m}_1 - g(\bar{f}(m_2))$$

Example 1.1.13: Milnor Patching A Module

Take S^n and $(R/I)^n$ with $1 \in \text{GL}_n(S/I)$. Consider $\bar{f} : R/I \rightarrow S/I$ Then:

$$\left\{ (\vec{s}, r + I) \mid \vec{s} = \bar{f}(r + I) \right\}$$

Hence, we have patched together two vector-spaces of the same dimension over different rings that can glue over R/I , or geometrically we have two trivial bundles over R/I and S and we glued those bundles to create a bundle over R .

The most important properties of Milnor patching that we shall use is summarized in the following:

Theorem 1.1.14: Milnor patching

Let

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ R/I & \longrightarrow & S/I \end{array}$$

be a commutative diagram where $I \subseteq R$ is identified with its image in S and also represented as I (i.e. a Milnor square). Then:

1. If P is obtained by patching together a finitely generated projective S -module P_1 and a finitely generated projective R/I -module P_2 , then P is a finitely generated projective R -module.
2. $P \otimes_R S \cong P_1$ and $P/IP \cong P_2$.
3. Every finitely generated projective R -module arises in this way.
4. If P is obtained by patching free modules along $g \in GL_n(S/I)$ and Q is obtained by patching free modules along g^{-1} , then $P \oplus Q \cong R^{2n}$.

Proof:

1. Let's start with a Milnor square, which is a pullback diagram:

$$\begin{array}{ccc} R & \rightarrow & S \\ \downarrow & & \downarrow \\ R/I & \rightarrow & S/I \end{array}$$

Suppose we have a finitely generated projective S -module P_1 and a finitely generated projective R/I -module P_2 , along with an isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$. Let

$$P = \{(x, y) \in P_1 \times P_2 \mid \bar{x} = g(\bar{f}(y))\}$$

To show P is finitely generated, Since P_1 and P_2 is a finitely generated, there exists a surjection $S^n \rightarrow P_1$ for some n and a surjection $(R/I)^m \rightarrow P_2$ for some m . Then it is certainly the case that we can construct a surjection $R^{m+n} \rightarrow P$, showing that P is finitely generated.

For showing it is projective, since P_1 is projective over S , there exists an S -module Q_1 such that $P_1 \oplus Q_1 \cong S^n$ for some n . Similarly, there exists an R/I -module Q_2 such that $P_2 \oplus Q_2 \cong (R/I)^m$ for some m .

The isomorphism $g : P_1 \otimes_S S/I \cong P_2/IP_2$ extends $\tilde{g} : S^n \otimes_S S/I \cong (R/I)^m/I(R/I)^m$, namely it is the isomorphism induced by taking $Q_1 \otimes_S S/I$ and Q_2/IQ_2 . Now, define F by:

$$F = \{(x, y) \in S^n \times (R/I)^m \mid \bar{x} = g(\bar{f}(y))\}$$

By the properties of the Milnor square, F is a free R -module. The inclusion maps $P_1 \hookrightarrow S^n$ and $P_2 \hookrightarrow (R/I)^m$ induce an inclusion $P \hookrightarrow F$. Moreover, the projections $S^n \rightarrow P_1$ and $(R/I)^m \rightarrow P_2$ induce a projection $F \rightarrow P$ such that the composition $P \rightarrow F \rightarrow P$ is the identity on P . But then P is projective,

2. This follows by our above construction
3. If M is any R -module, the Milnor square gives a natural map from M to the R -module M' by patching $M_1 = M \otimes_R S$ and $M_2 = M \otimes_R (R/I) = M/IM$ along the natural isomorphism

$$(M/IM) \otimes_{R/I} (S/I) \cong M \otimes_R (S/I) \cong (M \otimes_R S)/I(M \otimes_R S)$$

Now, tensor M with $0 \rightarrow R \rightarrow (R/I) \oplus S \rightarrow S/I \rightarrow 0$. This gives an exact sequence (recall that tensoring is right-exact, but not necessarily left-exact).

$$\mathrm{Tor}_1^R(M, S/I) \rightarrow M \rightarrow M' \rightarrow 0$$

so in general M' is just a quotient of M . However, if M is projective, the Tor term is zero and $M \cong M'$. Thus every projective R -module may be obtained by patching.

1.2 Picard Group of a Commutative Ring

Let us now tackle the connection between projective modules and vector bundles more closely. To fully understand the interplay between the algebraic and the geometric, the following theorem must be quoted:

Theorem 1.2.1: Serre-Swan's Theorem

1. **(Serre's Theorem)** Let R be a commutative Noetherian ring. Then the category of finitely generated projective R -modules is equivalent to the category of algebraic vector bundles^a on $\mathrm{Spec} R$
2. **(Swan's Theorem)** Let X be a compact Hausdorff space. Then the category of finitely generated projective R -modules over the continuous functions $C(X)$ is equivalent to the category of finite-rank vector bundles on X , the equivalence being established by sending vector bundles its module of continuous section.

^alocally free sheaves of structure sheaf-modules of constant finite rank

Proof:

See [Serre's paper](#) and [Swan's paper](#).

Due to these connections, we have the following definition:

Definition 1.2.2: Algebraic Line Bundle

Let R be a commutative ring. Then an *algebraic line bundle* L over R is a finitely generated projective R -module of constant rank 1.

In this section, we shall just say *line bundle*. As line bundles are locally 1 dimensional, it is easy to verify that the tensor product of line bundles is again a line bundle (hint: $\dim(V \otimes W) = \dim(V) \cdot \dim(W)$). Furthermore, it should be immediate to the current reader that $L \otimes_R R \cong L$ for any line bundle

(even projective module). Then $\otimes_R$ forms a commutative monoid. If we show there is an inverse, we have a group, and indeed:

Lemma 1.2.3: Inverse Line Bundle

Let L be a line bundle and $L^\vee = \text{Hom}_R(L, R)$. Then

$$L \otimes_R L^\vee \cong R$$

Proof:

First, $L^\vee$ is line bundle as $\text{rank}(L) = \text{rank}(L^\vee)$. Next, take:

$$f \otimes_R \ell \mapsto f(\ell)$$

If $L = R$, this is certainly an isomorphism. Hence in the general case, localizing at $\mathfrak{p}$ we get:

$$(L^\vee \otimes_R L)_\mathfrak{p} \cong L_\mathfrak{p}^\vee \otimes_{R_\mathfrak{p}} L_\mathfrak{p} \rightarrow R_\mathfrak{p}$$

which is then certainly an isomorphism. As being an isomorphism is a local property, we get the global isomorphism, completing the proof.

Definition 1.2.4: Picard Group

Let R be a commutative ring. Then the *picard group* of R is the set of isomorphism classes of line bundles on R with the operation being the tensor product $\otimes_R$ and inverses $L^{-1} = L^\vee$.

Proposition 1.2.5: Picard Group Homomorphism

Let $R \rightarrow S$ be a ring homomorphism between commutative rings. Then $\text{Pic}(R) \rightarrow \text{Pic}(S)$ is a group homomorphism mapping $L \mapsto L \otimes_R S$

Proof:

Certainly $L \otimes_R S$ is a line bundle over S . Then it is an exercise in algebra to see that:

$$(L \otimes_R M) \otimes_R S \cong (L \otimes_R S) \otimes_R (M \otimes_R S)$$

showing that $\otimes_R S$ is a group-homomorphism, as we sought to show.

We prove the following representation of line bundles for later:

Lemma 1.2.6: Representation of Line Bundle

Let L be a line bundle over R . Then

$$\text{End}_R(L) \cong R$$

Proof:

Multiplication by elements of r give a map $R \rightarrow \text{End}_R(L)$. This map is locally an isomorphism, and hence an isomorphism, completing the proof.

Example 1.2.7: Picard Groups of Commutative Rings

1. Let $R = k$. Then as k is a field, every projective k -module is free. But then any projective k module of rank 1 is a dimension 1 vector-space, namely it is isomorphic to k . Hence:

$$\text{Pic}(k) = 0$$

2. Similarly, as every module over a PID is free, if $R = \mathbb{Z}$, then:

$$\text{Pic}(\mathbb{Z}) = 0$$

3. Let R be a Dedekind domain (usually a number ring). We shall heavily be relying on the algebra presented in [6, chapter 22.6]. First, recall that each ideal $I \subseteq R$ is a projective R -module, and it is certainly locally of rank 1 (every ideal is generated by at most 2 elements, and localizing at the appropriate prime makes the ideal principal). Next, recall that any torsion-free R -module of rank n is isomorphic to:

$$M \cong R^{n-1} \oplus I$$

for an ideal I that can be replaced by $J = (a)I$ for $a \in R$. As all projective modules are torsion free, and we are looking at rank 1 modules, we see that the line bundles on R are in correspondence to the ideals of R . As principal ideals are isomorphic to R , we have that:

$$\text{Pic}(R) \cong \text{Cl}(R)$$

4. (If you know some divisor theory) Building off the work from the last point, If X is a nonsingular affine variety over an algebraically closed field $\bar{k}$ then $\text{Pic}(k[X])$ (the Picard group of its coordinate ring) is isomorphic to the divisor class group of X .
5. it was proven by Quillen and Suslin that every projective module over polynomial ring is free. Thus, the picard group of $k[x_1, \dots, x_n]$ is always trivial:

$$\text{Pic}(k[x_1, \dots, x_n]) = 0$$

6. (foreshadowing for elliptic curve theory). An elliptic curve is a smooth algebraic curve (over an algebraically closed field $\bar{k}$) whose genus is 1^a. We shall see that there is a natural subgroup of the Picard group, Pic^0 , and that if E is an elliptic curve associated to $R = \bar{k}[x, y, z]/I$, then:

$$\text{Pic}^0(R) \cong R$$

Which is a rather interesting idempotence! See cite:AlgGeo for more details.

^aEvery nonsingular curve maps onto a compact Riemann surface of genus g

We shall next define a natural notion known as the *determinant line bundle*. Its key feature of

interest to us is that all finitely generated projective R -modules are determined by their rank and determinant. First, recall the following properties of the exterior product:

Proposition 1.2.8: Properties of The Exterior Product

Let R be a commutative ring and M an R -module. Then:

1. $\Lambda^k(R^n)$ is a free R -module of rank $\binom{n}{k}$ generated by $e_{i_1} \wedge \cdots \wedge e_{i_n}$ where $i_1 \leq \cdots \leq i_n$.
Note that $\Lambda^n(R^n) \cong R$

2. If $R \rightarrow S$ is a ring map, there is an isomorphism:

$$(\Lambda^k M) \otimes_R S \cong \Lambda^k(M \otimes_R S)$$

3. There is a natural isomorphism:

$$\bigwedge^k (P \oplus Q) \cong \bigoplus_{i=0}^k \left(\bigwedge^i P \otimes \bigwedge^{k-i} Q \right)$$

4. If P is projective, $\Lambda^k(P)$ is projective (as it is locally free).

Proof:

exercise, or see [6, chapter 15.4]

Recall that $\Lambda^n(M)$ for a n -rank R -module is generated by a single element, and by moving all the coefficients to the front we get the determinant. This motivates the following terminology:

Definition 1.2.9: Determinant Bundle

Let P be a finitely generated projective R -module of constant rank n . Then $\Lambda^n(P)$ is called the *determinant line bundle*, and is denoted $\det(P)$.

If P is not constant rank, we define it component-wise. If $g : P \rightarrow P$ is an endomorphism, $\det(-)$ is a functor $\deg(g) : \det(P) \rightarrow \det(P)$, and by lemma 1.2.6 we can think of $\deg(g) \in R$.

We shall state, but not prove, the following useful classification result:

Proposition 1.2.10: Classification of Projective Module Over 1-dimensional Ring

Let R be a commutative Noetherian 1-dimensional ring. Then all finitely generated projective R -modules are completely classified by their rank and determinant. In particular, every finitely generated projective R -module P of rank ≥ 1 is isomorphic to $L \oplus R^f$, where $L = \det(P)$ and $f = \text{rank}(P) - 1$.

Proof:

hint: Show that R is the only stably free-line bundle.

Let us next focus on line bundles over integral domains. We shall see that all line bundles on integral domains are given by *invertible ideal*:

Proposition 1.2.11: Classification of Projective Modules over Integral Domains

Let R be a commutative integral domain. L is a line bundle if and only if it is (isomorphic to) an invertible ideal. If I and J are fractional ideals and I is invertible, then $I \otimes_R J \cong IJ$. Furthermore, there is an exact sequence of abelian groups:

$$1 \rightarrow R^\times \rightarrow F^\times \xrightarrow{\text{Div}} J_K(R) \rightarrow \text{Pic}(R) \rightarrow 0.$$

where $J_K(R)$ is the ideal group.

Proof:

If I and J are invertible ideals such that $IJ \subseteq R$, then we can interpret elements of J as homomorphisms $I \rightarrow R$. If $IJ = R$, then we can find $x_i \in I$ and $y_i \in J$ so that $x_1y_1 + \cdots + x_ny_n = 1$. The $\{x_i\}$ assemble to give a map $R^n \rightarrow I$ and the $\{y_i\}$ assemble to give a map $I \rightarrow R^n$. The composite $I \rightarrow R^n \rightarrow I$ is the identity, because it sends $r \in I$ to $\sum x_iy_ir = r$. Thus I is a summand of R^n , i.e., I is a finitely generated projective module.

As R is an integral domain and $I \subseteq F$, $\text{rank}(I)$ is the constant $\dim_F(I \otimes_R F) = \dim_F(F) = 1$. Hence I is a line bundle.

This construction gives a set map $J_K(R) \rightarrow \text{Pic}(R)$; to show that it is a group homomorphism, it suffices to show that $I \otimes_R J \cong IJ$ for invertible ideals. Suppose that I is a submodule of F which is also a line bundle over R . As I is projective, $I \otimes_R -$ is an exact functor. Thus if J is an R -submodule of F , then $I \otimes_R J$ is a submodule of $I \otimes_R F$. The map $I \otimes_R F \rightarrow F$ given by multiplication in F is an isomorphism because I is locally free and F is a field. Therefore the composite

$$I \otimes_R J \subseteq I \otimes_R F \xrightarrow{\text{multiply}} F$$

sends $\sum x_i \otimes y_i$ to $\sum x_iy_i$. Hence $I \otimes_R J$ is isomorphic to its image $IJ \subseteq F$. This proves the third assertion.

The kernel of $J_K(R) \rightarrow \text{Pic}(R)$ is the set of invertible ideals I having an isomorphism $I \cong R$. If $f \in I$ corresponds to $1 \in R$ under such an isomorphism, then $I = fR = \text{div}(f)$. This proves exactness of the sequence at $J_K(R)$.

Clearly the units $R^\times$ of R inject into $F^\times$. If $f \in F^\times$, then $fR = R$ if and only if $f \in R$ and f is in no proper ideal, i.e., if and only if $f \in R^\times$. This proves exactness at $R^\times$ and $F^\times$.

Finally, we have to show that every line bundle L is isomorphic to an invertible ideal of R . Since $\text{rank}(L) = 1$, there is an isomorphism $L \otimes_R F \cong F$. This gives an injection $L \cong L \otimes_R R \subseteq L \otimes_R F \cong F$, i.e., an isomorphism of L with an R -submodule I of F . Since L is finitely generated, I is a fractional ideal. Choosing an isomorphism $\tilde{L} \cong J$, Lemma 3.1 yields

$$R \cong L \otimes_R \tilde{L} \cong I \otimes_R J \cong IJ.$$

Hence $IJ = fR$ for some $f \in F^\times$, and $I(f^{-1}J) = R$, so I is invertible.

There is further discussion different interesting divisor groups, class groups, and their relation to the Picard group, but we leave that material to cite:algGeo.

2

K_0 Group

We saw the Picard group allowed us to treat the collection of (isomorphism classes of) line bundles of constant rank one as a group under $\otimes$. We shall now be looking at a similar situation with the direct sum.

2.1 Algebraic Build-up

Let M be an abelian monoid. Then we universally augment it into an abelian group, $M^{-1}M$ through the usual free-quotient construction. In particular, take $\mathcal{F}(M \times M) / ([m+n] - [m] - [n])$. This is called the *Grothendieck Completion*. A *Cancellation monoid* is a monoid where $m+a = n+a$ if and only if $m = n$.

Proposition 2.1.1: Properties of Monoid Completion

Let M be a monoid, $M^{-1}M$ be its completion. Then:

1. Every element of $M^{-1}M$ can be written as $[m] - [n]$ for $m, n \in M$
2. For each $[m], [n] \in M^{-1}M$, $[m] = [n]$ if and only if there exists a p such that $m+p = n+p$.
3. The map $M \times M \rightarrow M^{-1}M$ where $(m, n) \mapsto [m] - [n]$ is surjective
4. M injects into $M^{-1}M$ if and only if M is a cancellation monoid.
5. The Grothendieck Completion is a functor that is left adjoint to the forgetful functor from the category of abelian group to the category of abelian monoids.

Proof:

these are all simple exercises, but you can take a look at [6, chapter 15].

If we augment a monoid with a 2nd operation that satisfies the ring axioms, then we get a *semiring*. The semiring $\mathbb{N}$ is the initial semiring in this category. Then the Grothendieck functor can also be seen as a functor from semirings to rings.

When we shall be studying $K_0(R)$ groups, we shall find that the free R -modules play an important role. These elements are important enough so that we consider the following special type of subgroup of a monoid

Definition 2.1.2: Cofinal Monoid

Let M be a monoid. Then $L \subseteq M$ is called *cofinal* if for every $m \in M$, there exists a $m' \in M$ such that $m + m' \in L$.

The terminology *cofinal* comes from order theory, where a subset $A \subseteq B$ of an preorder $(B, \leq)$ is cofinal if for every $b \in B$, there is an $a \in A$ such that $b \leq a$.

Lemma 2.1.3: Cofinal Monoids

Let $L \subseteq M$ be a cofinal monoid. Then:

1. $L^{-1}L \subseteq M^{-1}M$ is a subgroup
2. Every element $[m] \in M^{-1}M$ can be written as $[m'] - [\ell]$ for $m' \in M$ and $\ell \in L$.
3. If $[m] = [m'] \in M^{-1}M$, then $m + \ell = m' + \ell$ for some $\ell \in L$.

Proof:

this is a matter of checking definitions

Example 2.1.4: Grothendieck Completion

1. some good examples in the book!

2.2 K_0 of a ring

Definition 2.2.1: First K-ring: $K_0(R)$

Let R be a [not necessarily commutative] ring, $P(R)$ the collection of (isomorphism classes of) finitely generated projective R -modules. Then combining it with $\oplus$ and appending an identity element 0, we get an abelian monoid. Its Grothendieck completion is the group $K_0(R)$.

When R is commutative, then along with $\otimes_R$, this forms a ring.

We restrict to finitely generated modules simply because the usual behaviour of infinitely generated modules trivializes the structure, for example if P is any finitely generated projective R -module then, then if we had to consider the infinitely generated module R^∞ we get:

$$P \oplus R^\infty \cong R^\infty$$

Therefore $[P] = 0$ and $K_0(R) = 0$, and so we naturally stick to the finitely generated case.

If R is commutative, then $K_0(R)$ is commutative with identity $1 = [R]$. This is indeed the identity, for $\oplus$ and $\otimes$ distribute, and $P \otimes_R R \cong P$. The canonical trivial example is when $R = k$ (or any division ring). Then $K_0(k) \cong \mathbb{Z}$. Similarity, if R is a local ring or a PID, then $K_0(R) \cong \mathbb{Z}$ (as these are sufficient conditions for a projective module to be free). We shall give non-trivial example after we have developed some more theory to be able to identify which properties lead to structural results for $K_0(R)$.

It should be noted that $K_0(-)$ is a functor from **Ring** to **Ab** or from **CRing** to **CRing**, namely if $\varphi : R \rightarrow S$ is a ring homomorphism, then $\otimes_R S : P(R) \rightarrow P(S)$ is a monoid map which induces the map $K_0(R) \rightarrow K_0(S)$. If R, S are commutative, then $\otimes_R S : P(R) \rightarrow P(S)$ is a semiring map as:

$$(P \otimes_R Q) \otimes_R S \cong (P \otimes_R S) \otimes_R (Q \otimes_R S)$$

and hence we get a ring homomorphism. This shall become handy when trying to translate properties from known $K_0(R)$ groups to one's we are trying to study.

In $K_0(R)$, the free-modules play a particularly important role. First of all, they are *cofinal* in $P(R)$ (see definition 2.1.2). Then every element of $K_0(R)$ can be written as:

$$[P] - [R^n]$$

Furthermore, by lemma 2.1.3, we have $[P] = [Q]$ if and only if:

$$P \oplus R^n \cong Q \oplus R^m$$

that is, P, Q are stably isomorphic, and $[P] = [R^n]$ if P is stably free. If R satisfies the IBP, then $L \subseteq P(R)$ is isomorphic to $\mathbb{N}$. If R respect IBP, we get the following useful characterization of trivial K_0 groups:

Lemma 2.2.2: Ibp and Trivial K_0

Let $\mathbb{N} \rightarrow P(R)$ be the monoid map sending $n \rightarrow R^n$ inducing the group homomorphism $\mathbb{Z} \rightarrow K_0(R)$. Then:

1. this map is injective if and only if R satisfies IBP
2. If R satisfies IBP, then

$$K_0(R) \cong \mathbb{Z} \text{ if and only if every finitely generated projective } R\text{-module is stably free}$$

Proof:

1. This is immediate by IBP
2. $P \oplus R^n \cong R^m$, and so each $[P] = [R^m]$ for some n .

Example 2.2.3: K_0 of some rings

1. As mentioned before, fields/division rings, PIDs, and local rings all have $K_0(R) = \mathbb{Z}$
2. If $R = R_1 \times R_2$, then by [6, chapter 23] we have the equivalence of categories:

$$(R \times S)\text{-Mod} \cong R\text{-Mod} \times S\text{-Mod}$$

Thus we get $P(R) = P(R_1) \times P(R_2)$, and its completion gives $K_0(R) = K_0(R_1) \times K_0(R_2)$. We may thus compute $K_0(R)$ component-wise.

3. Let $R = M_n(F)$. Then R is a simple ring, and so every $M_n(F)$ -module is projective. Recall that an $M_n(F)$ -module is some direct sums of F , and the *length* of an $M_n(F)$ -module is the number of copies. Then the length gives a natural map:

$$K_0(M_n(F)) \cong \mathbb{Z}$$

Note that $M_n(F)$ has length n , and so the subgroup generated by the free modules have index n , so the natural inclusion map $\mathbb{Z} \rightarrow K_0(M_n(F))$ does not split!

4. Let R be a semi-simple ring so that by Artin-Wedderburn:

$$R \cong M_{n_1}(D_1) \times \cdots \times M_{n_k}(D_k)$$

Thus, we get:

$$K_0(R) = \prod_i K_0(M_{n_i}(D_i)) \cong \mathbb{Z}^k$$

5. If R is a commutative ring, we have a natural map from R to a field, $R \rightarrow R/\mathfrak{m}$ for some maximal ideal $\mathfrak{m}$; call this field F . Then this induces a natural map $K_0(R) \rightarrow K(F)$, that is $K_0(R) \rightarrow \mathbb{Z}$ which maps $[R]$ to 1. Note that there doesn't always exist, such a map, we shall see an example where $K_0(R) \cong \mathbb{Q}$.
6. (If you know some algebraic geometry) Recall that a sheaf is *flasque* if all restriction maps are surjective. The key interesting property of Flasque sheaves is that they have trivial cohomology, $H^i(X, \mathcal{F}) = 0$ for $i > 0$ and $\mathcal{F}$ a flasque sheaf. We shall link the $K_0(R)$ group to (co)homological considerations, and will find the following definition to be the appropriate purely algebraic counter-part. We say a ring R is *flasque* if there is an R -bimodule M that is finitely generated and projective as a right R -module, such that there is a R -bimodule isomorphism:

$$\varphi : R \oplus M \xrightarrow{\sim} M$$

From this, we get that $K_0(R) = 0$. To see this, note that:

$$P \oplus (P \otimes_R M) \cong P \otimes_R (R \oplus M) \cong (P \otimes_R M)$$

Hence $[P] + [P \otimes_R M] = [P \otimes_R M]$, implying $[P] = [0]$. If R is flasque and the right R -module structure of M is R , then R is said to be an *infinite sum ring*. The isomorphism φ becomes $R^2 \cong R$. An example of such a ring is $\text{End}_R(R^\infty)$.

7. (There is an example on p.76 of the K-theory textbook I'm following on Von Neumann regular rings, which are rings where for every $x \in R$, there exists a $y \in R$ such that $xyx = x$. These are rings in which inverse need not exist, but they almost do, for example each field is Von Neumann regular because we can let $y = x^{-1}$. No integral domain is Von Neumann regular, but $M_n(F)$ is Von Neumann regular. There is a natural lattice structure that comes out of its idempotence that $K_0(R)$ is isomorphic to if R is a Von Neumann regular rings)
8. let $R = \varinjlim_i R_i$ where $\{R_i\}$ is a filtered system. Then every finitely generated projective R -module is of the form:

$$R \otimes_{R_i} P_i$$

where P_i is some finitely generated projective R_i -module. Any isomorphism:

$$R \otimes_{R_i} P_i \cong R \otimes_{R_i} P'_i$$

must set a finite set of generators to a finite set of generators, and so for j large enough this reduces to:

$$R_j \otimes_{R_i} P_i \cong R_j \otimes_{R_i} P'_i$$

But that means that $P(R)$ is then a filtered colimit of $P(R_i)$! We thus have the following wonderful result about colimits passing through K -groups:

$$K_0(R) = \varinjlim K_0(R_i)$$

If R is commutative, we can show that every R is the colimit of finitely generated subrings! As every finitely generated commutative ring is Noetherian with finite normalization (the integral closure is finitely generated over the ring), we may reduce studying $K_0(R)$ to the study of $K_0(R_i)$ where the R_i have nicer properties!

The following is very useful and allows for interesting geometric interpretations. For the following, we shall say that I is *complete* if every Cauchy sequence $\sum_i^\infty a_i$ where $a_i \in R/I^n$ converges to a unique element in I .

Theorem 2.2.4: K-group and Nilpotence

Let R be a ring and $I \subseteq R$ be nilpotent, or more generally complete (see definition above). Then:

$$K_0(R) \cong K_0(R/I)$$

And if R is commutative, then:

$$K_0(R) \cong K_0(R_{\text{red}})$$

Proof:

We'll show $P(R) \cong P(R/I)$, which shall complete the result. We first need to build-up some important result:

lemma 1: Let P_1, P_2 be finitely generated projective R -modules and $I \subseteq R$ a radical idea. Then $P_1 \cong_R P_2$ if and only if $P_1/IP_1 \cong_R P_2/IP_2$

Proof. Certainly $P_1 \cong_R P_2$ implies $P_1/IP_1 \cong_R P_2/IP_2$, so we do the converse. There is a natural map $P_1 \rightarrow P_1/IP_1$, and then there is an isomorphism $\varphi : P_1/IP_1 \rightarrow P_2/IP_2$. Finally, there is a

natural projection $\pi : P_2 \rightarrow P_2/IP_2$:

$$\begin{array}{ccc} P_1 & & P_2 \\ \pi \downarrow & \searrow \varphi \circ \pi & \downarrow \pi \\ P_1/IP_1 & \xrightarrow{\varphi} & P_2/IP_2 \end{array}$$

As P_1 is projective, there is a natural lift making the following diagram commute:

$$\begin{array}{ccc} P_1 & \xrightarrow{\tilde{\varphi}} & P_2 \\ \pi \downarrow & & \downarrow \pi \\ P_1/IP_1 & \xrightarrow{\varphi} & P_2/IP_2 \end{array}$$

If we show $\tilde{\varphi}$ is an isomorphism, we're done. Note that as φ is an isomorphism, in particular it is surjective, then as I is radical it is a consequence of Nakayama's lemma that $\varphi : P_1 \rightarrow P_2$ is surjective (see exercises in [6, Chapter 20.3] or checkout [13]). Let's thus show injectivity. To that end, let us localize at each $\mathfrak{p} \subseteq R$ and consider:

$$\begin{array}{ccc} P_{1,\mathfrak{p}} & \xrightarrow{\tilde{\varphi}_{\mathfrak{p}}} & P_{2,\mathfrak{p}} \\ \pi \downarrow & & \downarrow \pi \\ P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}} & \xrightarrow{\varphi_{\mathfrak{p}}} & P_{2,\mathfrak{p}}/I_{\mathfrak{p}}P_{2,\mathfrak{p}} \end{array}$$

As P is a finitely generated projective module, its localization at a prime ideal is free, hence each $P_{i,\mathfrak{p}}$ has a basis. The key is that the projection onto $P_{1,\mathfrak{p}}/I_{\mathfrak{p}}P_{1,\mathfrak{p}}$ is still a basis. If we show then as $\varphi_{\mathfrak{p}}$ is an isomorphism, by commutativity the image of the basis in $\tilde{\varphi}_{\mathfrak{p}}$ must be linearly independent which shows the map is injective. To that end, if $\{e_1, \dots, e_n\}$ is a basis for $P_{1,\mathfrak{p}}$, its projection spans so let's show it is linearly independent. Say

$$\sum_i r_i \bar{e}_i = 0$$

Then $\sum_i r_i e_i \in I_{\mathfrak{p}}P_{1,\mathfrak{p}}$. By properties of ideals, $r_i \in I_{\mathfrak{p}}$, but then we must have that in the quotient $r_i = 0$ for all i , which by our previous reasoning allows us to conclude injectivity.

Hence, $\tilde{\varphi}_{\mathfrak{p}}$ is injective for each prime $\mathfrak{p} \subseteq R$, and so it is a classical algebra result that $\tilde{\varphi}$ is injective (see [6, chapter 20.3]). $\square$

Now, let us show that if I is a nilpotent ideal, then there is a bijection between the finitely generated projective R -modules and the finitely generated projective R/I -modules, which shall then complete the proof. Without loss of generality we can assume that $I^2 = 0$ (we can then use induction to for the result for $I^n = 0$ for $n \geq 2$)^a. Let us first show that an idempotent $\bar{e} \in R/I$ lifts to an idempotent $e \in R$. If r is a lift of e , then we can verify that:

$$e = r + rxr + (1 - r)y(1 - r)$$

where $x, y \in I$. Verify that $e^2 = e$ and it indeed maps to $\bar{e}$. This lift is unique up to conjugation by an element $u \in 1 + I$. With this lift, we shall show that every finitely generated R/I -module $\bar{P}$ is isomorphic to P/IP for an appropriate finitely generated projective module P . The key

is that idempotent matrices correspond to summands of free modules. In particular if $\bar{P}$ is a finitely generated projective R/I -module, then there exists an idempotent $\bar{e} \in M_n(R/I)$ such that $\text{im}(\bar{e}) = \bar{P}$. Then this e can be lifted to $e \in M_n(R)$ where $e^2 = e$ and $e \equiv \bar{e} \pmod{I}$, so by taking $\text{im}(e) = P$ we get:

$$\bar{P} \cong P/IP$$

Finally, we are at the stage where we reduced our problem of finitely generated projective R -modules to modules of the form P/IP . Then by our lemma, $P_1/IP_1 \cong P_2/IP_2$ if and only if $P_1 \cong P_2$, showing that the isomorphism classes are in bijective corresponds, completing the proof.

^aThis is where the completeness generalization also comes in, as we can consider $\varprojlim R/I^n$

Example 2.2.5: 0-dimensional Commutative Ring

Let R be a commutative ring and R_{red} be its reduction. Then it is an easy exercise to show that R_{red} is Artinian if and only if $\text{Spec}(R)$ is finite and discrete. More generally (in the “non-finite case”), if a ring is von Neumann regular (for each $x \in R$, there exists a $y \in R$ such that $xyx = x$; this is a “weak inverse”), then R has Krull dimension 0 and is reduced (in fact, this is an equivalent characterization of von-Neumann regular rings, giving a good geometric characterization of such rings^a)

^aNote that there are many non-commutative examples, such matrix rings, left principal ideals generated by idempotents, and $\text{End}_k(V)$ for a vector space, and as they are closed under direct product that makes every semi-simple von-Neumann regular ring semi-simple

2.3 Connection to H_0

We start connecting the K_0 ring to (co)homological results, starting with the most simple one of the $H_0(R)$ group, which we shall take as $\text{Hom}(\text{Spec}(R), \mathbb{Z})$. Recall that for a ring R , there is a natural map $\text{Spec}(R) \rightarrow \mathbb{Z}$ sending $\mathfrak{p} \mapsto \dim_{\kappa(\mathfrak{p})} M \otimes_R \kappa(\mathfrak{p})$. To see this connection, consider the following warm-up result:

Theorem 2.3.1: Pierce’s Theorem

Let R be a 0-dimensional commutative ring. Then:

$$K_0(R) = \text{Hom}(\text{Spec}(R), \mathbb{Z})$$

Note that no assumption was made on R be finitely generated, for example it may be an infinite product of fields.

Proof:

Recall that as R is 0-dimensional all its prime ideals are maximal, and $\text{Spec}(R)$ has the discrete topology. Then by generalizing the classification of Artinian rings to 0-dimensional rings (in particular, every finitely generated projective R -module will be a finite product of modules of the form $R/\mathfrak{m}$), we can consider the rank of each $R/\mathfrak{m}$ individually which gives us the exact information of the $K_0(R)$ group, completing the proof.

We shall thus push this one step further and see the relation for more general commutative rings:

Definition 2.3.2: H_0 Group

Let R be a commutative ring. Then:

$$H_0(R) = [\text{Spec}(R), \mathbb{Z}] = \text{Hom}(\text{Spec}(R), \mathbb{Z})$$

As $\text{Spec}(R)$ is (quasi)compact, there are at most finitely many components, and so $H_0(R)$ will always be a free abelian group (ex. it can't be $\mathbb{Q}$). If furthermore R is Noetherian, then:

$$H_0(R) \cong \mathbb{Z}^n$$

If on top of that R is an (integral) domain (more generally $\text{Spec}(R)$ is connected), then:

$$H_0(R) \cong \mathbb{Z}$$

Lemma 2.3.3: H_0 direct summand of K -group

The group $H_0(R)$ is a direct summand of $K_0(R)$. If $\dim R = 0$, then $K_0(R) \cong H_0(R)$.

Proof:

Take the monoid L of the componentwise free modules R^n . Then as we saw, L is cofinal, and $L \rightarrow P(R)$ is a semiring map which naturally gives a map $L^{-1}L \rightarrow K_0(R)$. Then as L is isomorphic to $\text{Hom}(\text{Spec}(R), \mathbb{N})$, we must have $L^{-1}L \cong H_0(R)$, and it a subgroup.

Now, there is a natural map $K_0(R) \rightarrow H_0(R)$ given by the rank, namely there is a natural map from $\text{rank} : P(R) \rightarrow [\text{Spec}(R), \mathbb{N}]$, which is a semiring homomorphism. Thus, we get $H_0(R) \rightarrow K_0(R) \rightarrow H_0(R)$, is the identity, and so $H_0(R)$ is a direct summand of $K_0(R)$, as we sought to show.

For the special case of $\dim(R) = 0$, By Pierce's Theorem we have $K_0(R) \cong H_0(R)$.

As $K_0(R)$ splits with $H_0(R)$, the other component is the kernel of the rank map. We shall denote it $\tilde{K}_0(R)$ so that:

$$K_0(R) \cong H_0(R) \oplus \tilde{K}_0(R)$$

It shall be shown later that $\tilde{K}_0(R)$ is a nilradical, and as $H_0(R)$ is reduced already we would have that it is the nilradical of $K_0(R)$.

Lemma 2.3.4: Limit and Kernel

Let R be a commutative, let $P_n(R)$ denote the subset of all modules in $P(R)$ consisting of projective modules of constant rank n . There is a map $P_n(R) \rightarrow K_0(R)$ sending

$$P \mapsto [P] - [R']$$

This map is compatible with the stabilization map $P_n(R) \rightarrow P_{n+1}(R)$ sending $P \mapsto P \oplus R$, and the induced map is an isomorphism:

$$\varinjlim_n P_n(R) \cong \tilde{K}_0(R)$$

Proof:

Follows from 2.1.3.

Though we could take the colimit, by the base cancellation theorem we know that the interesting “geometric” structure of a finitely generated projective R -module over a ring of rank d can only happen for modules of rank $\leq d$:

Corollary 2.3.5: Higher Dimension Nilradical

Let R be a commutative Noetherian ring where $\text{kdim} R = d$. Then for every $n > d$,

$$P_n(R) \cong \tilde{K}_0(R)$$

Proof:

Let P, Q be finitely generated projective modules of rank $> d$. Then by the Base cancellation Theorem (theorem 1.1.12), we get that:

$$[P] = [Q] \text{ if and only if } P \cong Q$$

giving the desired result.

The H_0 group is based off of the spectrum of R , and hence is dependent on its commutativity. Let us now try to bring this interpretation to non-commutative rings. To do this, another interpretation of the kernel that is rather nice is that it is the intersection of all the kernels of map $K_0(R) \rightarrow K_0(F)$ for all maps $R \rightarrow F$ into a field: as rank is a natural map, $\tilde{K}_0(F) = 0$, and $H_0(R)$ captures the information at each prime ideals (i.e. each stalk), where we can get this information by going over all fields for which we can have $R \rightarrow F$.

We may thus take the non-commutative equivalent of a field: a simple Artinian ring. As maximal commutative subrings of a simple Artinian ring S is a finite product of 0-dimensional local rings, we get that this is the correct generalization. We can define $\tilde{K}_0(R)$ as the intersection of the kernels of $K_0(R) \rightarrow K_0(S)$ where S is a simple Artinian ring. If no map $R \rightarrow S$ exists, then we say $\tilde{K}_0(R) = K_0(R)$. We define:

$$H_0(R) = \frac{K_0(R)}{\tilde{K}_0(R)}$$

We saw that $H_0(R)$ is a free-abelian group in the commutative case. In the non-commutative case it is at least still torsion-free. To see this, let X be the set of homomorphisms $R \rightarrow S_x$ through which any other $R \rightarrow S'$ factors (such a set exists by using Zorn’s lemma). Then as $K_0(S_x) \rightarrow K_0(S')$ is an isomorphism. Hence, $\tilde{K}_0(R)$ is the intersection of all kernels of maps $K_0(R) \rightarrow K_0(S_x)$. We also then have that $H_0(R)$ is the image of $K_0(R)$ in the torsion free group:

$$\prod_{x \in X} K_0(S_x) \cong \prod_{x \in X} \mathbb{Z}$$

(The author of the textbook I’m reading talks a bit about whitehead groups, which I am omitting for now)

2.3.1 Some Important Functionals

We are in the realm of representation theory, hence we have our usual trace and determinant map that shows up. In this section, we shall see how these maps give us some interesting information on our K_0 and H_0 group.

Hattori-Stallings Trace Map

Let R be an associative ring, and let $[R, R] = \{rs - sr \mid r, s \in R\}$ be the commutator subgroup. Then the trace provides a natural map:

$$\text{tr} : M_n(R) \rightarrow \frac{R}{[R, R]}$$

where we quotient by $[R, R]$ as tr should be independent of conjugation, which shall require $a_{i\ell}b_{\ell i}$ to commute. Then it should be a simple computation to see that all traceless matrices are in $[M_n(R), M_n(R)]$ which gives us the isomorphism:

$$\frac{M_n(R)}{[M_n(R), M_n(R)]} \cong \frac{R}{[R, R]}$$

Naturally, not all elements of $\text{End}_R(P)$ of a finitely generated projective R -module P is in matrix form, however it close enough to still define the trace: as P is finitely generated and projective choose Q such that $P \oplus Q \cong R^n$. Then we have an idempotent element $e \in M_n(R)$ such that $e(R^n) = P$, so:

$$\text{End}_R(P) = eM_n(R)e$$

It is not too hard to verify that any other choices of idempotent will yield an e_1 which is idempotent and conjugate to e in some $M_m(R)$. Thus, we have that the trace map is independent of representative and we have a well-defined map:

$$\text{tr} : \text{End}_R(P) \rightarrow \frac{R}{[R, R]}$$

The trace of the identity map on P will be the trace of the idempotent element e ; we call this particular value the *trace* of P .

Now, if P' is a finitely generated projective R -module represented by an idempotent f , then $P \oplus P'$ is represented by $\begin{pmatrix} e & 0 \\ 0 & f \end{pmatrix}$, and so we have its trace is $\text{tr}(P) + \text{tr}(P')$! Thus, the trace is an additive map on $P(R)$, and so we get a natural map:

Definition 2.3.6: Hattori-Stallings Trace Map

Let R be a (not necessarily commutative ring). Then by the above discussion, the natural map:

$$K_0(R) \rightarrow \frac{R}{[R, R]}$$

is called the *Hattori-Stallings Trace Map*.

In the case where R is commutative, we have that $R = \frac{R}{[R, R]}$. This gives a more explicit description of $K_0(R)$ and $H_0(R)$ (as we have $\text{Spec}(R)$ well-defined). To see this, start by taking the trace map

on $K_0(R) \rightarrow R$ and restrict to $H_0(R)$. Then notice that each $\text{Spec}(R) \rightarrow \mathbb{Z}$ gives a decomposition of $\text{Spec}(R)$ into components $R = R_1 \times \cdots \times R_c$ given by idempotents $e_1, \dots, e_c$. As $\text{tr}(e_i R) = e_i$, we have that for any $f \in R$,

$$\text{tr}(f) = \sum_i f(i) e_i$$

And so we get:

$$\text{tr}(fg) = \text{tr}(f)\text{tr}(g)$$

We thus get the following natural map:

Proposition 2.3.7: Hattori-stallings Map Factorization

Let R be commutative. Then the Hattori-Stallings trace map factors through:

$$K_0(R) \xrightarrow{\text{rank}} H_0(R) \rightarrow R$$

Proof:

Consider the following diagram:

$$\begin{array}{ccc} K_0(R) & \longrightarrow & \prod_{\mathfrak{p}} K_0(R_{\mathfrak{p}}) \\ \text{tr} \downarrow & & \downarrow \text{tr} \\ R & \xrightarrow{\text{diag}} & \prod_{\mathfrak{p}} R_{\mathfrak{p}} \end{array}$$

As the kernel of the top arrow is $\tilde{K}_0(R)$ by construction, we get the left arrows map to $H_0(R)$, which gives the desired factorization.

Example 2.3.8: Trace and Characters on Representations

1. Let k be commutative and $R = k[G]$ of a group G . Then if $g, h \in G$ are conjugate, then $g - h \in [R, R]$ (as $g = xhx^{-1}$ so $xhx^{-1} - h \in [xh, x^{-1}]$). Then we see that $\frac{R}{[R, R]} \cong \oplus k[g]$ for $g \in G/\sim$ (i.e. up to conjugacy class). Given such a basis, we then get:

$$\text{tr}(P) = \sum_{g \in G} r_P(g)[g]$$

where $r_P(g)$ are coefficients seen as functions of the set $G/\sim$, for each P .

2. In the special case where G is finite, then any finitely generated projective $k[G]$ -module P is also a projective k -module, and so we can form the trace map $\text{End}_k(P) \rightarrow k$! Hence, we get the character $\chi_P : G \rightarrow k$ given by the formula:

$$\chi_P(g) = \text{tr}(g)$$

It was proven by a mathematician called *Hattori* that we can write:

$$\chi_P(g) = |Z_G(g)| r_P(g^{-1})$$

which gives a very interesting connection between characters and the general theory of trace we've been building up! We shall not look into this more deeply, but it is an interesting result to point out.

One very interesting result we get out of this is that $\mathbb{Z}[G]$ does not split, i.e. it has no non-trivial idempotents:

Proof. Let $e \in \mathbb{Z}[G]$ be an idempotent element. Then $\chi_P(1)$ is the rank of the $\mathbb{Z}$ -module $P = e\mathbb{Z}[G]$. Then by construction its rank is $\leq |G| = \text{rank}\mathbb{Z}[G]$. As $r_P(1) \in \mathbb{Z}$, this contradicts Hattori's formula as $\chi_P(1) = |G|r_P(1)$, completing the proof. $\square$

3. If k is a field of characteristic 0 and G a finite group, then by Maschke's Theorem we have $k[G]$ is the simple product of k -algebras $S_1 \times \cdots \times S_c$ where c represents the conjugacy classes of G . Hence we get:

$$\frac{k[G]}{[k[G], k[G]]}$$

Then we saw in example 2.2.3 that

$$K_0(K[G]) \cong \mathbb{Z}^c$$

By Hattori's formula, we can see that the trace map from $K_0(K[G])$ is isomorphic to $\mathbb{Z}^c \subseteq k^c$.

Determinant

Recall definition 1.2.9 where we can define the determinant in a natural way if R is commutative on a finitely generated projective R -module. We saw in proposition 1.2.11 that we may classify all finitely generated projective R -modules over a 1-dimension ring, and in proposition 1.2.11 we saw that the determinant of a finitely generated projective R -module P can be seen as elements of $\text{Pic}(R)$. These results work when reducing to K -groups:

Proposition 2.3.9: K_0 Group and Determinant

Let R be a commutative ring. Then the determinant determines a natural map:

$$\det : K_0(R) \rightarrow \text{Pic}(R)$$

Proof:

The key is that the non-top wedge products disappear. By the universal properties of monoids, it suffices to show that:

$$\det(P \oplus Q) \cong \det(P) \otimes \det(Q)$$

Without loss of generality, we may assume that P, Q have constant rank n, m respectively. Then $\bigwedge^{n+m} P \oplus Q$ is the sum over all i, j such that $i + j = n + m$ in $(\bigwedge^i P) \otimes (\bigwedge^j Q)$. Then if $i > n$ and $j > m$ then either $\bigwedge^i P = 0$ or $\bigwedge^j Q = 0$. But then:

$$\bigwedge^{n+m} P \oplus Q \cong \left(\bigwedge^n P \right) \otimes \left(\bigwedge^m Q \right)$$

as we sought to show.

Corollary 2.3.10: Rank and Determinant Represents K -group

Let R be a commutative ring. Then $H_0(R) \oplus \text{Pic}(R)$ is a ring with a square-zero ideal $\text{Pic}(R)$ and there is a surjective homomorphism:

$$K_0(R) \twoheadrightarrow H_0(R) \oplus \text{Pic}(R)$$

with kernel consists of equivalence classes representable by $[P] - [R^n]$ where P has constant rank n and $\bigwedge^n P \cong R$.

Proof:

Let $SK_0(R)$ represent the kernel. Notice first that this is naturally the kernel of $\det : K_0(R) \rightarrow \text{Pic}(R)$ by the above proposition. Next, $SK_0(R)$ is an ideal of $K_0(R)$. After showing this, the rest is combining earlier results.

Corollary 2.3.11: Rank and Determinant: One Dimensional Case

Let R be a commutative Noetherian 1-dimensional ring. Then from the above we have:

$$K_0(R) \cong H_0(R) \oplus \text{Pic}(R)$$

Proof:

This follows from the above corollary and proposition 1.2.10.

2.4 Morita Equivalence

An important invariant in the study of modules is the one given by the *Morita Equivalence*, which tells you when a ring action produces the same theory:

Definition 2.4.1: Morita Equivalence

Let R, S be (not necessarily commutative) rings. Then $R\text{-Mod}$ and $S\text{-Mod}$ are *Morita equivalent* if they are equivalent, namely there exists functors F, G such that $GF \cong \text{id}_R$ and $FG \cong \text{id}_S$ where id_R and id_S are the identity functors on their respective categories

We have in fact already seen many examples of Morita equivalence without invoking it, as we shall see in examples 2.4.4.

Note that F, G being inverses implies they preserve limits and colimits, hence the structure of in equivalence categories share many similarities (for example, direct sums and products pass through F, G).

We can in fact have a precise idea of what is the image of F, G . Let $F(R) = P$ and $G(S) = Q$. Then notice we have the natural maps:

$$R \cong \text{End}_R(R)^{\text{op}} \xrightarrow{F} \text{End}_S(P)^{\text{op}} \quad \text{and} \quad S \cong \text{End}_S(S)^{\text{op}} \xrightarrow{G} \text{End}_R(Q)^{\text{op}} \quad (2.1)$$

which make P into an S, R -bimodule and Q into a R, S -bimodule (if this is unfamiliar to you, see [6, chapter 23.2] for a review of the relation between R and its endomorphisms). Now, since F, G preserve limits/colimits, $F(R^\oplus) = P^\oplus$ and $G(S^\oplus) = Q^\oplus$, and any relation is preserved as equalizers are preserved, hence we have that for any $M \in \text{Obj}({}_R\mathbf{Mod}), N \in \text{Obj}({}_S\mathbf{Mod})$

$$F(M) \cong P \otimes_R M \quad \text{and} \quad G(N) \cong Q \otimes_S N$$

as bimodules, and:

$$GF(R) = G(P) = Q \otimes_S P \stackrel{!}{\cong} R \quad \text{and} \quad FG(S) = F(Q) = P \otimes_R Q \stackrel{!}{\cong} S$$

giving a very structured result for Morita equivalent categories! The $\stackrel{!}{\cong}$ comes from how we can think of the endomorphisms as being the “negative dimensional” lines bundles, or for more rigor you can checkout [6, chapter 24.2]. The following summarizes our above discussion and adds some useful insights the user may appreciate:

Theorem 2.4.2: Structure Theorem For Morita Equivalence

Let R, S be Morita Equivalent rings (represented by the functors F, G) and $F(R) = P$, $G(S) = Q$. Then:

1. P, Q are finitely generated projective both as R -modules and S -modules
2. $\text{End}_S(P) \cong R \cong \text{End}_S(Q)^{\text{op}}$ and $\text{End}_R(Q) \cong S \cong \text{End}_R(P)^{\text{op}}$
3. P, Q are dual to each other as S -modules:

$$P \cong \text{Hom}_S(Q, S) \quad \text{and} \quad Q \cong \text{Hom}_S(P, S)$$

The tensor gives us the R -tensor equivalent. Furthermore, $\text{Hom}_S(P, -)$ and $\text{Hom}_R(Q, -)$ (note it is over R) are faithful functors from $\mathbf{Mod}$ to $\mathbf{Ab}$ (for the respective rings).

4. The image of F, G are isomorphic (as bimodules) to:

$$F(M) \cong P \otimes_R M \quad \text{and} \quad G(N) \cong Q \otimes_S N$$

Proof:

1. This comes from $Q \otimes_S P \cong_R R$ and $P \otimes_R Q \cong_S S$
2. Follows from equation 2.1.
3. summarizes the above discussion
4. ibid

Corollary 2.4.3: Morita Equivalence and K Group

Let R, S be Morita Equivalent. Then

$$K_0(R) \cong K_0(S)$$

Proof:

Since P, Q are finitely generated and projective, and F, G represent the equivalence, we get

$$P(R) \cong P(S)$$

giving the desired result

Naturally, having isomorphic K -groups does not imply Morita equivalence (think $K_0(\mathbb{Z})$ and $K_0(\mathbb{Q})$).

Example 2.4.4: Morita Equivalence

The prototypical example is R and $M_n(R)$ for $n \geq 1$. In this case, $F(R) = P$ is the bimodule R^n of “column vectors” while $Q(M_n(R)) = (R^n)^t$ is the bimodule of “row vectors”.

More generally, by theorem 2.4.2(3), if P is a faithful finitely generated projective R -module, then $\text{End}_R(P)$ is Morita Equivalent to R .

2.5 *Mayer-Vietoris Sequence

In algebraic topology, the Mayer-Vietoris sequence is a useful sequence for computing homotopic groups by gluing together pieces. We bring over this discussion for computing K groups when gluing user Milnor squares. As we saw via the Hattori-Stallings Trace Map allows for some very important decompositions:

$$R \rightarrow \frac{R}{[R, R]}$$

As we just saw, $M_n(R)$ produces equivalent categories. More generally, in representation theory we are interested in $\text{GL}_n(R)$, the general linear group. Which n should we pick? Can we relate the n ’s in Chapter 3, we shall take the limit over all the n to define the group $\text{GL}(R)$. For now, we relate these through the natural map:

$$g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

These groups shall be central in understanding K_1 in the next section. One important construction we shall use is patching together information using Milnor patching: consider the Milnor square:

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow & & \downarrow \\ R/I & \xrightarrow{\bar{f}} & S/I \end{array}$$

Recall theorem 1.1.14(3) that any finitely generated projective R -module can derive from patching, hence define

$$\partial_n : \text{GL}_n(S/I) \rightarrow K_0(R)$$

where $\partial_n(g) = [P] - [R^n]$. By theorem 1.1.14(4), we see that:

$$\partial_n(g) = \partial_{n+1} \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

Hence, taking the sequence of $\{\partial_n\}_{n=1}^\infty$ give a homomorphism:

$$\partial : GL(S/I) \rightarrow K_0(R)$$

Then from Milnor-Patching

Theorem 2.5.1: Mayer-Vietoris

Given a Milnor square as above, the sequence

$$GL(S/I) \xrightarrow{\partial} K_0(R) \xrightarrow{\Delta} K_0(S) \oplus K_0(R/I) \xrightarrow{\pm} K_0(S/I)$$

is exact. The image of ∂ is the double coset space

$$GL(S) \backslash GL(S/I) / GL(R/I) = GL(S/I) / \sim$$

where $x \sim gxh$ for $x \in GL(S/I)$, $g \in GL(S)$, and $h \in GL(R/I)$.

Proof:

Follows from theorem 1.1.14(4) and our above discussion.

The Mayer-Vietoris sequence is a useful computation tool, so let's show how it can be used:

Example 2.5.2: Mayer-Vietoris

Consider the coordinate ring of an irreducible curve with a node (which can be shown is unique up to projective equivalence^a). Let us compute K_0 . As this is a 1-dimensional Noetherian ring, by corollary 2.3.11 we have:

$$K_0(R) \cong H_0(R) \oplus \text{Pic}(R)$$

Then the result follows from the following exact sequence known as the *Units-Pic sequence*:

Let Δ denotes the diagonal map, $\pm$ the difference map sending $(s, \bar{r})$ to $\bar{s}f(\bar{r})^{-1}$, resp. (L', L) to $L' \otimes_S S/I \otimes_{R/I} L^{-1}$. Then given a Milnor square, the following is exact:

$$\begin{aligned} 1 \rightarrow R^\times &\xrightarrow{\Delta} S^\times \times (R/I)^\times \xrightarrow{\pm} (S/I)^\times \xrightarrow{\partial} \\ \text{Pic}(R) &\xrightarrow{\Delta} \text{Pic}(S) \times \text{Pic}(R/I) \xrightarrow{\pm} \text{Pic}(S/I). \end{aligned}$$

Proof. As R is the pullback of S and R/I , exactness at the first two places is immediate. Theorem 1.1.14 immediately gives exactness at the last two places, leaving only exactness at $(S/I)^\times$. Given $s \in S^\times$ and $\bar{r} \in (R/I)^\times$, set $\beta = \pm(s, \bar{r}) = \bar{s}f(\bar{r})^{-1}$, where $\bar{s}$ denotes the reduction of s modulo I . Then we see that $\lambda = (s, \bar{r}) \in L_\beta \subset S \times R/I$, and every element of L_β is a multiple of λ . It follows that $L_\beta \cong R$. Conversely, suppose given $\beta \in (S/I)^\times$ with $L_\beta \cong R$. If $\lambda = (s, \bar{r})$ is a generator of L_β , then I claim that s and $\bar{r}$ are units, which implies that $\beta = \bar{s}f(\bar{r})^{-1}$ which completes the proof. To that end, if $s' \in S$ maps to $\beta \in S/I$, then $(s', 1) \in L_\beta$, namely since $(s', 1) = (xs, x\bar{r})$ for some $x \in R$, this implies that $\bar{r} \in (R/I)^\times$. If $t \in S$ maps to $f(\bar{r})^{-1}\beta^{-1} \in S/I$, then $st \equiv 1$ modulo I . But then $I \subset sR$ because $I \times 0 \subset L_\beta$, so $st = 1 + sx$ for some $x \in R$. So $s(t - x) = 1$, and hence $s \in S^\times$ giving the desired result. $\square$

Hence, we get that:

$$K_0(R) \cong \mathbb{Z} \oplus k^\times$$

^aEquivalence after a linear isomorphism

K_1 and K_2 Group

The K_0 group let us classify the vector bundles over a ring or (paracompact) topological space. As shown in the topological K -theory of [5], for (say complex) vector bundles over X :

$$K_0(X) = [X, \mathbb{Z} \times BU]$$

Showing there is a relation between homotopy groups and $K_0(X)$. From the algebraic perspective, we found the isomorphism classes of the projective modules. It is now natural to ask for the automorphism classes. We shall focus on the algebraic perspective, but shall see that it will naturally tie with the topological by taking suspension (recall from [10, chapter 5] that there is a natural map $\pi_k(X) \rightarrow \pi_{k+1}(\Sigma X)$).

3.1 Defining K_1

Let R be a ring (with unit). Take the maps:

$$G_n(R) \rightarrow G_{n+1}(R) \quad g \mapsto \begin{pmatrix} g & 0 \\ 0 & 1 \end{pmatrix}$$

The limit of these maps shall be denoted $GL(R)$. Then:

Definition 3.1.1: K_1 Group

Let R be a ring (not necessarily commutative). Then:

$$K_1(R) = \frac{GL(R)}{[GL(R), GL(R)]}$$

Given a ring homomorphism $R \rightarrow S$, there is a naturally induced map $\text{GL}(R) \rightarrow \text{GL}(S)$ by mapping the elements of R to the diagonal, and as this respects the abelianization we have a natural map:

$$K_1(R) \rightarrow K_1(S)$$

And hence, $K_1(-)$ is a functor as well.

Before looking at examples, let us have a better grasp of the commutator. As it turns out, these shall always be generated by elementary matrices.

Definition 3.1.2: Elementary Matrices Group of $\text{GL}(R)$

Let $E_n(R)$ denote the elementary matrices of $\text{GL}_n(R)$. Then $E(R)$ denotes the limit of the stabilization morphisms.

Lemma 3.1.3: Elementary Matrix Group is Perfect

For $n \geq 3$, $E_n(R)$ is a perfect group, that is $E_n(R) = [E_n(R), E_n(R)]$

For $n = 2$, we get $\text{SL}_2(\mathbb{F}_2) \cong S_2$ (symmetric group of order 2).

Proof:

For distinct i, j, k , we get:

$$[e_{ij}(r), e_{k\ell}(s)] = \begin{cases} 1 & \text{if } j \neq k \text{ and } i \neq \ell, \\ e_{i\ell}(rs) & \text{if } j = k \text{ and } i \neq \ell, \\ e_{kj}(-sr) & \text{if } j \neq k \text{ and } i = \ell. \end{cases}$$

and so:

$$e_{ij}(r) = [e_{ik}(r), e_{kj}(1)]$$

Theorem 3.1.4: Whitehead's Lemma

$$K_1(R) = \frac{\text{GL}(R)}{E(R)}$$

Proof:

The commutator subgroup contains $E(R)$ by lemma 3.1.3. Conversely, every commutator in $\text{GL}_n(R)$ can be expressed as a product in $\text{GL}_{2n}(R)$:

$$[g, h] = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & h^{-1} \end{pmatrix} \begin{pmatrix} (hg)^{-1} & 0 \\ 0 & hg \end{pmatrix}. \quad (3.1)$$

And certainly each of these terms is in $E_{2n}(R)$, giving the desired equivalence.

Example 3.1.5: K_1 Groups

1. Let R be commutative. Then we have a commutative group homomorphism:

$$\det : \mathrm{GL}(R) \rightarrow R^\times$$

This induces a surjective map $\det : K_1(R) \rightarrow R^\times$ whose kernel is written as $SK_1(R)$ (for reasons explained in a paragraph). Now, we also have a natural inclusion $R^\times \rightarrow \mathrm{GL}(R)$ mapping to the diagonals. We see that this map is a section map, i.e. it split by the determinant homomorphism. Taking $\mathrm{SL}(R) = \varprojlim_n \mathrm{SL}_n(R)$, as we are working over groups we have the semi-direct product (see [6, chapter 5.3]):

$$\mathrm{GL}(R) \cong \mathrm{SL}(R) \rtimes R^\times$$

Hence, when abelianizing we get:

$$K_1(R) \cong R^\times \oplus SK_1(R)$$

2. If we specifically let $R = k$ be a field where $k \notin \{\mathbb{F}_2, \mathbb{F}_3\}$, then we can show that $\mathrm{SL}(F) = [\mathrm{GL}(F), \mathrm{GL}(F)]$, and so we reduce to:

$$K_1(F) \cong F^\times$$

To see this, we first certainly see that $[\mathrm{GL}(R), \mathrm{GL}(R)] \leq \mathrm{SL}(R)$ as each commutator $[A, B] = ABA^{-1}B^{-1}$ as determinant 1. For the other way around, we must show that each matrix in $\mathrm{SL}_n(R)$ can be written as a product of commutators. As we are working over a field, it suffices to show that all elementary matrices can be written as a product of commutators. But this is now a matter of computations, for example type I elementary matrices $A = I + tE_{ij}$ and $B = I + E_{jk}$ where $k \neq i, j$ (which is possible since $n \geq 3$), when taking their commutator we get:

$$[A, B] = ABA^{-1}B^{-1} = (I + tE_{ij})(I + E_{jk})(I - tE_{ij})(I - E_{jk}) = I + tE_{ik}$$

notice how it was importance for $n \geq 3$ to insure the computations follow.

3. If $R = R' \times R''$ is a product of rings, then $\mathrm{GL}(R) = \mathrm{GL}(R') \times \mathrm{GL}(R'')$, and the commutator decompose too giving us that products are respected by the K_1
4. The K_1 group is invariant under Morita equivalence. First, as R and $M_n(R)$ are Morita equivalent, we have that:

$$M_{mn}(R) = \mathrm{End}_R(R^m \otimes R^n) \quad M_m(M_n(R)) = \mathrm{End}_{M_n(R)}((M_n(R))^m)$$

And so, we have a natural isomorphism:

$$\mathrm{GL}_{mn}(R) \cong \mathrm{GL}_m(\mathrm{GL}_n(R))$$

which give the same “stabilization” morphism (the map between $\mathrm{GL}_n(R)$ and $\mathrm{GL}_{n+1}(R)$), and so:

$$K_1(R) \cong K_1(M_n(R))$$

5. Let $R = D$ be a division ring. Then for the same reason as field, every invertible matrix may be reduced to:

$$\text{diag}(r, 1, \dots, 1)$$

and so we have:

$$\frac{\text{GL}_n(R)}{E_n(D)} \cong D^\times / N$$

By lemma 3.1.8, we can reduce to:

$$\frac{\text{GL}_n(R)}{E_n(D)} \cong D^\times / [D^\times, D^\times]$$

for all $n > 1$.

6. Let $F/\mathbb{Q}$ be a finite field extension and $R \subseteq F$ the integrally closed subring (i.e. the algebraic integers). Then it is a result by Bass and Milnor that

$$SK_1(R) = 1$$

and so $K_1(R) = R^\times$. Then by Dirichlet Unit theorem (see [6, chapter 22]), we have that $R^\times$ is isomorphic to:

$$\mu(F) \oplus \mathbb{Z}^{s-1}$$

where $\mu(F)$ is the cyclic group of all roots of unity in F and s is the places at infinity of R , i.e. the number of real embeddings.

Let us take a moment to look at the K_1 group of semi-local rings due to their importance in representation theory (they are the “local rings” in the non-commutative case).

Definition 3.1.6: Semi-local Ring

Let R be a ring. Then R is *semilocal* if $R/J(R)$ is semisimple where $J(R)$ is the (left) Jacobian ideal.

Note that all division rings are semi-local, hence the discussion about semi-local rings generalizes the above discussion.

We shall find a representation of $[R^\times, R^\times]$ for semilocal rings. If $r, s \in R$ are such that $1 + rs$ is a unit, then so is $1 + sr$ because $(1 + sr)(1 - s(1 + rs)^{-1}r) = 1$. Then define subgroup $W(R) \leq R^\times$ generated by the $(1 + rs)(1 + sr)^{-1}$. It is not too hard to show that this subgroup belongs to $E_2(R)$. For $R = M_2(\mathbb{F}_2)$, $W(R) = R^\times \cong \Sigma_3$ and $K_1(R) = 1$ but $R^\times \neq [R^\times, R^\times]$. If T is the subring of upper triangular matrices in $M_2(\mathbb{F}_2)$, its group of units is abelian ($T^\times \cong \mathbb{Z}/2$), but $K_1(T) = 1$ since

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = (1 + rs)(1 + sr)^{-1} \text{ for } r = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } s = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Now, a mathematician called *Vaserstein* showed that that $W(R) = [R^\times, R^\times]$ if $\Lambda = R/\text{rad}(R)$ is a product of matrix rings, none of which is $M_2(\mathbb{F}_2)$, and at most one of the factors in Λ is $\mathbb{F}_2$. In particular, $W(R) = [R^\times, R^\times]$ for every local ring R . The same applies for semilocal rings:

Proposition 3.1.7: K_1 of Semilocal Rings

If R is a semilocal ring, then the natural inclusion of $R^\times = GL_1(R)$ into $GL(R)$ induces an isomorphism $K_1(R) \cong R^\times / W(R)$.

If R is a commutative semilocal ring, then

$$SK_1(R) = 1 \quad \text{and} \quad K_1(R) = R^\times.$$

Proof:

By example 3.1.5 it suffices to prove that $R^\times$ maps onto $K_1(R)$. This will follow by induction after showing that $GL_n(R) = E_n(R)GL_{n-1}(R)$. Let J denote the Jacobson radical of R , so that R/J is a finite product of matrix algebras over division rings. We will show that $GL_n(R) \twoheadrightarrow GL_n(R/J)$. Indeed, if $I \subseteq R$ is an ideal, each homomorphism $E_n(R) \rightarrow E_n(R/I)$ is onto as the generators $e_{ij}(r)$ of $E_n(R)$ map onto the generators of $E_n(R/I)$ as I is radical. Hence, $(R/J)^\times$ maps onto $K_1(R/J)$; every $\bar{g} \in GL_n(R/J)$ is a product $\bar{e}\bar{g}_1$, where $\bar{e} \in E_n(R/J)$ and $\bar{g}_1 \in GL_1(R/J)$.

Now, given $g \in GL_n(R)$, its reduction $\bar{g}$ in $GL_n(R/J)$ may be decomposed into $\bar{g} = \bar{e}\bar{g}_1$. Then we can lift $\bar{e}$ to an element $e \in E_n(R)$. The matrix $e^{-1}g$ is congruent to the diagonal matrix $\bar{g}_1$ modulo J , so its diagonal entries are all units and its off-diagonal entries lie in J . Using elementary row operations $e_{ij}(r)$ with $r \in J$, it is a matter of computation to reduce $e^{-1}g$ to a diagonal matrix, say to $D = \text{diag}(r_1, \dots, r_n)$. But then the matrix $\text{diag}(1, \dots, 1, r_n, r_n^{-1})$ is in $E_n(R)$. Multiplying D by this matrix yields a matrix in $GL_{n-1}(R)$, giving the induction and the proof.

Lemma 3.1.8: Determinant on semilocal Rings

Let R be a noncommutative semilocal ring ($R/J(R)$ is a semisimple ring). Then there exists a unique determinant map from

$$\det : GL_n(R) \twoheadrightarrow R^\times / W(R)$$

where:

1. $\det(e) = 1$ for every elementary matrix e ,
2. if $\rho = \text{diag}(r, 1, \dots, 1)$ and $g \in GL_n(R)$, then $\det(\rho \cdot g) = r \cdot \det(g)$. Then show that $\det$ is a group homomorphism: $\det(gh) = \det(g)\det(h)$.

In particular, when reducing to $K_1(R)$:

$$K_1(R) \cong R^\times / W(R)$$

Proof:

Since R is a semilocal ring, every matrix in $GL_n(R)$ can be expressed as a product of elementary matrices. Define the map $\det : GL_n(R) \rightarrow R^\times / W(R)$ as follows. For any elementary matrix e , we set $\det(e) = 1$. For a diagonal matrix $\rho = \text{diag}(r_1, r_2, \dots, r_n)$, define

$$\det(\rho) = r_1 r_2 \cdots r_n \cdot W(R)$$

For a general matrix $g \in GL_n(R)$, express g as a product of elementary matrices and diagonal matrices, and define $\det(g)$ as the product of the determinants of these factors. As any matrix can be written as product of elementary matrices, this map is the unique determinant map (if it satisfies the determinant properties). And indeed, let $\rho = \text{diag}(r, 1, \dots, 1)$ and $g \in GL_n(R)$. Express g as a product $g = e_1 e_2 \cdots e_k \cdot d$, where each e_i is an elementary matrix and d the diagonal matrix. Then $\rho \cdot g = \rho \cdot e_1 e_2 \cdots e_k \cdot d$. Then this is equivalent to multiplying the first row of g by r , which multiplies the determinant by r . Hence,

$$\det(\rho \cdot g) = r \cdot \det(g)$$

Next, we must show $\det$ is a group homomorphism

$$\det(gh) = \det(g) \det(h) \text{ for all } g, h \in GL_n(R)$$

It suffices to establish this for elementary and diagonal matrices. For elementary matrices e_1 and e_2 , we have $\det(e_1 e_2) = 1 = \det(e_1) \det(e_2)$. For diagonal matrices $d_1 = \text{diag}(r_1, \dots, r_n)$ and $d_2 = \text{diag}(s_1, \dots, s_n)$, we have $\det(d_1 d_2) = \det(\text{diag}(r_1 s_1, \dots, r_n s_n)) = r_1 s_1 \cdots r_n s_n \cdot W(R) = (r_1 \cdots r_n \cdot W(R))(s_1 \cdots s_n \cdot W(R)) = \det(d_1) \det(d_2)$.

Finally, by definition $K_1(R)$ is the abelianization of $GL(R) = \varinjlim_n GL_n(R)$. The determinant map we constructed gives a homomorphism from $GL_n(R)$ to $R^\times / W(R)$ for each n , which are compatible under stabilization. Since $R^\times / W(R)$ is abelian and the determinant map captures exactly as we saw in example 3.1.5

$$K_1(R) \cong R^\times / W(R)$$

as we sought to show

With the basic properties of K_1 established on rings. Let us start by applying it to finitely generated projective R -modules. As $K_1(R)$ is defined as an amalgamation of all the “automorphisms” representable as matrices, one first thing we may want to find is a natural relation between a finitely generated projective R -module, say P , is the relation between $\text{Aut}(P)$ and $K_1(R)$. To that end, as P is projective, we know there exists a finitely generated projective R -module Q such that

$$P \oplus Q = R^n$$

Then we use this to define a natural map $\text{Aut}(P) \rightarrow GL_n(R)$ where $\alpha \mapsto \alpha \oplus 1_Q$. This is the basis the map to $K_1(R)$:

Lemma 3.1.9: Automorphism Map

There exists a well-defined map $\text{Aut}(P) \rightarrow GL(R)$ up to inner automorphism of $GL(R)$, and hence a well-defined homomorphism:

$$\text{Aut}(P) \rightarrow K_1(R)$$

Proof:

First, let's show it is defined up to inner automorphism. Fix Q and n . Then Two different isomorphisms between $P \oplus Q$ and R^n must differ by an automorphism of R^n , i.e., by an element

$g \in GL_n(R)$. Thus if $\alpha \in \text{Aut}(P)$ maps to the matrices A and B , respectively, we have:

$$A = gBg^{-1}$$

Next we observe that we can stabilize; replacing Q by $Q \oplus R^m$ and $P \oplus Q \cong R^n$ by $P \oplus (Q \oplus R^m) \cong R^{n+m}$, then $GL_n(R) \rightarrow GL(R)$ factors through $GL_{n+m}(R)$. Finally, suppose given a second isomorphism $P \oplus Q' \cong R^m$. Since $Q \oplus R^m \cong R^n \oplus Q'$, we may stabilize both Q and Q' to make them isomorphic and invoke the above argument.

But then, we get a map $\text{Aut}(P) \rightarrow GL(R)$, and as its defined up to inner automorphism we have a well-defined map:

$$\text{Aut}(P) \rightarrow K_1(R)$$

as we sought to show.

Corollary 3.1.10: Tensor and $K_1(S)$ as $K_0(R)$ module

Let R, S be rings. Then there is a natural tensor product:

$$K_0(R) \otimes K_1(S) \rightarrow K_1(R \times S)$$

If R is commutative and S an R -algebra, then we further have a natural map:

$$K_0(R) \otimes K_1(S) \rightarrow K_1(S)$$

Hence, $K_1(S)$ is a module over the ring $K_0(R)$.

Proof:

For each finitely generated projective R -module P and each m , lemma 3.1.9 gives us:

$$\text{Aut}(P \otimes S^m) \rightarrow K_1(R \otimes S)$$

Next, for each $\beta \in GL_m(S)$, let $[P] \cdot \beta$ denote the image of the automorphism $1_P \otimes \beta$ of $P \otimes S^m$ under this map. Fixing β and m , the isomorphism

$$(P \oplus P') \otimes S^m \cong (P \otimes S^m) \oplus (P' \otimes S^m)$$

yields the identity

$$[P \oplus P'] \cdot \beta = [P] \cdot \beta + [P'] \cdot \beta$$

in $K_1(R \otimes S)$. Hence $P \mapsto [P] \cdot \beta$ is an additive function of $P \in \mathbf{P}(R)$, and so it factors through $K_0(R)$.

Now fix P . then the map $GL_m(S) \rightarrow K_1(R \otimes S)$ given by $\beta \mapsto [P] \cdot \beta$ is compatible with stabilization in m . Thus it factors through a map $GL(S) \rightarrow K_1(R \otimes S)$ and through a map $K_1(S) \rightarrow K_1(R \otimes S)$. Hence, we see the product is well-defined and bilinear.

When R is commutative, $K_0(R)$ is a ring (as we saw in chapter 2. If S is an R -algebra, there is a ring map $R \otimes S \rightarrow S$. Composing the external product with $K_1(R \otimes S) \rightarrow K_1(S)$ yields a natural product operation $K_0(R) \otimes K_1(S) \rightarrow K_1(S)$. But then we see that that

$$[P \otimes_R Q] \cdot \beta = [P] \cdot ([Q] \cdot \beta)$$

completing the proof.

Recall from section 2.3 that we connected K_0 with the H_0 homology. We now do the same with K_1 . Recall first from [6, chapter 28] that $H_1(G; \mathbb{Z})$ is naturally isomorphic to $G/[G, G]$. This is great for us: if $G = \mathrm{GL}(R)$, we get:

$$K_1(R) = H - 1(\mathrm{GL}(R); \mathbb{Z}) = \lim_{n \rightarrow \infty} H - 1(\mathrm{GL}_n(R); \mathbb{Z})$$

By lemma 3.1.9, we get the composition:

$$H_1(\mathrm{Aut}(P); \mathbb{Z}) \rightarrow H_1(\mathrm{GL}_n(R); \mathbb{Z}) \rightarrow K_1(R)$$

Next, we want to show that $K_1(-)$ is Morita-invariant. To that end, we need to view the homological interpretation in a new light. Let $\mathbf{P}(R)$ be the category of finitely generated projective R -modules and let $t\mathbf{P}$ (the objects are the same, the morphisms between P and P' are the isomorphism classes of Q such that $P \oplus Q = P'$). Then $P \mapsto H - 1(\mathrm{Aut}(P) : \mathbb{Z})$ is a well-defined functor from $t\mathbf{P}$ to $\mathbf{Ab}$. Thus, we can take filtered colimits of the functor (recall [6, chapter 26]! But then, by taking advantage of the fact that free modules are cofinal in $t\mathbf{P}$, we get:

Lemma 3.1.11: Morita Buildup (Bass)

$$K_1(R) \cong \lim_{P \in t\mathbf{P}} H_1(\mathrm{Aut}(P); \mathbb{Z})$$

Proof:

outlined above

Now, as $\mathbf{P}(R)$ and $\mathbf{P}(S)$ are Morita equivalent, we apply the above corollary to get:

Proposition 3.1.12: Morita Invariance of K_1

Let R, S be Morita-equivalent rings. Then:

$$K_1(R) \cong K_1(S)$$

Proof:

Looking at the usual case of R and $M_n(R)$ using the corollary 3.1.11, and that we have the Morita-equivalent categories to get:

$$K_1(R) \cong K_1(M_n(R))$$

Next, let us look at the structure given a ring homomorphism $f : R \rightarrow S$. We can define the same map as we've done with K_0 , however we shall see that there is another arguably more interesting map that has better applications in cohomology. We shall show that whenever SS has a finite R -module resolution by finitely generated projective R -modules, then we can define:

$$f_* : K_1(S) \rightarrow K_1(R)$$

which is an interesting difference from the K_0 where we simply took $f \otimes S : \mathbf{P} \rightarrow \mathbf{S}$.

Lemma 3.1.13: Simple Transfer Functor

Let $T : \mathbf{S} \rightarrow \mathbf{R}$ be an additive functor. Then it induces a homomorphism

$$K_1(T) : K_1(S) \rightarrow K_1(R)$$

And $T_1 \oplus T_2$ induces the sum $K_1(T_1) + K_1(T_2)$.

Proof:

First, the functor T induces an evident functor $t\mathbf{P}(S) \rightarrow t\mathbf{P}(R)$. Next, if P is a finitely generated projective S -module, by cohomology theory T also induces a homomorphism $\text{Aut}_S(P) \rightarrow \text{Aut}_R(TP)$ and hence $H_1(\text{Aut}_S(P); \mathbb{Z}) \rightarrow H_1(\text{Aut}_R(TP); \mathbb{Z})$. As P varies, these assemble to give a natural transformation of functors from the translation category $t\mathbf{P}(S)$ to abelian groups. Since $K_1(S) = \varinjlim_{P \in \mathbf{P}(S)} H_1(\text{Aut}_S(P); \mathbb{Z})$ by corollary 3.1.11 taking the direct limit over $t\mathbf{P}(S)$ yields:

$$K_1(S) \rightarrow \varinjlim_{P \in \mathbf{P}(S)} H_1(\text{Aut}_R(TP); \mathbb{Z}) \rightarrow \varinjlim_{Q \in \mathbf{P}(R)} H_1(\text{Aut}_R(Q); \mathbb{Z}) = K_1(R)$$

as we sought to show.

Corollary 3.1.14: $K_1(-)$ Functor

Suppose that S is finitely generated projective as an R -module. Then the forgetful functor $\mathbf{P}(S) \rightarrow \mathbf{P}(R)$ induces a natural transfer homomorphism $f_* : K_1(S) \rightarrow K_1(R)$. If R is commutative, the composite

$$K_1(R) \xrightarrow{f^*} K_1(S) \xrightarrow{f_*} K_1(R)$$

is multiplication by $[S] \in K_0(R)$.

Proof:

Let T be the forgetful map. Then the map $K_1(S) \xrightarrow{f_*} K_1(R)$ is the transfer map. We compute the composite $f_* \circ f^*$ by computing its effects on $\alpha \in GL_n(R)$. We see that the matrix $f^*(\alpha) = 1_S \otimes_R \alpha$ lies in $GL_n(S)$. To apply f_* we consider $1_S \otimes_R \alpha$ as an element of the group $\text{Aut}_R(S^n) = \text{Aut}_R(S \otimes_R R^n)$, which we then map into $GL(R)$. But this is just the product $[S] \cdot \alpha$, completing the proof.

3.1.1 Relative K-group: Connection to Homology

The next process should be familiar to any reader with some knowledge of relative homology, and will give us our first deep connection between the K_n groups and homology. Consider an ideal $I \subseteq R$. Then given a map $R \rightarrow R/I$, we get a natural map $GL(R) \rightarrow GL(R/I)$. Let $GL(I)$ be its kernel. Now, let $E(R, I)$ be the smallest normal subgroup containing the elementary matrices $e_{ij}(x)$ with $x \in I$. Note that $E(R, I) = \bigcup_n E_n(R, I)$. Then we have

Theorem 3.1.15: Whitehead's Lemma for Relative Groups

Let R be a ring and $I \subseteq R$ an ideal. Then $E(R, I) \trianglelefteq GL(I)$ and contains the commutator subgroup of $GL(I)$.

Proof:

For any matrix $g = 1 + \alpha \in GL_n(I)$, the identity

$$\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \alpha & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & g^{-1}\alpha \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -g\alpha & 1 \end{pmatrix} \quad (3.2)$$

shows that the matrix $\begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}$ is in $E_{2n}(R, I)$. (The product of the first three matrices is in $E_{2n}(R, I)$.) Hence if $h \in E_n(R, I)$, then the conjugate

$$\begin{pmatrix} ghg^{-1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \begin{pmatrix} h & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} g^{-1} & 0 \\ 0 & g \end{pmatrix} \quad (3.3)$$

is in $E(R, I)$. Finally, if $g, h \in GL_n(I)$, then $[g, h]$ is in $E_{2n}(R, I)$ by equation (3.1).

Thus, the following is well-defined:

Definition 3.1.16: Relative K_1 -group

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_1(R, I) = \frac{GL(I)}{E(R, I)}$$

Proposition 3.1.17: Snake Lemma For K-theory

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_1(R, I) \hookrightarrow K_1(R) \rightarrow K_1(R/I) \xrightarrow{\partial} K_0(I) \rightarrow K_0(R) \rightarrow K_0(R/I)$$

Proof:

First, we shall prove a simpler exact sequence:

$$1 \rightarrow GL(I) \rightarrow GL(R) \rightarrow GL(R/I) \xrightarrow{\partial} K_0(I) \rightarrow K_0(R) \rightarrow K_0(R/I). \quad (3.4)$$

First, consider the short exact sequence:

$$0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0 \quad (3.5)$$

where I is an ideal in R . This induces a sequence of group homomorphisms

$$1 \rightarrow GL(I) \rightarrow GL(R) \rightarrow GL(R/I) \quad (3.6)$$

where $GL(I)$ consists of matrices of the form $1 + \alpha$ with α having entries in I . This sequence is exact since any matrix that maps to the identity in $GL(R/I)$ must have the form $1 + \alpha$ with α having entries in I .

To extend this to the K -theory groups, we define the boundary map $\partial : GL(R/I) \rightarrow K_0(I)$ as follows. For any $\bar{g} \in GL(R/I)$, we can lift it to a matrix $g \in M_n(R)$ (not necessarily invertible). The matrix g induces an R -module homomorphism $R^n \rightarrow R^n$. Let C be the cokernel of this homomorphism. Since $\bar{g}$ is invertible, the reduction of C modulo I is zero, which means C is an I -module. We define $\partial(\bar{g}) = [C] \in K_0(I)$. Verifying exactness is now a standard procedure.

Thus, we have the complete exact sequence

$$1 \rightarrow GL(I) \rightarrow GL(R) \rightarrow GL(R/I) \xrightarrow{\partial} K_0(I) \rightarrow K_0(R) \rightarrow K_0(R/I). \quad (3.7)$$

Since the K_1 groups are quotients of the GL groups and $E(R)$ maps onto $E(R/I)$, this gives exactness except at $K_1(R)$. Suppose $g \in GL(R)$ maps to zero under $GL(R) \rightarrow K_1(R) \rightarrow K_1(R/I)$. Then the reduction $\bar{g}$ of g mod I is in $E(R/I)$. Since $E(R)$ maps onto $E(R/I)$, there is a matrix e in $E(R)$ mapping to $\bar{g}$, i.e., ge^{-1} is in the kernel $GL(I)$ of $GL(R) \rightarrow GL(R/I)$. Hence the class of ge^{-1} in $K_1(R, I)$ is defined and maps to the class of g in $K_1(R)$, giving exactness as we sought to show.

3.2 A word on K_2 Groups

The K_0 and K_1 group we have covered so far had natural definitions motivated outside of homology. However, they both naturally were related to homology, namely they were tied to H_0 and H_1 of the group homology for matrix rings (where R can be seen as a 1-dimensional matrix ring). We may ask if we can further extend this and define higher K -groups via this method. We shall explore this in more detail in the next chapter, but in this chapter we shall cover more “naturally” how we may discover finding K_2 without simply saying $K_2(R) = H_2(E(R); \mathbb{Z})$. Given the group $E(R)$, a natural question to ask is what extensions we may have. Given we want to quotient by the centralizer, we are particularly interested in *central extensions*. Even better if we can find a *universal central extension*. Topologically, this would be associated to the *universal cover* of $BE(R)$ (similarly as to what we saw in topological K -theory). The group associated to this is called the *Steinberg Group*, and is often denoted $st(R)$. To stay concrete, we shall define everything “low-brow” directly with elements instead of as the consequence of universal properties. The advantage of this approach is to make it easier to “play” with the objects and get a feeling to the information they store:

Definition 3.2.1: Steinberg Group

Let R be a ring. Then for $n \geq 3$ the *Steinberg group* $\text{st}_n(R) \leq M_n(R)$ is given by the generators $x_{ij}(r)$ respecting the Steinberg relation:

1.

$$x_{ij}(r)x_{ij}(s) = x_{ij}(r+s)$$

2.

$$[x_{ij}(r), x_{kl}(s)] = \begin{cases} 1 & j \neq k, i \neq l \\ x_{il}(rs) & j = k, i \neq l \\ x_{kj}(rs) & j \neq k, i = l \end{cases}$$

Recall from lemma 3.1.3 that the elementary matrices of $E(R)$ satisfy the same relationships, hence there is a canonical surjection

$$\text{st}_n(R) \rightarrow E_n(R) \quad x_{ij}(r) \mapsto e_{ij}(r)$$

The Group $\text{st}_n(R)$ naturally embeds in $\text{st}_{n+1}(R)$ in the usual way, hence we may define $\text{st}(R)$. By looking at the kernel of this map, we get the following:

Definition 3.2.2: K_2 Group

Let R be a ring. Then the $K_2(R)$ is defined as the kernel of the map $\text{st}(R) \rightarrow E(R)$.

Thus, it fits in the exact sequence:

$$1 \rightarrow K_2(R) \rightarrow \text{st}(R) \rightarrow E(R) \rightarrow K_1(R) \rightarrow 1$$

We mentioned that $K_2(R)$ must come from the center of a group. We start by showing exactly this

Theorem 3.2.3: K_2 and Center of Steinberg

Let R be a ring. Then $K_2(R)$ is an abelian group and is the center of $\text{st}(R)$.

Proof:

If $x \in \text{st}(R)$ commutes with every element of $\text{st}(R)$, then $\varphi(x)$ must commute with all of $E(R)$. But the center of $E(R)$ is trivial (as it is a perfect group) so $\varphi(x) = 1$, that is $x \in K_2(R)$. Thus the center of $\text{st}(R)$ is contained in $K_2(R)$.

Conversely, suppose that $y \in \text{st}(R)$ satisfies $\varphi(y) = 1$. Then in $E(R)$ we have

$$\varphi([y, p]) = \varphi(y)\varphi(p)\varphi(y)^{-1}\varphi(p)^{-1} = \varphi(p)\varphi(p)^{-1} = 1 \quad (3.8)$$

for every $p \in \text{st}(R)$. Choosing an integer n large enough so that y can be expressed as a word in the symbols $x_{ij}(r)$ with $i, j < n$, for each element $p = x_{kn}(s)$ with $k < n$ and $s \in R$, the Steinberg relations implies that the commutator $[y, p]$ is an element of the subgroup P_n of $\text{st}(R)$ generated by the symbols $x_{in}(r)$ with $i < n$. On the other hand, we can see that φ maps P_n injectively into $E(R)$. Since $\varphi([y, p]) = 1$, this implies that $[y, p] = 1$. Hence y commutes with every generator $x_{kn}(s)$ with $k < n$.

By symmetry, we get that y also commutes with every generator $x_{nk}(s)$ with $k < n$. Hence y commutes with all of $St_n(R)$, since it commutes with every $x_{kl}(s) = [x_{kn}(s), x_{nl}(1)]$ with $k, l < n$. Since n can be arbitrarily large, this proves that y is in the center of $st(R)$ showing it is indeed the center, completing the proof.

Example 3.2.4: K_2 groups

The computation of K_2 group becomes a little non-trivial, however here are some interesting results:

1. Let $R = \mathbb{Q}$. Then it is somewhat remarkable that:

$$K_2(\mathbb{Q}) = (\mathbb{Z}/4\mathbb{Z})^* \times \prod_{p \text{ odd prime}} (\mathbb{Z}/p\mathbb{Z})^*$$

where this was proven following a proof similar to Gauss's proof of the Quadratic Reciprocity Law!

2. It was proven by R.K. Dennis and J. Sylvester that $K_2(\mathbb{F}[t]) = K_2(\mathbb{F})$, showing that $K_2(-)$ is invariant under extension by an algebraically independent element, and hence we can think of it as extracting "relational" or "number-theoretic" information.
3. More generally, Matsumoto's theorem (see [11]) states that:

$$K_2(k) \cong k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1-a) \mid a \in k^\times \setminus \{1\} \rangle$$

The proof is complicated, but due to our concrete definition given above we can see the outline of the proof. We shall draw on some results from section 3.2.1 to show that this group shall satisfy the same universal property of central extensions. For $a \in k^\times$, define the element $w_{ij}(a) \in st(k)$ by:

$$w_{ij}(a) = x_{ij}(a)x_{ji}(-a^{-1})x_{ij}(a)$$

For $a, b \in k^\times$, define the Steinberg symbol $\{a, b\} \in K_2(k)$ as:

$$\{a, b\} = h_{ij}(a)h_{ij}(b)h_{ij}(ab)^{-1}$$

where $h_{ij}(a) = w_{ij}(a)w_{ij}(1)^{-1}$. Then by the linearity properties given in definition 3.2.1, $\{a, b\}$ satisfy the following bilinearity relations:

$$\begin{aligned} \{a_1 a_2, b\} &= \{a_1, b\} \{a_2, b\} \\ \{a, b_1 b_2\} &= \{a, b_1\} \{a, b_2\} \end{aligned}$$

The crucial step is proving that: $\{a, 1-a\} = 1$ for all $a \in k^\times \setminus \{1\}$. To establish this, we work with explicit matrices in $SL_3(k)$ and their lifts to the Steinberg group. Consider the elements:

$$\begin{aligned} u(a) &= x_{12}(a)x_{23}(1)x_{12}(-a)x_{23}(-1) \\ v(a) &= x_{13}(a(1-a))x_{23}(-a)x_{12}(-(1-a)) \end{aligned}$$

Through direct calculation in $\text{st}(k)$, one can verify that $u(a)v(a)^{-1} \in K_2(k)$ and equals $\{a, 1-a\}^{-1}$. Furthermore, one can show that $u(a)v(a)^{-1} = 1$, establishing the Steinberg relation. Next, let $\mathcal{M}(k)$ denote the abstract group:

$$\mathcal{M}(k) = k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1-a) \mid a \in k^\times \setminus \{1\} \rangle$$

Define a homomorphism $\varphi : \mathcal{M}(k) \rightarrow K_2(k)$ by mapping $a \otimes b \mapsto \{a, b\}$. The bilinearity relations and the Steinberg relation ensure that φ is well-defined. If φ is an isomorphism, we're done. To that end, we construct a homomorphism $\psi : K_2(k) \rightarrow \mathcal{M}(k)$ and show that $\psi \circ \varphi = \text{id}_{\mathcal{M}(k)}$ and $\varphi \circ \psi = \text{id}_{K_2(k)}$.

The construction of ψ relies on the universal property of the Steinberg group as the universal central extension of $\text{SL}(k)$. Specifically, we use the fact that any central extension of $\text{SL}(k)$ factors uniquely through $\text{St}(k)$ (see section 3.2.1). The proof would then be done by showing that $\mathcal{M}(k)$ satisfies the same universal property as $K_2(k)$ with respect to central extensions of $\text{SL}(k)$. This involves constructing a central extension:

$$1 \rightarrow \mathcal{M}(k) \rightarrow E \rightarrow \text{SL}(k) \rightarrow 1$$

and proving that any central extension of $\text{SL}(k)$ factors uniquely through E (i.e. that it satisfies the universal property). By the uniqueness of universal central extensions, we conclude that

$$\mathcal{M}(k) \cong K_2(k)$$

and hence $K_2(k) \cong k^\times \otimes_{\mathbb{Z}} k^\times / \langle a \otimes (1-a) \mid a \in k^\times \setminus \{1\} \rangle$. We can verify this works by seeing that this matches our earlier claim of $K_2(\mathbb{Q})$!

4. Let $R = \mathbb{Z}$. Then $K_2(R) \cong \mathbb{Z}/2\mathbb{Z}$. Indeed, by Matsumoto's theorem, we have:

$$K_2(\mathbb{Q}) \cong \mathbb{Q}^\times \otimes_{\mathbb{Z}} \mathbb{Q}^\times / \langle a \otimes (1-a) \mid a \in \mathbb{Q}^\times \setminus \{1\} \rangle$$

Every element in $\mathbb{Q}^\times$ can be written as $\pm p_1^{n_1} \cdots p_k^{n_k}$ where p_i are distinct primes and $n_i \in \mathbb{Z}$. Thus, $K_2(\mathbb{Q})$ is generated by symbols of the form:

$$\{-1, -1\}, \{-1, p\}, \{p, q\}$$

where p, q are primes. For the next step, we shall combine theorem 3.2.10 which is a generalization of the snake lemma, and [12, chapter 5] from class field theory to get:

$$K_2(\mathcal{O}_F) \rightarrow K_2(F) \xrightarrow{\partial} \bigoplus_{\mathfrak{p}} \kappa(\mathfrak{p})^\times \rightarrow K_1(\mathcal{O}_F) \rightarrow K_1(F)$$

For $\mathbb{Z} \subset \mathbb{Q}$, this becomes:

$$K_2(\mathbb{Z}) \rightarrow K_2(\mathbb{Q}) \xrightarrow{\partial} \bigoplus_{p \text{ prime}} \mathbb{F}_p^\times \rightarrow K_1(\mathbb{Z}) \rightarrow K_1(\mathbb{Q})$$

where ∂ is the “generalized quadratic reciprocity”:

$$\partial_p(\{a, b\}) = (-1)^{v_p(a)v_p(b)} \cdot \left(\frac{a^{v_p(b)}}{b^{v_p(a)}} \right) \bmod p$$

where v_p is the p -adic valuation. For the generators of $K_2(\mathbb{Q})$, we compute:

$$\begin{aligned}\partial_p(\{-1, -1\}) &= 1 \text{ for all primes } p \\ \partial_p(\{-1, q\}) &= \begin{cases} -1 \bmod p & \text{if } p = q \\ 1 \bmod p & \text{if } p \neq q \end{cases} \\ \partial_p(\{p, q\}) &= \begin{cases} q \bmod p & \text{if } p = p \\ p^{-1} \bmod q & \text{if } p = q \\ 1 \bmod r & \text{if } p \neq p, q \end{cases}\end{aligned}$$

By definition, $K_2(\mathbb{Z})$ is the kernel of ∂ . An element $\{a, b\} \in K_2(\mathbb{Q})$ is in $K_2(\mathbb{Z})$ if and only if $\partial_p(\{a, b\}) = 1$ for all primes p .

Using the quadratic reciprocity law and properties, we see that the only non-trivial element in $K_2(\mathbb{Q})$ that maps to 1 under is $\{-1, -1\}$. To show its in $K_2(\mathbb{Z})$, we need to determine the order of $\{-1, -1\}$ in $K_2(\mathbb{Q})$. Using the properties of Steinberg symbols:

$$\begin{aligned}\{-1, -1\}^2 &= \{(-1)^2, -1\} \\ &= \{1, -1\} \\ &= 1\end{aligned}$$

Thus, $\{-1, -1\}$ has order 2 in $K_2(\mathbb{Q})$ and hence in $K_2(\mathbb{Z})$.

From these examples, we also see how $K_2(-)$ is directly tied to number theory!

3.2.1 Universal Central Extensions

As described earlier, $\text{st}(R)$ can naturally be seen as being the *universal central extension* of $E(R)$. Let us dive into what this means.

Let G be a group and A an abelian group. Then a *central extension* of G by A^1 if there exists a short exact sequence

$$1 \rightarrow A \rightarrow X \rightarrow G \rightarrow 1$$

such that A maps into the center of X . The extension splits if there is a section map $G \rightarrow X$, or equivalently that it is isomorphic to the extension:

$$1 \rightarrow A \rightarrow A \times G \xrightarrow{\pi_2} G \rightarrow 1$$

By theorem 3.2.3, we have a central extension:

$$1 \rightarrow K_2(R) \rightarrow \text{st}(R) \rightarrow E(R) \rightarrow 1$$

If X, Y are extensions of G , they are said to be equivalent if there is an isomorphism $f : X \rightarrow Y$ which is the identity on A and induces the identity map on G . Then it is a known result that $H^2(G; A)$ classifies all the isomorphism classes (see [1] or [9]). A *universal central extension* of G would then be a central extension $X \rightarrow G$ such that for any other central extension $Y \rightarrow G$ it must

¹remember that in group theory we usually reverse the order when we say “ X ” is extending “ Y ”, hence why we say G is extended by A

factor through $X \rightarrow G$ uniquely. Of course, we have yet to show that a universal central extension always exists. We build-up to this result now:

Lemma 3.2.5: Universal Central Extension and Perfect Groups I

Let G have a universal central extension $X \rightarrow G$. Then G, X must be perfect groups (have no center).

Proof:

Assume $B = \frac{X}{[X, X]}$ is nontrivial. Then there would be two at least homomorphisms over G from X to the central extension:

$$1 \rightarrow B \rightarrow B \times G \rightarrow G \rightarrow 1$$

namely $(0, \pi)$ and (pr, π) where $pr : X \rightarrow B$ is the natural projection, contradicting the uniqueness of universality.

Lemma 3.2.6: Universal Central Extension and Perfect Groups II

Let G have two central extension X, Y , and assume X is a perfect group. Then there is at most one homomorphism over G from X to Y .

Proof:

If f and f' are two such homomorphisms, then for any x and x' in X we can write $f'(x) = f(x)c$, $f'(x') = f(x')c'$ for elements c and c' in the center of Y . Therefore $f'(xx'x^{-1}(x')^{-1}) = f(xx'x^{-1}(x')^{-1})$. Since the commutators $[x, x'] = xx'x^{-1}(x')^{-1}$ generate X , we must have $f' = f$, completing the proof.

Theorem 3.2.7: Universal Central Extensions of Perfect Groups

Let G be a perfect group. Then G has a universal central extension:

$$1 \rightarrow H_2(G; \mathbb{Z}) \rightarrow \frac{[F, F]}{[R, F]} \rightarrow G \rightarrow 1$$

where F is a free group mapping onto G and $R \subseteq F$ is the kernel of this map (i.e. the relations).

If X is any central extension of G , then the following are equivalent:

1. X is a universal central extension
2. X is perfect, and every central extension of X splits
3. $H_1(X; \mathbb{Z}) = H_2(X; \mathbb{Z}) = 0$

Proof:

First, let $F \twoheadrightarrow G$ be a presentation of G with kernel R . Then $[R, F] \leq F$, and so we have a short exact sequence:

$$1 \rightarrow \frac{R}{[R, F]} \rightarrow \frac{F}{[R, F]} \rightarrow G \rightarrow 1$$

By the universal properties of commutators with some diagram chasing, we obtain:

$$1 \rightarrow \frac{R \cap [F, F]}{[R, F]} \rightarrow \frac{[F, F]}{[R, F]} \rightarrow [G, G] \rightarrow 1$$

If G is perfect, $G = [G, G]$, and by homology theory (see [1] or [9]) we have $\frac{R \cap [F, F]}{[R, F]} \cong H_2(G; \mathbb{Z})$, giving us the exact sequence.

For the equivalences, (2) $\Rightarrow$ (1) is immediate as trivial 1st/2nd homology groups implies there are only split exact sequences and lemma 3.2.6 give us the perfect condition, Lemma 3.2.5 gives (1) $\Rightarrow$ (2). Hence we'll show (1) $\Leftrightarrow$ (3). It suffices to show that the above exact sequence is a universal central extension. Given any central extension X of G , the map $\varphi : F \rightarrow G$ lifts to a map $\Phi : F \rightarrow X$ (as F is projective, as its free). Then as $\Phi(R)$ is the center of X , $\Phi([R, F]) = 1$, and so by the universal property of commutators Φ induces a map $\frac{[F, F]}{[R, F]} \rightarrow X$ over G . This map is unique by lemma 3.2.6, and hence the above sequence is indeed universal giving the desired result.

Corollary 3.2.8: K_2 Group and Homology

The Steinberg group $\text{st}(R)$ is the universal central extension of $E(R)$, and hence:

$$K_2(R) \cong H_2(E(R); \mathbb{Z})$$

Proof:

We shall show that for $n \geq 5$, every central extension $Y \xrightarrow{\varphi} \text{st}_n(R)$ splits, which by theorem 3.2.7 gives the desired result.

First, let's show that if $j \neq k$ and $l \neq i$, then every two elements $y, z \in Y$ with $\varphi(y) = x_{ij}(r)$ and $\varphi(z) = x_{kl}(s)$ must commute in Y . Pick t distinct from i, j, k, l and choose $y', y'' \in Y$ with $\varphi(y') = x_{it}(1)$ and $\varphi(y'') = x_{tj}(r)$. Then the Steinberg relations imply that both $[y', z]$ and $[y'', z]$ are in the center of Y , and since $\varphi(y) = \varphi([y', y''])$, this implies that z commutes with $[y', y'']$ and y .

Next, choose distinct indices i, j, k, l and elements $u, v, w \in Y$ with

$$\varphi(u) = x_{ij}(1), \quad \varphi(v) = x_{jk}(s), \quad \text{and} \quad \varphi(w) = x_{kl}(r). \quad (3.9)$$

If $G \leq Y$ is generated by u, v, w , then its commutator subgroup $[G, G]$ is generated by elements mapping under φ to $x_{ik}(s)$, $x_{jl}(sr)$, or $x_{il}(sr)$. From the first paragraph of this proof it follows that $[u, w] = 1$ and that $[G, G]$ is abelian. By group theory, we have $[[u, v], w] = [u, [v, w]]$, and so if $\varphi(y) = x_{ik}(s)$ and $\varphi(z) = x_{jl}(sr)$, we have $[y, w] = [u, z]$. Taking $s = 1$, this identity proves that the element

$$y_{il}(r) = [u, z], \quad \text{where } \varphi(u) = x_{ij}(1), \varphi(z) = x_{jl}(r), \quad (3.10)$$

does not depend upon the choice of j , nor upon the lifts u and z of $x_{ij}(1)$ and $x_{jl}(r)$.

Finally, I claim that the elements $y_{ij}(r)$ satisfy the Steinberg relations, so that there is a group homomorphism $St_n(R) \rightarrow Y$ sending $x_{ij}(r)$ to $y_{ij}(r)$. Such a homomorphism will provide the desired splitting of the extension φ . By the first paragraph of this proof, if $j \neq k$ and $l \neq i$, then $y_{ij}(r)$ and $y_{kl}(s)$ commute. The identity $[y, w] = [u, z]$ above may be rewritten as

$$[y_{ik}(r), y_{kl}(s)] = y_{il}(rs) \quad \text{for } i, k, l \text{ distinct.} \quad (3.11)$$

The final relation $y_{ij}(r)y_{ij}(s) = y_{ij}(r+s)$ follow similarly, giving us the desired result.

We shall not prove it here, but another important result is that $K_2(-)$ is stable under morita-equivalence (see [8] for the details). Instead, we shall finish off this section by extending the snake lemma for its use in example 3.2.4. First, for any ideal $I \subseteq R$, we may defined the augmented ring $R \oplus I$ via:

$$(r, x)(s, y) = (rs, ry + xs + xy)$$

There are two natural maps $\pi, + : R \oplus I \rightarrow R$ where $\pi(r, x) = r$ and $+(r, x) + r + x$. Let $\text{st}'(R, I)$ be the normal subgroup of $\text{st}(R \oplus I)$ generated by all $x_{ij}(0, v)$ where $v \in I$. Then there is certainly a map

$$\text{st}'(R, I) \rightarrow E(R \oplus I, 0 \oplus I)$$

where $E(R \oplus I, 0 \oplus I) \leq \text{GL}(R \oplus I)$. From this define:

Definition 3.2.9: Relative Steinberg Grouhatrp

Let R be a ring and $I \subseteq R$ an ideal. Then the *relative Steinberg group* $\text{st}(R, I)$ is the quotient of $\text{st}'(R, I)$ by the normal subgroup generate by all cross-commutators:

$$[x_{ij}(0, u), x_{kl}(v, -v)] \quad u, v \in I$$

We thus get:

Theorem 3.2.10: Snake Lemma For K-theory Extended

Let R be a ring and $I \subseteq R$ an ideal. Then:

$$K_2(R, I) \rightarrow K_2(R) \rightarrow K_2(R/I) \rightarrow K_1(R, I) \rightarrow K_1(R) \rightarrow \cdots \rightarrow K_0(R) \rightarrow K_0(R/I) \rightarrow K_0(I)$$

where there is no $\rightarrow 0$ at the end due to possible negative K -groups^a.

^aWe have not covered negative K -theory, but you can checkout [8] for more details

Proof:

This follows by doing some diagram chasing on:

$$\begin{array}{ccccccc}
 K_2(R, I) & \longrightarrow & St(R, I) & \longrightarrow & GL(I) & \longrightarrow & K_1(R, I) \\
 \downarrow & & \downarrow + & & \downarrow \text{inj.} & & \downarrow \\
 K_2(R) & \longrightarrow & St(R) & \longrightarrow & GL(R) & \longrightarrow & K_1(R) \\
 \downarrow & & \downarrow \text{surj.} & & \downarrow & & \downarrow \\
 K_2(R/I) & \longrightarrow & St(R/I) & \longrightarrow & GL(R/I) & \longrightarrow & K_1(R/I).
 \end{array}$$

and using snake's lemma (with proposition 3.1.17).

Higher K-Theory

As we have been discovering in the last 2 chapters, K -groups seem to be representable as the homology of certain groups. In the 1960s and 70s, a large amount of effort has been put in trying to find the right way to unify these results and give a single cohomology theory to unify these groups. As it turns out, the “right” approach was in fact to see each $K_n(R)$ as $\pi_n K(R)$ for a topological space $K(R)$, that is the *homotopy* groups rather than the homology groups. It turns out that in low dimensions, the homology and homotopy groups shall coincide, and in high dimensions the homotopy groups shall embed in the homology groups. The connection is less surprising when we see that the universal spaces given in [5] were the homotopy groups of some universal spaces. In particular, there has been a push to represent the information many algebraic objects as “spaces” (in many ways, the current abstract definition of topology was made to accommodate the more odd geometric behaviour these “classifying” spaces had) and we extracted information from these spaces using algebraic topology. In this chapter, we shall go over the basics of how we may generalize the K -theory we have been developing into higher order K -theory, and give the beginning of how these results link to the K -groups we have studied earlier.

4.1 Defining the Space: $BGL+$

Let G be any group. In [10, section 2.2], we saw we can construct a connected topological space BG whose fundamental group $\pi_1(BG) \cong G$ and $\pi_n(BG) = 0$ for $n > 1$. The homology of BG (with coefficients in a G -module M) coincides with the algebraic homology of G (with coefficients in M). For $G = GL(R)$, this gives $BGL(R)$.

Definition 4.1.1: $K_n(R)$

Let $BGL(R)$ be the covering groups for $GL(R)$, and let $BGL(R)^+$ denote any CW complex X which has a distinguished map $BGL(R) \rightarrow BGL(R)^+$ such that the following holds:

1. $\pi_1(BGL(R)^+) \cong K_1(R)$ and the natural map from $GL(R) = \pi_1(BGL(R))$ to $\pi_1(BGL(R)^+)$ is surjective with kernel $E(R)$
2. $H_*(BGL(R); M) \cong H_*(BGL(R)^+; M)$ for every $K_1(R)$ -module M

The space X is sometimes called the *model* for $BGL(R)^+$.

For $n \geq 1$,

$$K_n(R) = \pi_n(BGL(R)^+)$$

We shall show that any two models are homotopy equivalent, and hence the space $BGL(R)^+$ is uniquely defined up to homotopy, and so the homotopy group $K_n(R)$ are well-defined up to a canonical isomorphism. We have not actually given a model X ; there are in fact many such models, and we shall construct one of them soon.

Notice we have not included $K_0(R)$ in the definition. To amend this, we define the following:

Definition 4.1.2: $K(R)$

Let R be a ring, $K_0(R)$ the K -group given in definition 2.2.1. Then:

$$K(R) := K_0(R) \times BGL(R)^+$$

Namely, $K(R)$ is the disjoint union of copies of the connected space $BGL(R)^+$ with each element of $K_0(R)$. Then by construction:

$$K_0(R) = \pi_0(K(R)) \quad \pi_n(K(R)) = \pi_n(BGL(R)^+) = K_n(R), \quad n \geq 1$$

recovering all the K -groups.

The maps $K_n(-)$ and $K(-)$ are functorial. The former is a functor from rings to abelian group, while the latter is a functor from rings to homotopy categories of topological spaces (a homotopy category is a category whose objects remain the same and the morphisms are weak homotopy equivalences with diagrams added to induce inverse morphisms; see [6, chapter 27.5]). Note that until an actual model X is concretely defined, we can only prove these results up to homotopy. To see this, let $R \rightarrow R'$ be a ring map. Then this induces a natural group map $GL(R) \rightarrow GL(R')$, which induces a natural map $BGL(R) \rightarrow BGL(R')$, and by construction we get a map $BGL(R)^+ \rightarrow BGL(R')^+$ up to homotopy (see theorem 4.1.7). Hence, we get that $K_n(R) \rightarrow K_n(R')$ is well-defined up to homotopy. A similar argument follows for composition, giving us that it is indeed a functor.

Let us now work towards a concrete model X known as *Volodin spaces*. This space shall be a special acyclic subspace¹ of $BGL(R)$, and so let us get a little more familiar with how these spaces relate to perfect groups

¹Recall a space is acyclic if it has trivial homology groups, i.e. $\tilde{H}_*(X; \mathbb{Z}) = 0$

Lemma 4.1.3: Acyclic and Perfect Groups

Let E be an acyclic space. Then E is connected, its fundamental group $G = \pi_1(E)$ is a perfect group, and $H_2(G; \mathbb{Z}) = 0$.

This proof shall involve the use of *Serre's Spectral Sequence* which is covered in [1] or [3].

Proof:

First, the acyclic space E must be connected, as $H_0(E) = \mathbb{Z}$. Indeed, $G/[G, G] = H_1(E; \mathbb{Z}) = 0$, so we have $G = [G, G]$, i.e., G is a perfect group. To calculate $H_2(G)$, observe that the universal covering space $\tilde{E}$ has $H_1(\tilde{E}; \mathbb{Z}) = 0$. Moreover, the homotopy fiber of the canonical map $E \rightarrow BG$ is homotopy equivalent to $\tilde{E}$. To see this, we shall construct a spectral sequence from fibers maps. If $E \rightarrow B$ is a continuous map, we can induce the maps $\pi_* E \rightarrow \pi_* B$. Then define the *homotopy fiber* $F(f)$ of f with respect to the base point $*_B$ to be the space of pairs (e, γ) , $e \in E$ and $\gamma : [0, 1] \rightarrow B$ with $\gamma(0) = *_B$ and $\gamma(1) = f(e)$. Then a sequence of based spaces $F \rightarrow E \xrightarrow{f} B$ where $F \rightarrow B$ is constant is called a *homotopy fibration sequence*. Then we get a long exact sequence of homotopy groups:

$$\cdots \rightarrow \pi_{n+1}(B) \xrightarrow{\partial} \pi_n(F(f)) \rightarrow \pi_n(E) \rightarrow \pi_n(B) \xrightarrow{\partial} \pi_{n-1}(F(f)) \rightarrow \cdots$$

Then the *Serre spectral sequence* for this homotopy fibration is $E_{pq}^2 = H_p(G; H_q(\tilde{E}; \mathbb{Z})) \Rightarrow H_{p+q}(E; \mathbb{Z})$ and the conclusion that $H_2(G; \mathbb{Z}) = 0$ follows from the associated exact sequence of low degree terms:

$$H_2(E; \mathbb{Z}) \rightarrow H_2(G; \mathbb{Z}) \xrightarrow{d^2} H_1(\tilde{E}; \mathbb{Z})^G \rightarrow H_1(E; \mathbb{Z}) \rightarrow H_1(G; \mathbb{Z})$$

Definition 4.1.4: Volodin Space

Let R be a ring. Then the *Volodin space* $X(R)$ is an acyclic subspace of $BGL(R)$ constructed as follows:

1. For each n , let $T_n(R) \leq GL_n(R)$ consist of upper triangular matrices with 1's on the diagonal. Let $T(R) = \varinjlim_{n \rightarrow \infty} T_n(R) \leq GL(R)$
2. View the permutation group S_n as the subgroups of $GL_n(R)$ by their representation as permutation matrices, and their union as S_∞ .
3. For each $\sigma \in S_n$, let $T_n^\sigma(R)$ denote the subgroup of $GL_n(R)$ obtained by conjugating $T_n(R)$ by σ . Then $BT_n(R)$ and $BT_n(R)^\sigma$ are subspaces of $BGL_n(R)$, and hence of $BGL(R)$
4. The *Volodin space* is defined to be:

$$X(R) = \bigcup_{n, \sigma} BT_n(R)^\sigma$$

We shall show that $BGL(R)/X(R)$ is a model for $K(R)$. The whole proof is beyond the scope of this book, but we shall at least show that it matches with K_1 and K_2 . First, the image of the map

$\pi_1(X(R)) \rightarrow GL(R)$ is $E(R)$. Indeed, $\pi_1(X)$ is generated by the image of $\pi_1(BT_n(R)^\sigma)$, and the image of the composition

$$\pi_1(BT_n^\sigma(R)) \rightarrow \pi_1(X) \rightarrow \pi_1(BGL(R)) = GL(R)$$

is the subgroup $T_n^\sigma(R) \leq E(R)$, and every generator $e_{ij}(r)$ is contained in some $T_n^\sigma(R)$. This gives us that it matches K_1 , and also the first condition for $K(R)$. We shall show π_2 of this group is K_2 . This requires some more information about acyclic maps:

Definition 4.1.5: Acyclic Maps

Let X, Y be based connected CW complexes. Then a map $f : X \rightarrow Y$ is called *acyclic* if the homotopy fiber $F(f)$ of f is acyclic (i.e. Has trivial homology, the homology of a point).

Note that this implies that $F(f)$ is connected and $\pi_1(F(f))$ is a perfect group. Then from the exact sequence $\pi_1(F(f)) \rightarrow \pi_1(X) \rightarrow \pi_1(Y) \rightarrow \pi_0(F(f))$ of homotopy groups, we get that if $X \rightarrow Y$ is acyclic, then $\pi_1(X) \twoheadrightarrow \pi_1(Y)$, and its kernel P is a perfect normal subgroup of $\pi_1(X)$. We shall give a name to such group

Definition 4.1.6: +-Construction

Let P be a perfect normal subgroup of $\pi_1(X)$ where X is a based connected CW complex. Then an acyclic map $f : X \rightarrow Y$ is called a *+-construction* on X (relative to P) if P is the kernel of $\pi_1(X) \rightarrow \pi_1(Y)$.

For any perfect normal subgroup of the fundamental group, there exists such a group.

Theorem 4.1.7: Existence of +-construction

Let $P \leq \pi_1(X)$ be a perfect normal subgroup. Then:

1. There exists a +-construction $f : X \rightarrow Y$ relative to P
2. Let $f : X \rightarrow Y$ be a +-construction relative to P , and let $g : X \rightarrow Z$ be a map st P vanishes in $\pi_1(Z)$. Then there is a map $h : Y \rightarrow Z$ such that $g = hf$ and h is unique up to homotopy
3. If g is another +-construction relative to P , then the map h in (2) is a homotopy equivalence, namely $h : Y \xrightarrow{\sim} Z$

Proof:

See [4]

Let us now show why these are called acyclic maps:

Lemma 4.1.8: Acyclic Maps and Homology

Let X, Y be connected CW complexes. Then $f : X \rightarrow Y$ is acyclic if and only if

$$H_*(X; M) \cong H_*(Y; M)$$

for every $\pi_1(Y)$ -module M

Proof:

Suppose first that f is acyclic, with homotopy fiber $F(f)$. Then $\pi_1 F(f) \rightarrow \pi_1(Y)$ is trivial, and $\pi_1(F(f))$ acts trivially upon M . By the Universal Coefficient Theorem (see [10, chapter 4]), $H_q(F(f); M) = 0$ for $q \neq 0$ and $H_0(F(f); M) = M$.

Therefore $E_{pq}^2 = 0$ for $q \neq 0$ in the Serre spectral sequence for f :

$$E_{pq}^2 = H_p(Y; H_q(F(f); M)) \Rightarrow H_{p+q}(X; M)$$

The spectral sequence collapses, yielding $H_p(X; M) \cong H_p(Y; M)$ for all p .

Conversely, we suppose first that $\pi_1(Y) = 0$ and

$$H_*(X; \mathbb{Z}) \cong H_*(Y; \mathbb{Z})$$

By the Comparison Theorem for the Serre spectral sequences (see [3]) for $F(f) \rightarrow X \xrightarrow{f} Y$ and $* \rightarrow Y \xrightarrow{\cong} Y$, we have $\tilde{H}_*(F(f); \mathbb{Z}) = 0$. Hence $F(f)$ and f are acyclic.

The general case reduces to this by the following trick: Let $\tilde{Y}$ denote the universal covering space of Y , and let $\tilde{X} = X \times_Y \tilde{Y}$ denote the corresponding covering space of X . Then there are natural isomorphisms

$$H_*(\tilde{Y}; \mathbb{Z}) \cong H_*(Y; \mathbb{Z}) \quad \text{and} \quad H_*(\tilde{X}; \mathbb{Z}) \cong H_*(X; \mathbb{Z})$$

where $M = \mathbb{Z}[\pi_1(Y)]$. The assumption that $H_*(X; M) \cong H_*(Y; M)$ implies that the map $\tilde{f} : \tilde{X} \rightarrow \tilde{Y}$ induces isomorphisms on integral homology. But $\pi_1(\tilde{Y}) = 0$, so by the special case above, the homotopy fiber $F(\tilde{f})$ of $\tilde{f}$ is an acyclic space. But by path lifting we have $F(\tilde{f}) \cong F(f)$, so $F(f)$ is acyclic. Thus f is an acyclic map.

Now, recall from section 3.2.1 that every perfect group P has a universal central extension $E \rightarrow P$ whose kernel is the abelian group $H_2(P; \mathbb{Z})$, namely theorem 3.2.7. From the above we now get:

Theorem 4.1.9: 2-homotopy group and K_2

let P be a perfect normal subgroup of G with corresponding $+$ -construction $f : BG \rightarrow BG^+$. Then if $F(f)$ is the homotopy fiber of f , then $\pi_1(F(f))$ is the universal central extension of P , and

$$\pi_2(BG^+) \cong H_2(P; \mathbb{Z})$$

IF $G = GL(R)$, we get:

$$K_2(R) \cong H_2(E(R); \mathbb{Z})$$

Proof:

Consider the exact sequence from homotopy theory:

$$\pi_2(BG) \rightarrow \pi_2(BG^+) \rightarrow \pi_1(F(f)) \rightarrow G \rightarrow G/P \rightarrow 1$$

Then by algebraic topology we know we have $\pi_2(BG) = 0$ and $\pi_2(BG^+)$ is in the center of $\pi_1(F(f))$. Thus, $\pi_1(F(f))$ is a central extension of P with kernel $\pi_2(BG^+)$. As $F(f)$ is acyclic, $\pi_1(F(f))$ is perfect, so $H_2(F; \mathbb{Z}) = 0$, and so by theorem 3.2.7, we have that $\pi_1(F(f))$ is the universal central extension of P .

The last result comes from applying this to $G = \mathrm{GL}(R)$ so that $P = E(R)$, completing the proof.

Corollary 4.1.10: Fundamental Group of Volodin Space

The fundamental group of $\pi_1(X(R))$ is the Steinberg group $\mathrm{st}(R)$

4.2 The Q -Construction

The plus construction of the previous section produced the higher K -groups $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$ of definition 4.1.1 by surgery on the classifying space of $\mathrm{GL}(R)$. That construction is tied to a single ring R and to the general linear group; it says nothing, on its face, about how K -theory behaves for an arbitrary class of modules glued by short exact sequences, and it offers no direct handle on $K_0(R)$, which had to be appended by hand in definition 4.1.2. Quillen's Q -construction repairs both deficiencies at once. It takes as input not a ring but an *exact category* (a category of modules, or sheaves, or bundles, together with a distinguished class of short exact sequences) and outputs a single space $BQ\mathcal{C}$ whose homotopy groups are the K -groups of that category. This section builds the Q -construction from scratch, forms its classifying space, and proves the first comparison theorem: the fundamental group $\pi_1(BQ\mathcal{P}(R))$ recovers the Grothendieck group $K_0(R)$ of definition 2.2.1. The agreement of these Quillen K -groups with the plus-construction groups of definition 4.1.1 is the subject of the next section.

The starting datum is an additive category in which one knows which short exact sequences are allowed. The axioms are due to Quillen and isolate exactly what is needed to run the construction; they are the same axioms governing exact categories in homological algebra, and the reader who has seen them there ([7]) will recognize the list.

Definition 4.2.1: Exact Category

An *exact category* is an additive category $\mathcal{E}$ together with a class $\mathcal{M}$ of *admissible* short exact sequences

$$0 \rightarrow M' \xrightarrow{i} M \xrightarrow{j} M'' \rightarrow 0$$

closed under isomorphism, in which i is a kernel of j and j is a cokernel of i , satisfying:

1. the split sequence $0 \rightarrow M' \rightarrow M' \oplus M'' \rightarrow M'' \rightarrow 0$ is admissible for all objects M', M'' of $\mathcal{E}$;
2. the class of *admissible monomorphisms* (those i occurring as the left map of an admissible sequence, written $i: M' \rightarrowtail M$) is closed under composition, and the pushout of an admissible monomorphism along an arbitrary map exists and is again an admissible monomorphism;
3. the class of *admissible epimorphisms* (those j occurring as the right map, written $j: M \twoheadrightarrow M''$) is closed under composition, and the pullback of an admissible epimorphism along an arbitrary map exists and is again an admissible epimorphism.

The axioms encode the operations a short exact sequence supports without committing to an ambient abelian category. The pushout and pullback clauses are the engine of the whole theory: they guarantee that a square built from one admissible mono and one admissible epi can always be completed, and it is precisely this completion that will define composition in the Q -construction below. An admissible monomorphism plays the role of a subobject inclusion and an admissible epimorphism the role of a quotient projection, but only those inclusions and quotients in the distinguished class $\mathcal{M}$ are visible to K -theory. Choosing a smaller class $\mathcal{M}$ remembers less; choosing the split sequences alone, as in the running example, remembers only the direct-sum decompositions.

Definition 4.2.2: The Exact Category $\mathcal{P}(R)$

Let R be a ring. The exact category $\mathcal{P}(R)$ has as objects the finitely generated projective R -modules and as morphisms all R -module homomorphisms. A short exact sequence of R -modules

$$0 \rightarrow P' \xrightarrow{i} P \xrightarrow{j} P'' \rightarrow 0$$

with P', P, P'' finitely generated projective is declared *admissible*.

Because every object of $\mathcal{P}(R)$ is projective, every short exact sequence of objects splits: the surjection j admits a section, so $P \cong P' \oplus P''$. The admissible monomorphisms are therefore exactly the split injections onto direct summands, and the admissible epimorphisms exactly the split projections off direct summands. The pullback and pushout axioms hold because the pullback of a split surjection is a split surjection and dually for pushouts, and the category is additive with $\oplus$ as biproduct. Thus $\mathcal{P}(R)$ is an exact category in the sense of definition 4.2.1, and it is the category whose K -theory will be the K -theory of the ring R .

Example 4.2.3: The Split Exact Category over a Field

Let F be a field. An object of $\mathcal{P}(F)$ is a finite-dimensional F -vector space, and up to isomorphism it is determined by its dimension, so the isomorphism classes are $\{F^n : n \geq 0\}$. Every short exact

sequence

$$0 \rightarrow F^a \rightarrow F^n \rightarrow F^b \rightarrow 0$$

of finite-dimensional vector spaces splits (a basis of the image lifts), and the rank–nullity relation forces $n = a + b$. Hence the admissible sequences of $\mathcal{P}(F)$ are precisely the sequences realizing $F^n \cong F^a \oplus F^b$ with $n = a + b$, and an admissible monomorphism $F^a \rightarrow F^n$ is the inclusion of an a -dimensional subspace as a direct summand, which (since F is a field) means any injective linear map. The single numerical invariant, dimension, is additive on admissible sequences. This is the smallest nontrivial exact category and the one against which every abstract statement below should first be tested.

With an exact category in hand, the Q -construction produces a new category. Its objects are unchanged, but a morphism is no longer a single map: it is a subobject-then-quotient datum, a way of relating M to P through a common intermediary N that sits inside P and surjects onto M .

Definition 4.2.4: The Q -Construction

Let $\mathcal{E}$ be an exact category. The category $Q\mathcal{E}$ has the same objects as $\mathcal{E}$. A morphism from M to P in $Q\mathcal{E}$ is an isomorphism class of diagrams

$$M \leftarrow N \rightarrow P$$

consisting of an admissible epimorphism $N \rightarrow M$ and an admissible monomorphism $N \rightarrow P$, where two such diagrams $M \leftarrow N \rightarrow P$ and $M \leftarrow N' \rightarrow P$ are identified when there is an isomorphism $N \xrightarrow{\sim} N'$ commuting with both legs. The composite of $M \leftarrow N \rightarrow P$ with $P \leftarrow N' \rightarrow S$ is the diagram $M \leftarrow N'' \rightarrow S$ in which N'' is the pullback

$$\begin{array}{ccc} N'' & \xrightarrow{\quad} & N' \\ \downarrow & & \downarrow \\ N & \xrightarrow{\quad} & P \end{array}$$

of $N \rightarrow P$ along $N' \rightarrow P$, with the leg $N'' \rightarrow M$ obtained by composing $N'' \rightarrow N \rightarrow M$ and the leg $N'' \rightarrow S$ obtained by composing $N'' \rightarrow N' \rightarrow S$. The identity morphism of M is the class of $M \leftarrow M \rightarrow M$ with both legs the identity.

The pullback in the composition law exists and has the stated legs precisely because of the closure axioms of definition 4.2.1: the top map $N'' \rightarrow N'$ is the pullback of the admissible monomorphism $N \rightarrow P$ along $N' \rightarrow P$, hence an admissible monomorphism, and the left map $N'' \rightarrow N$ is the pullback of the admissible epimorphism $N' \rightarrow P$ along $N \rightarrow P$, hence an admissible epimorphism. Composing $N'' \rightarrow N$ with the admissible epimorphism $N \rightarrow M$ gives an admissible epimorphism by axiom (3), and composing $N'' \rightarrow N'$ with the admissible monomorphism $N' \rightarrow S$ gives an admissible monomorphism by axiom (2). Thus the composite is again a legitimate morphism of $Q\mathcal{E}$. That this composition is associative up to the identifications by isomorphism follows from the universal property of pullbacks, which makes the iterated pullback of three composable morphisms canonically independent of the order of formation; this verification is recorded as part of the next proposition.

Proposition 4.2.5: $Q\mathcal{E}$ is a Category

Let $\mathcal{E}$ be an exact category. Then $Q\mathcal{E}$ of definition 4.2.4 is a category: composition is well-defined on isomorphism classes, associative, and unital.

Proof:

Step 1: Well-definedness. Suppose $M \leftarrow N \rightarrow P$ is replaced by an isomorphic representative $M \leftarrow \tilde{N} \rightarrow P$ via $\theta: N \xrightarrow{\sim} \tilde{N}$ commuting with both legs, and likewise $P \leftarrow N' \rightarrow S$ by $\theta': N' \xrightarrow{\sim} \tilde{N}'$. Forming the pullback of $\tilde{N} \rightarrow P$ along $\tilde{N}' \rightarrow P$ yields an object $\tilde{N}''$, and the pair (θ, θ') induces an isomorphism $N'' \xrightarrow{\sim} \tilde{N}''$ by the universal property of the pullback, since θ and θ' are compatible with the maps to P . This isomorphism commutes with the two legs to M and to S because θ, θ' do. Hence the composite class is independent of the chosen representatives.

Step 2: Unitality. Composing $M \leftarrow N \rightarrow P$ with the identity $P \leftarrow P \rightarrow P$ forms the pullback of $N \rightarrow P$ along $P \xrightarrow{\text{id}} P$, which is N itself with the identity legs; the resulting diagram is $M \leftarrow N \rightarrow P$ again. The same holds on the other side, so the class of $M \leftarrow M \rightarrow M$ is a two-sided identity.

Step 3: Associativity. Given three composable morphisms $M \leftarrow N_1 \rightarrow P$, $P \leftarrow N_2 \rightarrow S$, $S \leftarrow N_3 \rightarrow T$, both bracketings produce, as the intermediary, the limit of the diagram $N_1 \rightarrow P \leftarrow N_2 \rightarrow S \leftarrow N_3$. Forming the limit by first taking the left pullback and then the right, or first the right and then the left, yields canonically isomorphic objects, because iterated pullbacks compute the same limit regardless of order (the pasting law for pullback squares). The induced legs to M and to T agree under this canonical isomorphism, so the two composites are equal as isomorphism classes, as we sought to show.

The proposition says $Q\mathcal{E}$ is a genuine category, so it has a nerve. The reader should keep two special kinds of morphism in view. Taking $N = M$ with the left leg the identity gives an *injective* morphism $i_! : M \rightarrow P$, recording an admissible monomorphism $M \rightarrow P$; taking $N = P$ with the right leg the identity gives a *surjective* morphism $j^! : M \rightarrow P$ from an admissible epimorphism $P \rightarrow M$ (note the direction: a quotient map $P \rightarrow M$ becomes a Q -morphism from M to P). Every morphism of $Q\mathcal{E}$ factors as $i_! \circ j^!$ through its intermediary N , and these two families of morphisms are the edges from which the fundamental group will be assembled.

The next step passes to topology. Recall from [10, section 2.2] that the nerve $B\mathcal{C}$ of a (small) category $\mathcal{C}$ is the simplicial set whose n -simplices are the chains $C_0 \rightarrow C_1 \rightarrow \cdots \rightarrow C_n$ of n composable morphisms, with faces given by composing or dropping end maps and degeneracies by inserting identities; its geometric realization is a CW complex whose homotopy groups are the homotopy invariants of $\mathcal{C}$. This applies to $Q\mathcal{E}$.

Definition 4.2.6: Quillen K -Groups

Let $\mathcal{E}$ be a (small) exact category with a chosen zero object 0 . Its *classifying space* is the geometric realization $BQ\mathcal{E}$ of the nerve of the category $Q\mathcal{E}$ from definition 4.2.4, based at the vertex 0 . The *Quillen K -groups* of $\mathcal{E}$ are

$$K_n(\mathcal{E}) := \pi_{n+1}(BQ\mathcal{E}, 0) \quad (n \geq 0)$$

For a ring R , the Quillen K -groups of R are $K_n(R) := K_n(\mathcal{P}(R))$, the K -groups of the exact category $\mathcal{P}(R)$ of definition 4.2.2.

The degree shift is the price of the construction: $BQ\mathcal{E}$ is connected (every object is joined to 0 by a zigzag of injective and surjective morphisms), so π_0 carries no information, and the index $n+1$ realigns the homotopy groups so that K_0 lands where one wants it. The space $BQ\mathcal{E}$ is one connected space carrying all the K -groups simultaneously, in contrast to the plus construction, which built $BGL(R)^+$ for the higher groups and then formally adjoined $K_0(R)$ in definition 4.1.2. Here $K_0(\mathcal{E})$ emerges intrinsically as $\pi_1(BQ\mathcal{E})$, and the next theorem identifies it with the Grothendieck group already defined. The smallness hypothesis is a set-theoretic convenience: $\mathcal{P}(R)$ is equivalent to a small category (restrict to the projective summands of the modules R^n , $n \geq 0$), and one works with such a skeleton so that the nerve is a genuine simplicial set; the homotopy type is unaffected by the choice of skeleton.

Example 4.2.7: The Classifying Space $BQ\mathcal{P}(F)$ for a Field

For a field F , the isomorphism classes of objects of $\mathcal{P}(F)$ are $F^0, F^1, F^2, \dots$, one for each dimension. In $Q\mathcal{P}(F)$ there is an injective edge $i_! : F^a \rightarrow F^n$ for every admissible monomorphism, that is for every $a \leq n$ (realizing F^a as a summand of F^n), and a surjective edge $j^! : F^b \rightarrow F^n$ for every admissible epimorphism $F^n \twoheadrightarrow F^b$, that is for every $b \leq n$. The vertex $F^0 = 0$ is the basepoint. The one-dimensional skeleton of $BQ\mathcal{P}(F)$ already connects every F^n to the basepoint: the chain $0 \rightarrow F^n$ is a single injective edge from 0 to F^n . Loops at 0 are built from these edges, and the relations imposed by the 2-simplices of the nerve will, by theorem 4.2.9 below, collapse π_1 to the free abelian group on the one generator $[F^1]$, recovering $\mathbb{Z}$. The whole space is thus a concrete model whose fundamental group is computed entirely from the additive arithmetic of dimensions.

Before the comparison theorem, two structural facts about the maps that make $Q\mathcal{E}$ tractable must be isolated. Each records one of the two edge families, and each will contribute a family of loops to π_1 .

Lemma 4.2.8: Edges and Their Composition in $Q\mathcal{E}$

Let $\mathcal{E}$ be an exact category. In $Q\mathcal{E}$ the following hold.

1. Every morphism $M \rightarrow P$ factors as $i_! \circ j^!$, where $j^!: M \rightarrow N$ comes from the admissible epimorphism $N \twoheadrightarrow M$ and $i_!: N \rightarrow P$ from the admissible monomorphism $N \hookrightarrow P$, and the intermediary N is determined up to isomorphism.
2. For composable admissible monomorphisms $M \hookrightarrow N \hookrightarrow P$ one has $(i' i)_! = i'_! i_!$, and for composable admissible epimorphisms $P \twoheadrightarrow N \twoheadrightarrow M$ one has $(j j')^! = j^! j'^!$.
3. For an admissible short exact sequence $0 \rightarrow M' \xrightarrow{i} M \xrightarrow{j} M'' \rightarrow 0$ the injective edge $i_!: M' \rightarrow M$ and the surjective edge $j^!: M'' \rightarrow M$ are distinct edges of $Q\mathcal{E}$ with common target M , related by a 2-simplex of the nerve whose third edge runs between M' and M'' ; no edge $u: M' \rightarrow M''$ with $j^! \circ u = i_!$ exists when $M' \neq 0$, since $j \circ i = 0$.

Proof:

Part (1). A morphism $M \rightarrow P$ is by definition an isomorphism class of $M \leftarrow N \rightarrow P$. The datum $N \twoheadrightarrow M$ is a surjective morphism $j^!: M \rightarrow N$ in $Q\mathcal{E}$ (the diagram $M \leftarrow N \xrightarrow{\text{id}} N$), and $N \hookrightarrow P$ is an injective morphism $i_!: N \rightarrow P$ (the diagram $N \xleftarrow{\text{id}} N \rightarrow P$). Composing $j^!$ then $i_!$ forms the pullback of $N \xrightarrow{\text{id}} N$, which returns N with the original legs, so $i_! \circ j^! = (M \leftarrow N \rightarrow P)$. The intermediary N is determined up to isomorphism by the definition of equality of morphisms in $Q\mathcal{E}$.

Part (2). Composing $i_!: M \rightarrow N$ (diagram $M \xleftarrow{\text{id}} M \rightarrow N$) with $i'_!: N \rightarrow P$ (diagram $N \xleftarrow{\text{id}} N \rightarrow P$) forms the pullback of $M \rightarrow N$ along id_N , which is M , with composite leg $M \rightarrow N \rightarrow P$ the admissible monomorphism $i' i$; thus $i'_! i_! = (i' i)_!$. The epimorphism statement is dual, composing $j^!: M \rightarrow N$ and $j'^!: N \rightarrow P$ and forming the pullback along id .

Part (3). That $i_!$ and $j^!$ are edges with target M is immediate from their definitions. Suppose an edge $u: M' \rightarrow M''$ satisfied $j^! \circ u = i_!$. Writing $u = (M' \leftarrow N \rightarrow M'')$, the composite $j^! \circ u$ is, by definition 4.2.4, the pullback of the mono leg $N \hookrightarrow M''$ along the epi leg $j: M \twoheadrightarrow M''$ of $j^!$, with mono leg into M . For this to equal $i_! = (M' \xleftarrow{\text{id}} M' \xrightarrow{i} M)$ the intermediary must be M' with identity epi leg, forcing $N = M'$, and the mono leg $M' \hookrightarrow M''$ must satisfy $j \circ i =$ (that leg) under the pullback. But $j \circ i = 0$, so the leg is the zero map, which is not a monomorphism unless $M' = 0$; hence no such u exists for $M' \neq 0$. The relation among the three edges is therefore not a strict factorization but the homotopy relation imposed by a 2-simplex of the nerve whose three faces are $i_!$, $j^!$, and a third edge between M' and M'' determined by the sequence; this 2-simplex is the one whose presence forces the additivity relation computed in theorem 4.2.9, and we defer its precise simplicial description to [8, p. IV.6], completing the proof.

The content of the lemma is bookkeeping, but it is the bookkeeping that drives the computation of π_1 . Part (1) says the two edge families generate all morphisms; part (2) says each family composes functorially; part (3) records that an admissible sequence produces two distinct edges into M tied by a 2-simplex, the source of the additivity relation. The next theorem turns this into an isomorphism.

Theorem 4.2.9: π_1 of $Q\mathcal{E}$ Recovers K_0

Let $\mathcal{E}$ be a small exact category with zero object 0. Then there is a natural isomorphism

$$\pi_1(BQ\mathcal{E}, 0) \cong K_0(\mathcal{E})$$

where $K_0(\mathcal{E})$ is the Grothendieck group of $\mathcal{E}$: the free abelian group on the isomorphism classes $[M]$ of objects, modulo the relations $[M] = [M'] + [M'']$ for every admissible short exact sequence $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$. In particular, for a ring R ,

$$\pi_1(BQ\mathcal{P}(R), 0) \cong K_0(R)$$

the Grothendieck group of definition 2.2.1.

Proof:

The fundamental group of the realization of a nerve is computed from its 1- and 2-simplices: by the description of π_1 of a simplicial set as the edge-path group (see [8, p. IV.6] and [10, chapter 2]), $\pi_1(BQ\mathcal{E}, 0)$ is the group generated by the directed edges of the nerve, with each edge formally invertible, modulo the relations $\gamma_{02} = \gamma_{12} \gamma_{01}$ imposed by the 2-simplices $C_0 \xrightarrow{\gamma_{01}} C_1 \xrightarrow{\gamma_{12}} C_2$. The proof constructs mutually inverse homomorphisms Φ and Ψ between $K_0(\mathcal{E})$ and this group.

Step 1: The map Φ . For each object M let $\iota_M: 0 \rightarrow M$ be the injective edge from the unique admissible monomorphism $0 \rightarrow M$ (admissible by axiom (1) of definition 4.2.1 applied to $0 \rightarrow M \xrightarrow{\text{id}} M$), and let $\sigma_M: 0 \rightarrow M$ be the surjective edge from the admissible epimorphism $M \rightarrow 0$. As edges of the nerve both run between the vertices 0 and M , so the product

$$\ell_M := \iota_M^{-1} \sigma_M$$

is a based loop at 0. Define $\Phi[M] := [\ell_M]$. To see that this respects the defining relations of $K_0(\mathcal{E})$, let $0 \rightarrow M' \xrightarrow{i} M \xrightarrow{j} M'' \rightarrow 0$ be admissible. The injective edge $i_!: M' \rightarrow M$ and surjective edge $j^!: M'' \rightarrow M$ of lemma 4.2.8(3), together with the chosen edges ι and σ from 0, span 2-simplices of the nerve forcing

$$\iota_M = i_! \iota_{M'} \quad \text{and} \quad \sigma_M = j^! \sigma_{M''}$$

(by lemma 4.2.8(2) applied to $0 \rightarrow M' \rightarrow M$ and to $M \rightarrow M'' \rightarrow 0$), and a further 2-simplex spanned by $i_!$, $j^!$ and the third edge between M' and M'' of lemma 4.2.8(3). Assembling these simplices into the based loop ℓ_M yields the additivity relation

$$\ell_M = \ell_{M'} \ell_{M''}$$

The verification that these four 2-simplices combine to the displayed word is the homotopy-theoretic core of the construction: it is a 2-skeleton computation in the nerve of $Q\mathcal{E}$, carried out in full in [8, p. IV.6.3], and uses simplicial machinery beyond what is developed here, so it is cited rather than reproduced. This multiplicative relation among the loops holds in $\pi_1(BQ\mathcal{E}, 0)$ for every admissible sequence; once π_1 is shown abelian in Step 3 it reads additively as $[\ell_M] = [\ell_{M'}] + [\ell_{M''}]$, the defining relation of $K_0(\mathcal{E})$, so the assignment $\Phi[M] = [\ell_M]$ is then a well-defined homomorphism on $K_0(\mathcal{E})$.

Step 2: The map Ψ . Build a homomorphism in the reverse direction by assigning a K_0 -class to each edge and checking the 2-simplex relations. To an injective edge $i_!: N \rightarrow P$ assign

$0 \in K_0(\mathcal{E})$; to a surjective edge $j^!: M \rightarrow N$ coming from an admissible epimorphism $N \twoheadrightarrow M$ with admissible kernel $0 \rightarrow K \rightarrow N \rightarrow M \rightarrow 0$ assign $[K]$; and to a general edge $M \xrightarrow{(i_! j^!)} P$ assign the sum, namely $[\ker(N \twoheadrightarrow M)]$ for its intermediary N . Extend to formal inverses by negation. This assignment respects every 2-simplex relation $\gamma_{02} = \gamma_{12}\gamma_{01}$: composing two morphisms forms the pullback intermediary of definition 4.2.4, and the kernel of the composite admissible epimorphism is an extension of the two constituent kernels, so its class is their sum by the additivity of $K_0(\mathcal{E})$, while injective edges contribute trivial kernels. Therefore the assignment descends to a homomorphism

$$\Psi: \pi_1(BQ\mathcal{E}, 0) \rightarrow K_0(\mathcal{E})$$

(the target $K_0(\mathcal{E})$ is abelian, so the edge-path relations suffice to define Ψ). On a generator,

$$\Psi(\ell_M) = \Psi(\iota_M^{-1}\sigma_M) = -\Psi(\iota_M) + \Psi(\sigma_M) = -0 + [\ker(M \twoheadrightarrow 0)] = [M]$$

since the kernel of $M \twoheadrightarrow 0$ is M itself. Thus $\Psi \circ \Phi = \text{id}$ on the generators $[M]$ of $K_0(\mathcal{E})$, in particular Φ is injective.

Step 3: Surjectivity, commutativity, and conclusion. Every edge of $Q\mathcal{E}$ factors as $i_! \circ j^!$ by lemma 4.2.8(1), and the injective edges ι_N from 0 form a spanning tree of the 1-skeleton (each object is joined to 0 by the single chosen ι_N); contracting this tree, π_1 is generated by the loops $\ell_M = \iota_M^{-1}\sigma_M$ obtained from the remaining surjective edges σ_M . Hence Φ is surjective. Together with injectivity from Step 2, Φ is an isomorphism with inverse Ψ . Applying the additivity relation of Step 1 to the two split sequences $0 \rightarrow M' \rightarrow M' \oplus M'' \rightarrow M'' \rightarrow 0$ and $0 \rightarrow M'' \rightarrow M' \oplus M'' \rightarrow M' \rightarrow 0$ shows $[\ell_{M'}]$ and $[\ell_{M''}]$ commute, so $\pi_1(BQ\mathcal{E}, 0)$ is abelian, consistent with its being a copy of $K_0(\mathcal{E})$.

Naturality. An exact functor $F: \mathcal{E} \rightarrow \mathcal{E}'$ induces $QF: Q\mathcal{E} \rightarrow Q\mathcal{E}'$ (it preserves admissible monomorphisms, admissible epimorphisms, and the pullbacks of definition 4.2.4), hence a based map BQF carrying ℓ_M to ℓ_{FM} ; thus Φ commutes with the maps induced by F on K_0 and on π_1 , so the isomorphism is natural in $\mathcal{E}$.

The ring case. For $\mathcal{E} = \mathcal{P}(R)$ the Grothendieck group $K_0(\mathcal{P}(R))$ is the free abelian group on isomorphism classes of finitely generated projective R -modules modulo the admissible sequences. Every such sequence splits (each object is projective), so its relations are exactly $[P] = [P'] + [P'']$ whenever $P \cong P' \oplus P''$, which is the defining relation of $K_0(R)$ in definition 2.2.1. Hence $\pi_1(BQ\mathcal{P}(R), 0) \cong K_0(\mathcal{P}(R)) = K_0(R)$, completing the proof.

The theorem is the first payoff of the Q -construction: the Grothendieck group, defined in definition 2.2.1 as a purely algebraic completion of a monoid, reappears as the fundamental group of a space, and the defining relations $[M] = [M'] + [M'']$ are realized geometrically as the loops contributed by admissible short exact sequences. This is the pattern the whole theory follows. Each $K_n(\mathcal{E})$ measures the $(n+1)$ -dimensional loops of $BQ\mathcal{E}$, and the relations one would impose by hand at the level of K_0 are encoded once and for all in the simplicial structure of the nerve. The plus construction reached the same $K_0(R)$ only by the artificial device of definition 4.1.2; here it is intrinsic.

Functoriality, established inside the proof, deserves a separate statement because it is what lets K -theory transport along ring maps and along exact functors between categories of modules.

Corollary 4.2.10: Functoriality of the Quillen K -Groups

Let $F: \mathcal{E} \rightarrow \mathcal{E}'$ be an exact functor between small exact categories. Then F induces a based map $BQF: BQ\mathcal{E} \rightarrow BQ\mathcal{E}'$ and hence homomorphisms

$$K_n(F): K_n(\mathcal{E}) \rightarrow K_n(\mathcal{E}') \quad (n \geq 0)$$

functorial in F . In particular, a ring homomorphism $\varphi: R \rightarrow S$ induces the exact functor $-\otimes_R S: \mathcal{P}(R) \rightarrow \mathcal{P}(S)$, and hence homomorphisms $K_n(R) \rightarrow K_n(S)$ for all $n \geq 0$, agreeing on K_0 with the map of definition 2.2.1 under the isomorphism of theorem 4.2.9.

Proof:

An exact functor preserves admissible monomorphisms and admissible epimorphisms and commutes with the pullbacks defining composition in $Q\mathcal{E}$, so $M \leftarrow N \rightarrow P \mapsto FM \leftarrow FN \rightarrow FP$ is a functor $QF: Q\mathcal{E} \rightarrow Q\mathcal{E}'$. The nerve is functorial in categories ([10, section 2.2]), so QF realizes to a based map $BQF: BQ\mathcal{E} \rightarrow BQ\mathcal{E}'$ fixing the zero object, inducing $K_n(F) = \pi_{n+1}(BQF)$. Functoriality in F follows from functoriality of the nerve and of π_{n+1} . For a ring map $\varphi: R \rightarrow S$, the functor $-\otimes_R S$ sends a finitely generated projective R -module to a finitely generated projective S -module and preserves split exact sequences (it is additive and right exact, and split sequences are preserved by any additive functor), hence is exact on $\mathcal{P}(R)$. On K_0 the induced map sends $[P]$ to $[P \otimes_R S]$, which is the map of definition 2.2.1; under theorem 4.2.9 this is K_0 of the induced map, as we sought to show.

This is the structural counterpart, for the Q -construction, of the functoriality of $BGL(R)^+$ recorded after definition 4.1.2. There it required passing through homotopy because no model had been fixed; here the functor QF is honest, defined before any homotopy theory, and the induced maps on K_n are immediate. The price paid for this directness is the degree shift and the more elaborate morphisms of $Q\mathcal{E}$, but the reward is a theory that accepts any exact category as input, opening the door to the resolution, dévissage, and localization theorems of the next section, none of which has a natural formulation in the plus-construction picture.

We close the section by carrying out the computation promised in example 4.2.7 and recording the case $R = \mathbb{Z}$, the two computations the theorem most directly delivers.

Example 4.2.11: $\pi_1(BQ\mathcal{P}(F)) = \mathbb{Z}$ and $\pi_1(BQ\mathcal{P}(\mathbb{Z})) = \mathbb{Z}$

1. Let F be a field. Every finitely generated projective F -module is a finite-dimensional vector space F^n , and $[F^n] = n[F]$ in $K_0(F)$ because the standard flag $0 \subset F \subset F^2 \subset \dots \subset F^n$ gives admissible sequences forcing $[F^n] = [F^{n-1}] + [F]$ inductively. The dimension homomorphism $K_0(F) \rightarrow \mathbb{Z}$, $[F^n] \mapsto n$, is thus an isomorphism, with inverse $1 \mapsto [F]$. By theorem 4.2.9,

$$\pi_1(BQ\mathcal{P}(F), 0) \cong K_0(F) \cong \mathbb{Z}$$

generated by the loop ℓ_F attached to the line F .

2. Let $R = \mathbb{Z}$. Since $\mathbb{Z}$ is a principal ideal domain, every finitely generated projective $\mathbb{Z}$ -module is free, of the form $\mathbb{Z}^n$, and as in part (1) the rank gives $K_0(\mathbb{Z}) \cong \mathbb{Z}$ with $[\mathbb{Z}^n] \mapsto n$. By theorem 4.2.9,

$$\pi_1(BQ\mathcal{P}(\mathbb{Z}), 0) \cong K_0(\mathbb{Z}) \cong \mathbb{Z}$$

generated by $\ell_{\mathbb{Z}}$. The two computations are formally identical because both $\mathcal{P}(F)$ and $\mathcal{P}(\mathbb{Z})$ have the property that every object is free of a well-defined rank and every admissible sequence splits; the invariant computing π_1 is the rank in each case.

Exercise 4.2.1

1. Verify directly that $\mathcal{P}(R)$ of definition 4.2.2 satisfies the pullback axiom (3) of definition 4.2.1: given an admissible epimorphism $j: P \twoheadrightarrow P''$ of finitely generated projectives and an arbitrary map $f: Q \rightarrow P''$ with Q finitely generated projective, show the pullback $Q \times_{P''} P$ exists in $\mathcal{P}(R)$ and that its projection to Q is an admissible epimorphism.
2. In $Q\mathcal{P}(R)$, describe explicitly the composite of the surjective edge $j^!: R \rightarrow R^2$ coming from the projection $R^2 \twoheadrightarrow R$ onto the first coordinate with the injective edge $i_!: R^2 \hookrightarrow R^3$ coming from the inclusion of the first two coordinates. Identify the intermediary N and both legs of the resulting diagram $R \leftarrow N \rightarrow R^3$.
3. Let F be a field. Using theorem 4.2.9, compute $\pi_1(BQ\mathcal{P}(F \times F), 0)$. (Hint: a finitely generated projective module over $F \times F$ is a pair of vector spaces.)
4. Show that the loop ℓ_0 attached to the zero object is trivial in $\pi_1(BQ\mathcal{E}, 0)$, consistently with $[0] = 0$ in $K_0(\mathcal{E})$.
5. Let $\mathcal{E}$ be an exact category in which the only admissible sequences are the split ones. Prove that $K_0(\mathcal{E})$ is the Grothendieck group of the monoid $(\text{Obj}\mathcal{E}/\cong, \oplus)$, and conclude that $K_0(\mathcal{P}(R)) = K_0(R)$ matches definition 2.2.1 without invoking the general theorem 4.2.9 machinery.
6. An exact functor $F: \mathcal{E} \rightarrow \mathcal{E}'$ that is an equivalence of categories preserving and reflecting admissible sequences induces an isomorphism $K_n(\mathcal{E}) \cong K_n(\mathcal{E}')$ for all n . Use this to show that Morita equivalent rings R and $M_n(R)$ have isomorphic Quillen K -groups (section 2.4).

4.3 The Plus-Equals-Q Theorem

Two definitions of higher algebraic K -theory now stand side by side. The first, from definition 4.1.1, sets $K_n(R) = \pi_n(\text{BGL}(R)^+)$ using the plus construction on the classifying space of $\text{GL}(R)$. The second, from section 4.2, sets $K_n(R) = \pi_{n+1}(BQ\mathcal{P}(R))$ using the nerve of the Q -construction on the exact category $\mathcal{P}(R)$ of finitely generated projective modules. The plus construction is built from the linear group and sees only the matrices; the Q -construction is built from the category of modules and exact sequences, and sees the whole monoidal structure of direct sum together with the data of admissible monomorphisms and epimorphisms. That these two routes, starting from such different data, land on the same groups is the central structural theorem of the subject. This section states that agreement, lays out the architecture of its proof in full, and deduces from it that the higher theory genuinely extends the classical groups K_0 , K_1 , and K_2 computed in the previous chapters.

The agreement is what licenses the name “ $K_n(R)$ ” for both objects. Without it, the plus-construction groups and the Q -construction groups would be two unrelated invariants that happen to coincide in degrees 0, 1, and 2. The theorem promotes that low-degree coincidence into an isomorphism in every degree, and (this is the operative point) does so naturally in the ring R . From a computational standpoint the two definitions have complementary strengths: the plus construction is concrete enough to read off K_1 and K_2 by hand, while the Q -construction is the one that carries the long

exact sequences (localization, dévissage, resolution) that make higher K -groups computable. The theorem lets each definition lend its strengths to the other.

The full proof occupies a substantial part of Quillen's foundational paper and of any modern treatment; the present section gives the architecture in complete structural detail and defers the longest homotopy-theoretic verifications to [8], where they are carried out in full. This is a deferral to a complete published proof, not an omission: every step is named, the role of each tool is stated, and the only content sent elsewhere is the chain of simplicial homotopy equivalences whose verification is mechanical but lengthy.

Two tools from [10] drive the argument, and stating them precisely is the bulk of the structural work. The first is a criterion for when a functor between (nerves of) categories is a homotopy equivalence, and the second is a criterion identifying the homotopy fibre of such a functor. For a small category $\mathcal{C}$ write BC for the geometric realization of its nerve, the classifying space whose n -simplices are chains of n composable morphisms.

Theorem 4.3.1: Quillen's Theorems A and B

Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor between small categories. For an object d of $\mathcal{D}$, let F/d denote the comma category whose objects are pairs (c, u) with c an object of $\mathcal{C}$ and $u: F(c) \rightarrow d$ a morphism of $\mathcal{D}$, and whose morphisms are the evident triangles.

1. (*Theorem A.*) If $B(F/d)$ is contractible for every object d of $\mathcal{D}$, then $BF: BC \rightarrow BD$ is a homotopy equivalence.
2. (*Theorem B.*) If for every morphism $d \rightarrow d'$ in $\mathcal{D}$ the induced functor $F/d \rightarrow F/d'$ is a homotopy equivalence on classifying spaces, then for each object d the square

$$\begin{array}{ccc} B(F/d) & \longrightarrow & BC \\ \downarrow & & \downarrow_{BF} \\ B(\mathcal{D}/d) & \longrightarrow & BD \end{array}$$

is homotopy cartesian. Since $B(\mathcal{D}/d)$ is contractible, $B(F/d)$ is the homotopy fibre of BF over the point d .

Proof:

Both statements are theorems of homotopy theory whose proofs proceed by analysing the bisimplicial set associated to F and applying the realization lemma for bisimplicial sets; the complete arguments are given in [8, Chapter IV] and in [10]. The deferral is to those complete published proofs.

Theorem A says that to check a functor induces a homotopy equivalence it suffices to check that each comma category F/d is contractible, a condition that is often combinatorially accessible: contractibility of $B(F/d)$ frequently follows from the existence of an initial or terminal object, or from a natural transformation to a constant functor. Theorem B is the engine that produces fibrations: it identifies the homotopy fibre of BF with the classifying space of a comma category, which converts a long exact sequence of homotopy groups into a statement about categories one can manipulate algebraically. The hypothesis of Theorem B (invariance of F/d along morphisms of $\mathcal{D}$) is exactly

what is needed to make the fibre independent of the chosen point, so that the comma category deserves to be called “the” fibre.

The second ingredient is Quillen’s group-completion construction $S^{-1}S$, which performs at the level of categories the same Grothendieck completion that turned the monoid $\mathcal{P}(R)$ into $K_0(R)$ in chapter 2, but now homotopically rather than on π_0 alone. Let S be a symmetric monoidal category in which every morphism is an isomorphism, with monoidal product written $\oplus$; the case of interest is $S = \text{iso } \mathcal{P}(R)$, the category of finitely generated projective R -modules and their isomorphisms, under direct sum.

Definition 4.3.2: The $S^{-1}S$ Construction

Let $(S, \oplus)$ be a symmetric monoidal category whose morphisms are all isomorphisms. The category $S^{-1}S$ has objects the pairs (a, b) of objects of S . A morphism $(a, b) \rightarrow (a', b')$ is an equivalence class of data

$$(s, \alpha: s \oplus a \xrightarrow{\sim} a', \beta: s \oplus b \xrightarrow{\sim} b')$$

where s is an object of S , two such data being identified when they differ by an isomorphism of the chosen s . Composition is by $\oplus$ of the connecting objects. The monoidal product $\oplus$ makes $S^{-1}S$ a symmetric monoidal category, and translation by (a, a) is invertible up to natural isomorphism, so $\pi_0(B(S^{-1}S))$ is the group completion of the monoid $\pi_0(BS)$.

The construction formalizes “formal differences $a \ominus b$ ” as honest objects, with a morphism recording a common stabilization s that makes two such differences equal. On π_0 it is precisely the passage from the monoid of isomorphism classes to its Grothendieck group, so $\pi_0(B(S^{-1}S)) = K_0(R)$ when $S = \text{iso } \mathcal{P}(R)$. The content of group completion is that it does the right thing in higher degrees too: under a homology-stability hypothesis the higher homotopy of $B(S^{-1}S)$ computes the higher K -groups. The translation-invertibility clause is what distinguishes this from the bare monoid; it is the categorical shadow of inverting $[R]$ in the Grothendieck group, and it is what forces $B(S^{-1}S)$ to be an infinite loop space rather than an unstructured space.

Example 4.3.3: Group Completion of the Natural Numbers

Take R a ring satisfying the invariant basis property (definition 1.1.2) and let $S = \mathcal{F}(R)$ be the symmetric monoidal category of finitely generated free modules R^n ($n \geq 0$) with their isomorphisms, under $\oplus$. The set of isomorphism classes is $\mathbb{N}$, with $\oplus$ giving addition, so $\pi_0(BS) = \mathbb{N}$ and $\pi_0(B(S^{-1}S)) = \mathbb{Z}$, the Grothendieck group of $\mathbb{N}$. The automorphism group of the object R^n is $\text{GL}_n(R)$, so BS is the disjoint union $\coprod_{n \geq 0} B\text{GL}_n(R)$. Forming $S^{-1}S$ stabilizes the matrix size: the object (R^a, R^b) has automorphisms recording isomorphisms after adding a common R^s , and as $s \rightarrow \infty$ the automorphism monoid stabilizes to $\text{GL}(R) = \varinjlim_n \text{GL}_n(R)$. Concretely, the component of $B(S^{-1}S)$ containing $(0, 0)$ has fundamental group $\text{GL}(R)/E(R) = K_1(R)$, the abelianization of the stabilized linear group. The group-completion theorem (Step 2 of the proof of theorem 4.3.4) then identifies the whole space:

$$B(S^{-1}S) \simeq \mathbb{Z} \times B\text{GL}(R)^+$$

already in this free-module case, with the factor $\mathbb{Z} = K_0$ of the free theory and $B\text{GL}(R)^+$ the plus construction of definition 4.1.1. Replacing free modules by all finitely generated projectives upgrades $\mathbb{Z}$ to $K_0(R)$.

The example exhibits, in the simplest case, the two phenomena the general theorem packages: the $S^{-1}S$ space splits as π_0 times a connected piece, and that connected piece is the plus construction. The general statement replaces $\mathcal{F}(R)$ by $\mathcal{P}(R)$ and identifies the connected piece's homotopy with the Q -construction groups. The bridge between $S^{-1}S$ and the Q -construction is a homotopy equivalence $B(S^{-1}S) \simeq \Omega BQ\mathcal{P}(R)$, Quillen's $S^{-1}S$ theorem, proven by applying Theorem B of theorem 4.3.1 to an auxiliary functor with target $Q\mathcal{P}(R)$. With the architecture in place, the main theorem can be stated.

Theorem 4.3.4: The Plus-Equals- Q Theorem

Let R be a ring and let $\mathcal{P}(R)$ be the exact category of finitely generated projective R -modules of definition 4.2.2. There is a homotopy equivalence

$$\Omega BQ\mathcal{P}(R) \simeq K_0(R) \times \mathrm{BGL}(R)^+$$

natural in R , where the left side is the loop space of the nerve of the Q -construction and the right side is the space $K(R)$ of definition 4.1.2. Consequently, for every $n \geq 1$ there is an isomorphism, natural in R ,

$$\pi_n(\mathrm{BGL}(R)^+) \xrightarrow{\sim} \pi_{n+1}(BQ\mathcal{P}(R))$$

so that the plus-construction group $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$ of definition 4.1.1 coincides with the Q -construction group $K_n(R) = \pi_{n+1}(BQ\mathcal{P}(R))$ of definition 4.2.6. The two definitions of $K_n(R)$ agree for all $n \geq 0$.

The equivalence is the precise sense in which “plus equals Q ”. Looping the Q -construction space once recovers, up to homotopy, the space $K(R) = K_0(R) \times \mathrm{BGL}(R)^+$ that the plus construction produced. Taking π_n of both sides and using $\pi_n(\Omega X) = \pi_{n+1}(X)$ converts this into the degree-shifted isomorphism of homotopy groups: the loop shift Ω is exactly what accounts for the index gap between π_n on the plus side and π_{n+1} on the Q side. On π_0 both sides give $K_0(R)$, and on π_n for $n \geq 1$ the connected component $\mathrm{BGL}(R)^+$ supplies the higher groups; the splitting off of $K_0(R)$ as a discrete factor matches the way definition 4.1.2 appended K_0 to the plus construction by hand. Naturality in R is what makes the agreement usable: a ring map $R \rightarrow R'$ induces the same map on K_n whichever definition is used, so theorems proven in one model transfer to the other.

Proof Architecture, with the homotopy-theoretic core deferred to [8]:

The proof assembles the three tools above into a chain of equivalences. We record the structure of each link and indicate precisely which verifications are carried out in full in [8, Chapter IV].

Step 1: Group completion of the projective monoid. Let $S = \mathrm{iso} \mathcal{P}(R)$ be the symmetric monoidal category of finitely generated projective R -modules and their isomorphisms, under $\oplus$. Its classifying space is

$$BS \simeq \coprod_{[P]} B\mathrm{Aut}_R(P)$$

the disjoint union over isomorphism classes, and $\pi_0(BS)$ is the monoid of definition 2.2.1 whose group completion is $K_0(R)$. Forming $S^{-1}S$ of definition 4.3.2 produces a symmetric monoidal category with $\pi_0(B(S^{-1}S)) = K_0(R)$. The group-completion theorem identifies the homology of $B(S^{-1}S)$ with the localization of the homology of BS at the multiplicative action of π_0 ; this is the homological input and is proven in full in [8, Chapter IV].

Step 2: The plus construction appears as a component. Restricting to the cofinal subcategory of free modules and stabilizing, the automorphism groups assemble to $\mathrm{GL}(R) = \varinjlim_n \mathrm{GL}_n(R)$. The group-completion theorem of Step 1 yields a homology equivalence

$$\mathbb{Z} \times \mathrm{BGL}(R)^+ \xrightarrow{\sim} (B(\mathrm{iso} \mathcal{F}(R)))^{-1} (B(\mathrm{iso} \mathcal{F}(R)))$$

the plus construction entering precisely because it is, by definition 4.1.1 and theorem 4.1.7, the space with the homology of $\mathrm{BGL}(R)$ and abelianized fundamental group, which is what the localized homology computes. Passing from free to projective modules replaces the discrete factor $\mathbb{Z}$ by $K_0(R)$, since projectives are cofinal over frees and the extra components are indexed by $K_0(R)$. This gives

$$B(S^{-1}S) \simeq K_0(R) \times \mathrm{BGL}(R)^+$$

The verification that the relevant maps are homology isomorphisms and that the target is simple enough for a homology isomorphism to be a homotopy equivalence is carried out in [8, Chapter IV].

Step 3: $S^{-1}S$ as a loop space of the Q-construction. Quillen's $S^{-1}S$ theorem provides a homotopy equivalence

$$\Omega BQ\mathcal{P}(R) \simeq B(S^{-1}S)$$

The mechanism is Theorem B of theorem 4.3.1: one constructs an auxiliary functor whose target is $Q\mathcal{P}(R)$ and whose comma categories are, after passing to classifying spaces, models for $B(S^{-1}S)$, so that Theorem B identifies the homotopy fibre of the map to $BQ\mathcal{P}(R)$ with $B(S^{-1}S)$. Since $BQ\mathcal{P}(R)$ is connected, the homotopy fibre over the zero object is the based loop space $\Omega BQ\mathcal{P}(R)$, giving the displayed equivalence. The hypothesis of Theorem B (homotopy-invariance of the relevant comma categories along the morphisms of $Q\mathcal{P}(R)$) holds because every morphism in the Q -construction is built from admissible monics and epis, along which the comma categories are homotopy equivalences by an explicit functorial deformation; the construction of the auxiliary functor and this deformation are carried out in [8, Chapter IV].

Step 4: Assembly. Combining Steps 2 and 3,

$$\Omega BQ\mathcal{P}(R) \simeq B(S^{-1}S) \simeq K_0(R) \times \mathrm{BGL}(R)^+ = K(R)$$

which is the asserted equivalence. Applying π_n and using $\pi_n(\Omega X) \cong \pi_{n+1}(X)$ for the first space and $\pi_n(K_0(R) \times \mathrm{BGL}(R)^+) \cong \pi_n(\mathrm{BGL}(R)^+)$ for $n \geq 1$ (the discrete factor contributing only to π_0) gives

$$\pi_{n+1}(BQ\mathcal{P}(R)) \cong \pi_n(\Omega BQ\mathcal{P}(R)) \cong \pi_n(\mathrm{BGL}(R)^+)$$

for $n \geq 1$, which is the degree-shifted isomorphism. On π_0 both sides return $K_0(R)$, completing the agreement in all degrees. Each link is a homotopy equivalence natural in R because every functor used (group completion, the Q -construction, the comma-category identifications) is functorial in the ring, so the composite isomorphism on homotopy groups is natural, completing the proof.

The architecture should be read as a relay: Step 1 group-completes the monoid of projectives homologically, Step 2 recognizes the connected part of that group completion as the plus construction, Step 3 recognizes the same group completion as the loop space of the Q -construction via Theorem B, and Step 4 chains the two recognitions. The only content deferred to [8] is the homotopy-theoretic verification that the named maps are equivalences (the group-completion theorem and the explicit

deformation of comma categories); the logical skeleton is complete here. The reason the deferral is appropriate rather than a gap is that these verifications are simplicial-homotopy computations with no further conceptual content, and they are reproduced in full in the cited source.

A first consequence is that the higher theory is genuinely an extension of the classical groups. The plus construction was designed in definition 4.1.1 to reproduce K_1 on π_1 , and theorem 4.1.9 identified $\pi_2(\mathrm{BGL}(R)^+)$ with K_2 . The plus-equals- Q theorem now says the Q -construction reproduces these same groups, so both definitions extend the classical K_0, K_1, K_2 .

Corollary 4.3.5: Low-Degree Agreement of the Two Definitions

Let R be a ring. Under the isomorphism of theorem 4.3.4, the higher K -groups recover the classical groups in low degrees:

1. $K_0(R) = \pi_1(BQP(R)) = \pi_0(K(R))$ is the Grothendieck group of definition 2.2.1.
2. $K_1(R) = \pi_1(\mathrm{BGL}(R)^+) = \pi_2(BQP(R))$ is the Whitehead group $\mathrm{GL}(R)/E(R)$ of definition 3.1.1.
3. $K_2(R) = \pi_2(\mathrm{BGL}(R)^+) = \pi_3(BQP(R))$ is the group of definition 3.2.2, the kernel of $\mathrm{st}(R) \rightarrow E(R)$.

Proof:

We treat the three degrees in turn, in each case computing the plus-construction side from the previous chapters and transporting across theorem 4.3.4.

1. *Degree 0.* By theorem 4.3.4 the space $\Omega BQP(R)$ is homotopy equivalent to $K(R) = K_0(R) \times \mathrm{BGL}(R)^+$, so its set of path components is $\pi_0(K(R)) = K_0(R)$, the Grothendieck group of definition 2.2.1. Equivalently $\pi_1(BQP(R)) = \pi_0(\Omega BQP(R)) = K_0(R)$, since $BQP(R)$ is connected and looping drops the degree by one. This is the statement that the Q -construction recovers K_0 , recorded as theorem 4.2.9.

2. *Degree 1.* The plus construction was built in definition 4.1.1 so that $\pi_1(\mathrm{BGL}(R)^+)$ is the abelianization of $\pi_1(\mathrm{BGL}(R)) = \mathrm{GL}(R)$ relative to the perfect normal subgroup $E(R)$; by theorem 4.1.7 the map $\mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R)) \rightarrow \pi_1(\mathrm{BGL}(R)^+)$ is surjective with kernel $E(R)$, hence

$$\pi_1(\mathrm{BGL}(R)^+) \cong \mathrm{GL}(R)/E(R)$$

By theorem 3.1.4 the right side is exactly $K_1(R) = \mathrm{GL}(R)/[\mathrm{GL}(R), \mathrm{GL}(R)]$ of definition 3.1.1, the commutator subgroup and $E(R)$ coinciding by Whitehead's lemma. Transporting across theorem 4.3.4 with $n = 1$ gives $\pi_2(BQP(R)) \cong \pi_1(\mathrm{BGL}(R)^+) \cong \mathrm{GL}(R)/E(R)$, as claimed.

3. *Degree 2.* Apply theorem 4.1.9 with $G = \mathrm{GL}(R)$ and the perfect normal subgroup $P = E(R)$: the homotopy fibre of the plus construction $\mathrm{BGL}(R) \rightarrow \mathrm{BGL}(R)^+$ has fundamental group the universal central extension of $E(R)$, and

$$\pi_2(\mathrm{BGL}(R)^+) \cong H_2(E(R); \mathbb{Z})$$

Now theorem 3.2.3 identifies $K_2(R)$ of definition 3.2.2 (the kernel of $\mathrm{st}(R) \rightarrow E(R)$) with the center of $\mathrm{st}(R)$, and by theorem 3.2.7 the Steinberg group $\mathrm{st}(R)$ is the universal central

extension of the perfect group $E(R)$, whose kernel is $H_2(E(R); \mathbb{Z})$. Hence

$$\pi_2(\mathrm{BGL}(R)^+) \cong H_2(E(R); \mathbb{Z}) \cong K_2(R)$$

Transporting across theorem 4.3.4 with $n = 2$ gives $\pi_3(BQP(R)) \cong \pi_2(\mathrm{BGL}(R)^+) \cong K_2(R)$, as required.

In each degree the classical group of the earlier chapters is recovered on both sides of the plus-equals- Q isomorphism, so the higher theory extends K_0 , K_1 , and K_2 , completing the proof.

The corollary closes the loop opened at the start of chapter 4: the higher K -groups were introduced as homotopy groups of a space precisely so that K_0 , K_1 , K_2 would reappear as π_0 , π_1 , π_2 , and now both constructions of that space deliver on the promise. The degree-1 case is worth isolating, because it can be checked directly from both definitions without the full strength of theorem 4.3.4, and doing so makes the agreement concrete.

Example 4.3.6: Both Definitions Give $K_1(R) = \mathrm{GL}(R)/E(R)$

We verify directly that the plus construction and the Q -construction assign the same group to $K_1(R)$, namely the Whitehead group $\mathrm{GL}(R)/E(R)$.

The plus-construction side. By definition 4.1.1, $K_1(R) = \pi_1(\mathrm{BGL}(R)^+)$. The construction $\mathrm{BGL}(R) \rightarrow \mathrm{BGL}(R)^+$ is performed relative to the perfect normal subgroup $E(R) \leq \mathrm{GL}(R) = \pi_1(\mathrm{BGL}(R))$, perfect by lemma 3.1.3. The defining property in definition 4.1.1(1) and theorem 4.1.7 make $\pi_1(\mathrm{BGL}(R)) \rightarrow \pi_1(\mathrm{BGL}(R)^+)$ surjective with kernel exactly $E(R)$, so

$$\pi_1(\mathrm{BGL}(R)^+) \cong \mathrm{GL}(R)/E(R)$$

By theorem 3.1.4 this equals $\mathrm{GL}(R)/[\mathrm{GL}(R), \mathrm{GL}(R)]$, the K_1 of definition 3.1.1.

The Q -construction side. By definition 4.2.6, $K_1(R) = \pi_2(BQP(R))$. By theorem 4.3.4, $\Omega BQP(R) \simeq K_0(R) \times \mathrm{BGL}(R)^+$, so looping and reading π_1 of the connected factor,

$$\pi_2(BQP(R)) \cong \pi_1(\Omega BQP(R)) \cong \pi_1(\mathrm{BGL}(R)^+) \cong \mathrm{GL}(R)/E(R)$$

the discrete factor $K_0(R)$ contributing nothing to π_1 .

Agreement. Both routes terminate at $\mathrm{GL}(R)/E(R)$. For a concrete instance take $R = \mathbb{Z}$: then $E(\mathbb{Z}) = \mathrm{SL}(\mathbb{Z})$ and $\mathrm{GL}(\mathbb{Z})/\mathrm{SL}(\mathbb{Z}) \cong \{\pm 1\} = \mathbb{Z}^\times$ via the determinant, so both definitions give $K_1(\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, matching example 3.1.5.

The example isolates the mechanism: on the plus side K_1 is the abelianization of $\mathrm{GL}(R)$ killing $E(R)$, and on the Q side the loop-space equivalence of theorem 4.3.4 transports the same computation up one degree. The two are forced to agree because the equivalence is a homotopy equivalence of spaces, not just an isomorphism of one chosen homotopy group. The same transport, applied in degree n , yields the agreement of all higher groups, which is the content the example cannot reach by hand but theorem 4.3.4 supplies.

Exercise 4.3.1

1. Let R be a ring. Using theorem 4.3.4, prove that $\pi_n(\Omega BQP(R)) \cong K_n(R)$ for all $n \geq 1$ and that $\pi_0(\Omega BQP(R)) \cong K_0(R)$. Explain in one sentence why the loop Ω produces the index

shift between the two definitions of K_n .

2. The plus-equals- Q theorem asserts a homotopy equivalence $\Omega BQ\mathcal{P}(R) \simeq K_0(R) \times \mathrm{BGL}(R)^+$, not just an isomorphism of homotopy groups. Explain why the stronger statement (an equivalence of spaces) is needed to conclude that the two definitions of $K_n(R)$ agree *naturally* in R for all n simultaneously, whereas a degree-by-degree isomorphism would not suffice.
3. In the $S^{-1}S$ construction of definition 4.3.2, show that $\pi_0(B(S^{-1}S))$ is the group completion of the monoid $\pi_0(BS)$. For $S = \mathrm{iso} \mathcal{F}(\mathbb{Z})$ the category of finitely generated free abelian groups under $\oplus$, identify both $\pi_0(BS)$ and $\pi_0(B(S^{-1}S))$ explicitly.
4. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor of small categories such that $\mathcal{D}$ has a terminal object t and F/t has an initial object. Use Theorem A of theorem 4.3.1 to decide whether BF must be a homotopy equivalence, justifying each hypothesis you invoke. (Hint: a comma category with an initial object has contractible classifying space.)
5. Compute $K_1(R)$ from both definitions when $R = k$ is a field with $k \notin \{\mathbb{F}_2, \mathbb{F}_3\}$, and confirm the two answers agree. State which result from the K_1 chapter you use to evaluate $\mathrm{GL}(k)/E(k)$.
6. Quillen's Theorem B (theorem 4.3.1) requires that for each morphism $d \rightarrow d'$ of $\mathcal{D}$ the functor $F/d \rightarrow F/d'$ be a homotopy equivalence. Explain, in the language of Step 3 of the proof of theorem 4.3.4, where this hypothesis is used and why it is what makes the homotopy fibre of $* \rightarrow BQ\mathcal{P}(R)$ deserve the name " $\Omega BQ\mathcal{P}(R)$ ".

4.4 Fundamental Theorems and Computations

With the Q -construction in hand and the plus-equals- Q theorem identifying $\pi_{n+1}(BQ\mathcal{C}\mathcal{P}(R))$ with $K_n(R) = \pi_n(\mathrm{BGL}(R)^+)$, we now have a definition of higher K -theory that is flexible enough to admit genuine computation. The $+$ -construction is tied to one specific category, the finitely generated projective modules over a single ring, and offers little leverage for relating the K -groups of one situation to another. The Q -construction takes an arbitrary exact category as input, and this is precisely what is needed to state the three structural theorems that drive almost every computation in the subject: the *resolution theorem*, which lets one replace a category of modules by a smaller subcategory adapted to a homological dimension bound; *déviissage*, which strips an abelian category down to its simple objects; and the *localization sequence*, which fits the K -theory of a Serre subcategory, an abelian category, and its quotient into a single long exact sequence. We state all three relative to the Q -construction K -groups $K_n(\mathcal{C})$ of definition 4.2.6, prove what is in scope, and cite [8] for the homotopy-theoretic core of those proofs that rests on Quillen's Theorem A and Theorem B. We then put the machinery to work: the K -theory of a field, recovered in low degrees by hand, and Quillen's computation of $K_*(\mathbb{F}_q)$, the first infinite family of higher K -groups ever determined.

Throughout, $\mathcal{C}$ denotes a (small) exact category in the sense of definition 4.2.1, and $K_n(\mathcal{C}) = \pi_{n+1}(BQ\mathcal{C})$ are its Q -construction K -groups (definition 4.2.6). When $\mathcal{C} = \mathrm{CP}(R)$ is the category of finitely generated projective R -modules with the split exact sequences, $K_n(\mathrm{CP}(R)) = K_n(R)$ recovers the K -groups of the ring (definition 4.1.1, definition 4.1.2), by the plus-equals- Q theorem. The notation $G_n(R)$ is reserved for the K -theory of the abelian category $\mathrm{Mod}_{fg}(R)$ of all finitely generated R -modules (defined when R is Noetherian, so that this category is abelian), the so-called *G-theory* of R .

The first theorem answers a question that already arose for K_0 : when can the projective modules be

traded for a larger or smaller class without changing the K -groups? The prototype is the comparison between projective modules and modules of finite projective dimension. Over a regular ring every finitely generated module has a finite projective resolution, so one expects the K -theory built from projectives to agree with the K -theory built from all finitely generated modules. The resolution theorem is the general mechanism behind such comparisons.

Theorem 4.4.1: Resolution Theorem

Let $\mathcal{C}$ be an exact category and $\mathbb{CP} \subseteq \mathcal{C}$ a full subcategory, closed under extensions, and equipped with the exact structure inherited from $\mathcal{C}$. Suppose:

1. $\mathbb{CP}$ is closed under kernels of admissible epimorphisms in $\mathcal{C}$; that is, if $0 \rightarrow C' \rightarrow P \rightarrow P'' \rightarrow 0$ is an admissible exact sequence in $\mathcal{C}$ with $P, P'' \in \mathbb{CP}$, then $C' \in \mathbb{CP}$.
2. Every object $C \in \mathcal{C}$ admits a finite $\mathbb{CP}$ -resolution: an admissible exact sequence

$$0 \rightarrow P_n \rightarrow P_{n-1} \rightarrow \cdots \rightarrow P_1 \rightarrow P_0 \rightarrow C \rightarrow 0$$

with each $P_i \in \mathbb{CP}$.

Then the inclusion $\mathbb{CP} \hookrightarrow \mathcal{C}$ induces a homotopy equivalence $BQ\mathbb{CP} \xrightarrow{\sim} BQ\mathcal{C}$, and hence isomorphisms

$$K_n(\mathbb{CP}) \xrightarrow{\cong} K_n(\mathcal{C})$$

for all $n \geq 0$.

The content of the theorem is that a homological dimension bound is invisible to K -theory: enlarging $\mathbb{CP}$ to the category $\mathcal{C}$ of all objects of finite $\mathbb{CP}$ -dimension changes nothing. The two hypotheses encode exactly what a resolution argument needs. Closure under kernels of admissible epimorphisms guarantees that the syzygies appearing in a resolution stay inside $\mathbb{CP}$, so the resolution can be built one step at a time; the existence of finite resolutions guarantees that the enlargement is no bigger than $\mathbb{CP}$ allows the homological algebra to reach. The principal application is to a regular ring: every finitely generated module over a regular Noetherian ring has a finite resolution by finitely generated projectives, so $K_n(R) \cong G_n(R)$ for R regular, the identification that makes G -theory computable through the localization and dévissage theorems below.

The homotopy equivalence $BQ\mathbb{CP} \xrightarrow{\sim} BQ\mathcal{C}$ is proved by analysing the Q -construction with Quillen's Theorem A, filtering $\mathcal{C}$ by the length of the shortest $\mathbb{CP}$ -resolution and showing each filtration step induces an equivalence; the argument is given in full in [8, Ch. V, Thm. 3.1]. What is fully in scope, and what already carries the whole geometric idea, is the statement on K_0 , where the maps and relations can be written down explicitly.

Proof:

The homotopy equivalence $BQ\mathbb{CP} \xrightarrow{\sim} BQ\mathcal{C}$ requires Quillen's Theorem A applied to the inclusion functor, filtering by resolution length; this is carried out in [8, Ch. V, Thm. 3.1], and we do not reproduce the simplicial homotopy theory here. We give the full argument on $\pi_1(BQ\mathcal{C}) = K_0$, which exhibits the mechanism and suffices for all of the K_0 -level applications.

Step 1: The induced map on K_0 is surjective. Recall from theorem 4.2.9 that $K_0(\mathcal{C})$ is the free abelian group on isomorphism classes $[C]$ of objects of $\mathcal{C}$, modulo the relations $[C] = [C'] + [C'']$ for every admissible exact sequence $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$, and similarly for $K_0(\mathbb{CP})$ using

the admissible sequences with terms in $\mathbb{CP}$. Given $C \in \mathcal{C}$, choose a finite $\mathbb{CP}$ -resolution

$$0 \rightarrow P_n \rightarrow \cdots \rightarrow P_0 \rightarrow C \rightarrow 0$$

Breaking it into short exact sequences $0 \rightarrow Z_i \rightarrow P_{i-1} \rightarrow Z_{i-1} \rightarrow 0$ with $Z_{-1} = C$ and $Z_i = \ker(P_{i-1} \rightarrow Z_{i-1})$, the closure hypothesis (1) forces every syzygy Z_i into $\mathbb{CP}$ (inductively from the top, $Z_{n-1} = P_n \in \mathbb{CP}$, and each Z_i is the kernel of an admissible epimorphism between objects already known to lie in $\mathbb{CP}$). The defining relation of $K_0(\mathcal{C})$ then gives, after telescoping,

$$[C] = \sum_{i=0}^n (-1)^i [P_i] \quad (4.1)$$

in $K_0(\mathcal{C})$. The right-hand side lies in the image of $K_0(\mathbb{CP}) \rightarrow K_0(\mathcal{C})$, so the map is surjective.

Step 2: A well-defined inverse on classes. Define $\chi: \{\text{objects of } \mathcal{C}\} \rightarrow K_0(\mathbb{CP})$ by the same formula, $\chi(C) = \sum_i (-1)^i [P_i]$ computed from any chosen finite $\mathbb{CP}$ -resolution. Independence of the resolution is the standard Schanuel argument: any two finite $\mathbb{CP}$ -resolutions of C have a common refinement obtained from the comparison of resolutions, and the alternating sums agree in $K_0(\mathbb{CP})$ because each elementary comparison step alters the resolution by a split short exact sequence of $\mathbb{CP}$ -objects, which contributes zero to the alternating sum. Hypothesis (1) is what keeps every term of every comparison inside $\mathbb{CP}$. Thus χ depends only on the isomorphism class of C .

Step 3: χ respects the relations and inverts the inclusion. If $0 \rightarrow C' \rightarrow C \rightarrow C'' \rightarrow 0$ is admissible in $\mathcal{C}$, splice chosen $\mathbb{CP}$ -resolutions of C' and C'' via the horseshoe lemma (valid here because $\mathbb{CP}$ is closed under extensions) to obtain a $\mathbb{CP}$ -resolution of C whose terms are $P'_i \oplus P''_i$; the alternating sum gives $\chi(C) = \chi(C') + \chi(C'')$. Hence χ descends to a homomorphism $K_0(\mathcal{C}) \rightarrow K_0(\mathbb{CP})$. On an object $P \in \mathbb{CP}$ the one-term resolution $0 \rightarrow P \rightarrow P \rightarrow 0$ gives $\chi(P) = [P]$, so $\chi \circ (\text{inclusion}) = \text{id}$ on $K_0(\mathbb{CP})$; and (4.1) shows $(\text{inclusion}) \circ \chi = \text{id}$ on $K_0(\mathcal{C})$. The inclusion therefore induces an isomorphism on K_0 , completing the proof in the case $n = 0$, with the higher-degree statement supplied by [8, Ch. V, Thm. 3.1].

The alternating sum (4.1) is an Euler characteristic: the class of an object equals the Euler characteristic of any finite resolution by the preferred subcategory. This is the same formula by which a coherent sheaf is identified with the alternating sum of a locally free resolution in the K -theory of a variety, and the resolution theorem is the precise statement that the identification is an isomorphism of K -groups, not only a surjection.

Example 4.4.2: K -theory of a Regular Ring

Let R be a regular Noetherian ring, so every finitely generated R -module has finite projective dimension and a finite resolution by finitely generated projectives. Take $\mathcal{C} = \text{Mod}_{fg}(R)$, the abelian category of finitely generated R -modules with its usual exact structure, and $\mathbb{CP} = \mathbb{CP}(R)$, the finitely generated projectives. Hypothesis (1) of theorem 4.4.1 holds unconditionally: a sequence $0 \rightarrow C' \rightarrow P \rightarrow P'' \rightarrow 0$ with P, P'' both projective splits, since P'' is projective, so C' is a direct summand of P and hence again projective. Hypothesis (2) is exactly regularity: finite projective dimension of every finitely generated module is equivalent to regularity for a Noetherian ring, and gives the required finite projective resolutions. The resolution theorem gives

$$K_n(R) = K_n(\mathbb{CP}(R)) \xrightarrow{\cong} K_n(\text{Mod}_{fg}(R)) = G_n(R)$$

for all $n \geq 0$. For $R = \mathbb{Z}$ this reads $K_n(\mathbb{Z}) \cong G_n(\mathbb{Z})$: the K -theory of free abelian groups agrees

with the G -theory of all finitely generated abelian groups. The equality $K_0(\mathbb{Z}) = G_0(\mathbb{Z}) = \mathbb{Z}$ is visible directly, since the class of a finite abelian group is 0 in $G_0(\mathbb{Z})$ (it has a finite resolution $0 \rightarrow \mathbb{Z}^k \rightarrow \mathbb{Z}^k \rightarrow M \rightarrow 0$ with vanishing Euler characteristic), leaving only the rank.

Déviissage moves in the opposite direction from resolution. Where resolution trades a category for a smaller one inside it via resolutions, déviissage trades an abelian category for a subcategory of its *simple* layers, reached not by resolving but by filtering. The motivating computation is the G -theory of a ring modulo a nilpotent ideal, or of the torsion modules over a Dedekind domain: every such module is built from the residue-field modules by a finite filtration, and déviissage says the K -theory sees only those residue-field layers.

Theorem 4.4.3: Déviissage

Let $\mathcal{A}$ be a small abelian category and $\mathcal{B} \subseteq \mathcal{A}$ a nonempty full abelian subcategory that is closed under subobjects, quotient objects, and finite products in $\mathcal{A}$. Suppose every object $A \in \mathcal{A}$ admits a finite filtration

$$0 = A_0 \subseteq A_1 \subseteq \cdots \subseteq A_m = A$$

by subobjects with each successive quotient A_j/A_{j-1} lying in $\mathcal{B}$. Then the inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$ induces a homotopy equivalence $BQ\mathcal{B} \xrightarrow{\sim} BQ\mathcal{A}$, hence isomorphisms

$$K_n(\mathcal{B}) \xrightarrow{\cong} K_n(\mathcal{A})$$

for all $n \geq 0$.

The closure conditions are what make a filtration usable: closure under subobjects and quotients guarantees that the layers of any filtration, and the layers of any comparison between filtrations, stay inside $\mathcal{B}$, so the combinatorics of the proof never leaves the subcategory. The conclusion is that an abelian category is K -theoretically determined by its simple objects together with the way arbitrary objects filter through them. The canonical instance takes $\mathcal{A}$ to be the finite-length modules over a commutative ring supported at a single maximal ideal $\mathfrak{m}$ and $\mathcal{B}$ the modules killed by $\mathfrak{m}$, that is, the $R/\mathfrak{m}$ -vector spaces; every finite-length module at $\mathfrak{m}$ has a composition series with $R/\mathfrak{m}$ -quotients, so déviissage computes the local torsion G -theory as the K -theory of the residue field.

The homotopy equivalence is Quillen's Theorem 4 of the foundational paper; the proof analyses the map of Q -construction nerves using Theorem A, and is reproduced in [8, Ch. V, Thm. 4.1]. As with resolution, the K_0 statement is fully in scope and exhibits the mechanism: the filtration produces the surjectivity and the relation $[A] = \sum_j [A_j/A_{j-1}]$.

Proof:

The homotopy equivalence on the full Q -construction spaces is [8, Ch. V, Thm. 4.1]; it rests on Quillen's Theorem A applied to the inclusion of $Q\mathcal{B}$ into $Q\mathcal{A}$, and we do not reproduce that simplicial argument. We prove the statement on $K_0(\mathcal{A}) = \pi_1(BQ\mathcal{A})$, which is fully in scope.

Step 1: Surjectivity on K_0 . Let $A \in \mathcal{A}$ with a finite filtration $0 = A_0 \subseteq \cdots \subseteq A_m = A$ and quotients $S_j = A_j/A_{j-1} \in \mathcal{B}$. Each inclusion $A_{j-1} \hookrightarrow A_j$ is part of an admissible exact sequence $0 \rightarrow A_{j-1} \rightarrow A_j \rightarrow S_j \rightarrow 0$ in the abelian category $\mathcal{A}$. The defining relation of $K_0(\mathcal{A})$

(theorem 4.2.9) gives $[A_j] = [A_{j-1}] + [S_j]$, and summing,

$$[A] = \sum_{j=1}^m [S_j] \quad (4.2)$$

with every $S_j \in \mathcal{B}$. Hence $[A]$ lies in the image of $K_0(\mathcal{B}) \rightarrow K_0(\mathcal{A})$, and the inclusion is surjective on K_0 .

Step 2: A left inverse. Define $\theta: \{\text{objects of } \mathcal{A}\} \rightarrow K_0(\mathcal{B})$ by $\theta(A) = \sum_j [S_j]$, computed from a chosen filtration. To see this is independent of the filtration, note that any two filtrations of A admit a common refinement (the Jordan–Hölder / Schreier refinement argument, valid in any abelian category with the given closure), and refining a filtration replaces a layer S by a two-step filtration $0 \rightarrow S' \rightarrow S \rightarrow S'' \rightarrow 0$ with $S', S'' \in \mathcal{B}$ (closure under subobjects and quotients), which leaves the sum $\sum [S_j]$ unchanged because $[S] = [S'] + [S'']$ in $K_0(\mathcal{B})$. Thus θ is well-defined on isomorphism classes.

Step 3: θ is additive and inverts the inclusion. Given an admissible $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ in $\mathcal{A}$, a filtration of A' followed by the preimage of a filtration of A'' is a filtration of A whose layers are those of A' together with those of A'' ; hence $\theta(A) = \theta(A') + \theta(A'')$, so θ descends to a homomorphism $K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B})$. On $B \in \mathcal{B}$ the one-step filtration gives $\theta(B) = [B]$, so θ restricts to the identity on $K_0(\mathcal{B})$; and (4.2) shows the composite $K_0(\mathcal{A}) \rightarrow K_0(\mathcal{B}) \rightarrow K_0(\mathcal{A})$ is the identity. The inclusion is therefore an isomorphism on K_0 , with the higher-degree statement furnished by [8, Ch. V, Thm. 4.1], completing the proof.

Equation (4.2) is again an Euler characteristic, now read off a filtration rather than a resolution: the class of an object is the sum of the classes of its filtration layers. The two theorems are genuinely dual in flavour, resolution descending through projective resolutions and dévissage ascending through composition series, and together they reduce the G -theory of a Noetherian ring to the K -theory of its residue fields once the localization sequence is available to organize the supports.

Example 4.4.4: Torsion Modules over a Dedekind Domain

Let R be a Dedekind domain and let $\mathcal{A} = \text{Mod}_{fg, \text{tors}}(R)$ be the abelian category of finitely generated torsion R -modules. Every such module is a finite direct sum of modules $R/\mathfrak{p}^k$ over nonzero primes $\mathfrak{p}$, and $R/\mathfrak{p}^k$ carries the filtration

$$0 \subseteq \mathfrak{p}^{k-1}/\mathfrak{p}^k \subseteq \cdots \subseteq \mathfrak{p}/\mathfrak{p}^k \subseteq R/\mathfrak{p}^k$$

whose successive quotients are all isomorphic to the residue field $R/\mathfrak{p} = \kappa(\mathfrak{p})$. Let $\mathcal{B} \subseteq \mathcal{A}$ be the subcategory of modules annihilated by some nonzero ideal that are semisimple, equivalently the finite products of residue-field modules $\kappa(\mathfrak{p})$; it is closed under subobjects, quotients, and finite products, and the filtrations just exhibited show every object of $\mathcal{A}$ filters with layers in $\mathcal{B}$. As $\mathcal{B}$ decomposes over the primes, $\mathcal{B} \cong \coprod_{\mathfrak{p}} \text{Mod}_{fg}(\kappa(\mathfrak{p}))$, so dévissage gives

$$G_n(\mathcal{A}) \cong \bigoplus_{\mathfrak{p} \neq 0} K_n(\kappa(\mathfrak{p}))$$

the sum over all nonzero primes. For $n = 0$ this recovers the divisor group: G_0 of the torsion modules is $\bigoplus_{\mathfrak{p}} \mathbb{Z}$, the free abelian group on the nonzero primes, which is exactly the group of divisors of R .

The third theorem is the engine that connects the previous two. Resolution and dévissage each compare two categories by an inclusion; the localization sequence instead relates a category, a subcategory, and the quotient, producing a long exact sequence rather than an isomorphism. It is the K -theoretic shadow of the fact that the Q -construction sends a short exact sequence of categories to a fibration of spaces.

Theorem 4.4.5: Localization Exact Sequence

Let $\mathcal{A}$ be a small abelian category and $\mathcal{B} \subseteq \mathcal{A}$ a *Serre subcategory*: a nonempty full subcategory such that for every admissible exact sequence $0 \rightarrow A' \rightarrow A \rightarrow A'' \rightarrow 0$ in $\mathcal{A}$, the object A lies in $\mathcal{B}$ if and only if both A' and A'' lie in $\mathcal{B}$. Let $\mathcal{A}/\mathcal{B}$ denote the quotient abelian category, with exact quotient functor $\mathcal{A} \rightarrow \mathcal{A}/\mathcal{B}$. Then there is a long exact sequence of abelian groups

$$\cdots \rightarrow K_n(\mathcal{B}) \rightarrow K_n(\mathcal{A}) \rightarrow K_n(\mathcal{A}/\mathcal{B}) \xrightarrow{\partial} K_{n-1}(\mathcal{B}) \rightarrow \cdots \rightarrow K_0(\mathcal{B}) \rightarrow K_0(\mathcal{A}) \rightarrow K_0(\mathcal{A}/\mathcal{B}) \rightarrow 0$$

in which the map $K_n(\mathcal{B}) \rightarrow K_n(\mathcal{A})$ is induced by the inclusion $\mathcal{B} \hookrightarrow \mathcal{A}$, the map $K_n(\mathcal{A}) \rightarrow K_n(\mathcal{A}/\mathcal{B})$ is induced by the quotient functor, and ∂ is the connecting homomorphism.

The exactness encodes a conservation principle: a class in $K_n(\mathcal{A})$ dies in the quotient $\mathcal{A}/\mathcal{B}$ exactly when it comes from the subcategory $\mathcal{B}$, and the failure of the quotient map to be surjective in degree n is measured by the image of ∂ in $K_{n-1}(\mathcal{B})$. The Serre condition is precisely what guarantees that the quotient $\mathcal{A}/\mathcal{B}$ is again abelian and that the quotient functor is exact, so that the K -theory of all three categories is defined and the sequence makes sense. The defining geometric input is Quillen's Theorem B: the sequence of Q -construction spaces $BQ\mathcal{B} \rightarrow BQ\mathcal{A} \rightarrow BQ(\mathcal{A}/\mathcal{B})$ is a homotopy fibration, and the long exact sequence of the theorem is the long exact homotopy sequence of that fibration, reindexed by $K_n(\mathcal{C}) = \pi_{n+1}(BQC)$. The proof that this sequence of spaces is a fibration is the most demanding part of Quillen's foundational paper; it is given in full in [8, Ch. V, Thm. 5.1], and we cite it rather than reproduce the Theorem B argument.

Proof Key ideas:

The proof is beyond the homotopy theory developed here: it requires Quillen's Theorem B to identify $BQ\mathcal{B}$ with the homotopy fibre of $BQ\mathcal{A} \rightarrow BQ(\mathcal{A}/\mathcal{B})$ over the basepoint, after first checking that the quotient $\mathcal{A}/\mathcal{B}$ is abelian and the quotient functor exact (which is where the Serre condition enters). The long exact sequence is then the homotopy long exact sequence of the fibration $BQ\mathcal{B} \rightarrow BQ\mathcal{A} \rightarrow BQ(\mathcal{A}/\mathcal{B})$ under the identification $\pi_{n+1}(BQC) = K_n(\mathcal{C})$. The connecting map ∂ is the fibration boundary. The full argument is [8, Ch. V, Thm. 5.1]. The right-hand end of the sequence, exactness at $K_0(\mathcal{A}/\mathcal{B})$ and the surjectivity $K_0(\mathcal{A}) \rightarrow K_0(\mathcal{A}/\mathcal{B}) \rightarrow 0$, is elementary from theorem 4.2.9: the quotient functor sends generators to generators and the relations of $K_0(\mathcal{A}/\mathcal{B})$ are images of relations in $K_0(\mathcal{A})$, so every class in $K_0(\mathcal{A}/\mathcal{B})$ lifts. This is as far as the elementary methods reach, completing the discussion.

The reason this sequence is so productive is that, combined with resolution and dévissage, it computes G -theory along a stratification. For a Noetherian ring R and a nonzerodivisor s , the modules supported on $\{s = 0\}$ form a Serre subcategory of $\text{Mod}_{fg}(R)$ whose quotient is $\text{Mod}_{fg}(R[1/s])$; the localization sequence then reads

$$\cdots \rightarrow G_n(R/s) \rightarrow G_n(R) \rightarrow G_n(R[1/s]) \rightarrow G_{n-1}(R/s) \rightarrow \cdots$$

after dévissage identifies the K -theory of the supported modules with $G_n(R/s)$. This is the algebraic

K -theory analogue of the Gysin sequence relating a space, an open subspace, and a closed complement, and it is the workhorse for inductive computations of G_n over Dedekind domains and beyond.

We turn now from structure to computation, beginning with the simplest input of all, a field. Here the higher machinery is not yet needed for the bottom two groups, which can be extracted by hand from the definitions of K_0 and K_1 already established. This anchors the abstract theory to a concrete answer and serves as the base case against which the finite-field computation is checked.

Example 4.4.6: K_0 and K_1 of a Field

Let F be a field. We compute the two lowest K -groups directly.

$K_0(F) = \mathbb{Z}$. Every finitely generated F -module is a finite-dimensional vector space, hence free, and two such are isomorphic if and only if they have the same dimension. The monoid $\mathcal{CP}(F)$ of isomorphism classes under $\oplus$ is therefore $(\mathbb{N}, +)$, generated by $[F]$, and is a cancellation monoid. Its Grothendieck completion (definition 4.1.2, definition 2.2.1) is

$$K_0(F) = \mathbb{Z}$$

freely generated by the class $[F]$, with $[V] = \dim_F V$ under the isomorphism. The same computation runs through the Q -construction: $\mathcal{CP}(F)$ has the split exact structure, and theorem 4.2.9 presents $K_0(\mathcal{CP}(F))$ on classes $[V]$ modulo $[V] = [V'] + [V'']$ for short exact sequences, which over a field are all split, again giving $\mathbb{Z}$ via dimension.

$K_1(F) = F^\times$. By definition (definition 3.1.1) and Whitehead's lemma (theorem 3.1.4),

$$K_1(F) = \mathrm{GL}(F)/E(F)$$

The determinant $\det: \mathrm{GL}(F) \rightarrow F^\times$ is surjective (the diagonal matrix $\mathrm{diag}(u, 1, \dots, 1)$ has determinant u) and kills every elementary matrix, since an elementary transvection has determinant 1; hence it factors through $K_1(F) \rightarrow F^\times$. This map is split by $u \mapsto [\mathrm{diag}(u, 1, \dots, 1)]$, so it is surjective. For injectivity, over a field every matrix of determinant 1 is a product of elementary matrices, that is $\mathrm{SL}(F) = E(F)$ (Gaussian elimination over a field uses only row operations of elementary type once the determinant is normalized). Therefore $\ker(\det) = \mathrm{SL}(F)/E(F) = 1$, and

$$K_1(F) \cong F^\times$$

the multiplicative group of the field. For $F = \mathbb{F}_q$ this already gives $K_1(\mathbb{F}_q) \cong \mathbb{F}_q^\times$, a cyclic group of order $q - 1$, the first value we will see reappear in Quillen's general formula.

The computation of $K_1(F) = F^\times$ used that $\mathrm{SL}(F) = E(F)$ over a field, which is the field case of the general fact that SK_1 measures the failure of elementary generation; for a field it vanishes. With K_0 and K_1 of $\mathbb{F}_q$ in hand, the natural question is what happens above degree 1. The answer, due to Quillen, was the first complete determination of an infinite family of higher K -groups, and it set the template for the subject: identify the relevant classifying space with a space built from topological K -theory, then read off homotopy groups.

Theorem 4.4.7: Quillen's Computation of $K_*(\mathbb{F}_q)$

Let $\mathbb{F}_q$ be the finite field with q elements. Then

$$K_0(\mathbb{F}_q) = \mathbb{Z}, \quad K_{2i-1}(\mathbb{F}_q) \cong \mathbb{Z}/(q^i - 1) \quad (i \geq 1), \quad K_{2i}(\mathbb{F}_q) = 0 \quad (i \geq 1)$$

In particular the higher K -groups are finite cyclic in odd degrees and vanish in positive even degrees.

The pattern is concrete. In degree 1 the formula reads $K_1(\mathbb{F}_q) = \mathbb{Z}/(q - 1) = \mathbb{F}_q^\times$, agreeing with example 4.4.6; in degree 3, $K_3(\mathbb{F}_q) = \mathbb{Z}/(q^2 - 1)$; in degree 5, $K_5(\mathbb{F}_q) = \mathbb{Z}/(q^3 - 1)$; and every positive even group vanishes. The method is the second of the two great ideas of the paper (the Q -construction being the first), and rests on comparing the algebraic classifying space with a space governed by the Adams operations of topological K -theory.

Proof Key ideas:

The full proof determines the mod- ℓ homology of $\mathrm{GL}(\mathbb{F}_q)$ for all primes ℓ and assembles it into the homotopy type of $\mathrm{BGL}(\mathbb{F}_q)^+$; it is long and uses the representation theory of $\mathrm{GL}_n(\mathbb{F}_q)$ together with the homology of the topological space BU . We describe the architecture and defer the computation to [8, Ch. IV, Thm. 1.13].

Step 1: The Brauer lift. The complex representations of $\mathrm{GL}_n(\mathbb{F}_q)$ are studied through the Brauer character, which lifts the eigenvalues of an $\mathbb{F}_q$ -linear map (roots of unity in $\mathbb{F}_q$) to roots of unity in $\mathbb{C}$ via a chosen embedding of the multiplicative group of $\mathbb{F}_q$ into $\mathbb{C}^\times$. Applied to the standard representation, this Brauer lift produces a map of spaces

$$\mathrm{BGL}(\mathbb{F}_q)^+ \longrightarrow BU$$

to the classifying space of the infinite unitary group, the representing space for complex topological K -theory in [5]. The construction is natural enough to interact with the Frobenius.

Step 2: Frobenius and the Adams operation ψ^q . The Frobenius $x \mapsto x^q$ on $\mathbb{F}_q$ induces an operation on the representation ring that the Brauer lift carries to the q -th Adams operation ψ^q on the topological K -theory of BU . The fixed points of Frobenius are exactly $\mathbb{F}_q$, so the algebraic data over $\mathbb{F}_q$ are matched with the data on BU fixed by ψ^q . Concretely, Quillen identifies $\mathrm{BGL}(\mathbb{F}_q)^+$ with the homotopy fibre of the self-map

$$\psi^q - 1: BU \longrightarrow BU$$

an identification proved by comparing mod- ℓ homology on both sides for every prime ℓ , including $\ell = p$ where the source has the homology of a point in the relevant range.

Step 3: Reading off homotopy groups. The homotopy groups of BU are $\pi_{2i}(BU) = \mathbb{Z}$ for $i \geq 1$ and $\pi_{2i-1}(BU) = 0$ (Bott periodicity, [5]). The Adams operation ψ^q acts on $\pi_{2i}(BU) = \mathbb{Z}$ by multiplication by q^i . The relevant stretch of the long exact homotopy sequence of the fibration $\mathrm{Fib}(\psi^q - 1) \rightarrow BU \xrightarrow{\psi^q - 1} BU$ around degree $2i$ is

$$\pi_{2i+1}(BU) \rightarrow \pi_{2i}(\mathrm{Fib}) \rightarrow \pi_{2i}(BU) \xrightarrow{q^i - 1} \pi_{2i}(BU) \rightarrow \pi_{2i-1}(\mathrm{Fib}) \rightarrow \pi_{2i-1}(BU)$$

in which the two outer terms $\pi_{2i+1}(BU)$ and $\pi_{2i-1}(BU)$ vanish, so the sequence reduces to

$$0 \rightarrow \pi_{2i}(\mathrm{Fib}) \rightarrow \mathbb{Z} \xrightarrow{q^i - 1} \mathbb{Z} \rightarrow \pi_{2i-1}(\mathrm{Fib}) \rightarrow 0$$

Since $q^i - 1 \neq 0$, multiplication by $q^i - 1$ on $\mathbb{Z}$ is injective with cokernel $\mathbb{Z}/(q^i - 1)$. Hence $\pi_{2i}(\text{Fib}) = \ker(q^i - 1) = 0$ and $\pi_{2i-1}(\text{Fib}) = \text{coker}(q^i - 1) = \mathbb{Z}/(q^i - 1)$. Under the identification of Step 2, $K_{2i}(\mathbb{F}_q) = \pi_{2i}(\text{BGL}(\mathbb{F}_q)^+) = 0$ and $K_{2i-1}(\mathbb{F}_q) = \pi_{2i-1}(\text{BGL}(\mathbb{F}_q)^+) = \mathbb{Z}/(q^i - 1)$ for $i \geq 1$, with $K_0(\mathbb{F}_q) = \mathbb{Z}$ from example 4.4.6. The verification that the Brauer lift induces the claimed homology isomorphism is the substance of [8, Ch. IV, Thm. 1.13], completing the discussion.

The computation is the prototype for every later K -theory calculation by “homotopy fibre of an Adams operation”: the same shape recurs in the étale and motivic descriptions of the K -theory of more general fields. The vanishing of $K_{2i}(\mathbb{F}_q)$ for $i \geq 1$ reflects that BU has no odd homotopy and that $\psi^q - 1$ is injective on the even homotopy; the odd groups $\mathbb{Z}/(q^i - 1)$ record exactly the cokernel of $q^i - 1$, the order of the cyclic group of q^i -power roots of unity caught by the i -th Bott class.

Example 4.4.8: $K_*(\mathbb{F}_q)$ in Low Degrees

Tabulating theorem 4.4.7 for the first several degrees:

n	$K_n(\mathbb{F}_q)$
0	$\mathbb{Z}$
1	$\mathbb{Z}/(q - 1)$
2	0
3	$\mathbb{Z}/(q^2 - 1)$
4	0
5	$\mathbb{Z}/(q^3 - 1)$
6	0

The degree-1 entry is the consistency check. By example 4.4.6, $K_1(\mathbb{F}_q) = \mathbb{F}_q^\times$, the multiplicative group of the field, which is cyclic of order $q - 1$. The formula gives $K_1(\mathbb{F}_q) = K_{2 \cdot 1 - 1}(\mathbb{F}_q) = \mathbb{Z}/(q^1 - 1) = \mathbb{Z}/(q - 1)$, and a cyclic group of order $q - 1$ is exactly $\mathbb{Z}/(q - 1)$. The two computations agree.

As a numerical instance, take $q = 2$. Then $K_1(\mathbb{F}_2) = \mathbb{Z}/1 = 0$ (the field $\mathbb{F}_2$ has trivial unit group), $K_3(\mathbb{F}_2) = \mathbb{Z}/3$, $K_5(\mathbb{F}_2) = \mathbb{Z}/7$, $K_7(\mathbb{F}_2) = \mathbb{Z}/15$, and all even higher groups vanish. For $q = 3$: $K_1(\mathbb{F}_3) = \mathbb{Z}/2$, $K_3(\mathbb{F}_3) = \mathbb{Z}/8$, $K_5(\mathbb{F}_3) = \mathbb{Z}/26$. The orders $q^i - 1$ grow geometrically, so unlike a field of characteristic 0, all higher K -groups of a finite field are finite.

Exercise 4.4.1

1. Let $\mathcal{C}$ be an exact category and $\mathbb{CP} \subseteq \mathcal{C}$ a resolving subcategory as in theorem 4.4.1. Suppose $C \in \mathcal{C}$ has two finite $\mathbb{CP}$ -resolutions of lengths n and n' . Using only the K_0 -level argument of the resolution theorem, show that the two Euler characteristics $\sum_i (-1)^i [P_i]$ and $\sum_i (-1)^i [P'_i]$ agree in $K_0(\mathbb{CP})$, and deduce that the class $[C] \in K_0(\mathcal{C})$ is independent of the chosen resolution.
2. Verify Quillen’s formula in degree 1 against the elementary computation. That is, show directly from definition 3.1.1 that $K_1(\mathbb{F}_q) \cong \mathbb{F}_q^\times$ is cyclic of order $q - 1$, and confirm this matches $K_{2i-1}(\mathbb{F}_q) = \mathbb{Z}/(q^i - 1)$ at $i = 1$. Then compute $K_3(\mathbb{F}_4)$ and $K_5(\mathbb{F}_4)$ explicitly as cyclic groups, giving their orders.
3. Let R be a Dedekind domain with fraction field F . Applying the localization sequence (theorem 4.4.5) to the Serre subcategory of finitely generated torsion modules inside $\text{Mod}_{fg}(R)$,

together with dévissage (theorem 4.4.3), write down the resulting long exact sequence in terms of $G_n(R)$, $K_n(F)$, and the residue fields $\kappa(\mathfrak{p})$. Identify the degree-0 portion and explain how it recovers the exact sequence $0 \rightarrow \mathbb{Z} \rightarrow G_0(R) \rightarrow \text{Cl}(R) \rightarrow 0$ relating $G_0(R)$ to the class group, given that $G_0(R) \cong K_0(R)$ for R regular.

4. In the proof of theorem 4.4.7, the Adams operation ψ^q acts on $\pi_{2i}(BU) = \mathbb{Z}$ by multiplication by q^i . Using only this fact and the long exact homotopy sequence of the fibration $\text{Fib}(\psi^q - 1) \rightarrow BU \rightarrow BU$, rederive that $\pi_{2i}(\text{Fib}) = 0$ and $\pi_{2i-1}(\text{Fib}) = \mathbb{Z}/(q^i - 1)$. Where does the hypothesis $q \geq 2$ (equivalently $q^i - 1 \neq 0$) enter, and what would go wrong at the formal value $q = 1$?
5. Let R be a regular Noetherian ring. Combining example 4.4.2 with the resolution theorem, explain why $K_n(R) \cong G_n(R)$ for all n , and why this isomorphism can fail when R is not regular. Give an explicit non-regular example where $K_0(R) \neq G_0(R)$, computing both sides.

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